Chapter 1: Man, and Medicine

“This chapter has never been asked before and doesn’t hold exam-relevant content. Don’t waste time here — you can skip it and focus on high-yield areas.”

Chapter-2: Concept of health and disease

Changing Concepts of Health and Disease

Biomedical concept

- Means "_________ **of disease”** also has basis in the "______ **theory of disease".**
- If **one was _____ from disease**, then the person was considered healthy.

Ecological concept

- Health is viewed as a _________ equilibrium between man and his environment, and disease a ____________ of the human organism to environment in this concept.
- "Health implies the relative absence of pain and discomfort and a continuous adaptation and adjustment to the environment to ensure optimal function".

	Human Development Index (HDI)	Physical Quality of Life Index (PQLI)
Components	• • •	• • •
Range		
Measures		• • •

Psychosocial concept

- Thus, **health is both a biological and social phenomenon.**

Holistic concept

- It is the synthesis of all the above concepts and emphasis of this concept is on the **promotion and protection of health.**

ssssssss

Spectrum of health

- The lowest point health disease spectrum is ______ and highest point corresponds to WHO definition of health.

- The transition from optimum health to ill-health is often _________, and where one state ends and the other begins is a matter of judgment.

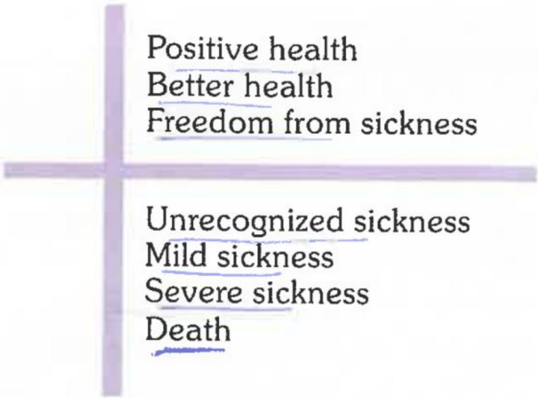


FIG. 2
The health sickness spectrum

Biological determinant of health

- It plays a particularly important role in __________ screening and _______ therapy.

- **Genetic stand-point, health defined** as that "state of the individual which is based upon the absence from the genetic constitution of such genes as correspond to characters that take the form of serious defect and derangement and to the absence of any aberration in respect of the total amount of chromosome material in the karyotype or stated in positive terms, from the presence in the genetic constitution of the genes

- Correspond to the ________ characterization and to the presence of a __________ karyotype".

- This is possible only when the person is allowed to live in ________________ with his environment an environment that transforms genetic potentialities into phenotypic realities.

Right to health

Key principles:

-
-
-

-
-

Characteristics of ideal health indicators

- ________, i.e., they should actually measure what they are supposed to measure
- ________ and _________, i.e., the answers should be the same if measured by different people in similar circumstances;
- _________, i.e., they should be sensitive to changes in the situation concerned,
- ________, i.e., they should reflect changes only in the situation concerned,

- _______, i.e., they should have the ability to obtain data needed, and;
- _______, i.e., they should contribute to the understanding of the phenomenon of interest.

Indicators of health

- _____________ **rate:** Indicates the rate at which people are dying.
- ____________ **rate:** Sensitive indicator of availability, utilization and effectiveness of health care.
- __________________ **rate:** Simplest measure of estimating disease burden in community.

Crude death rate
= (Number of deaths during the year / Mid-year population) × 1000

Infant mortality rate (IMR)
= (Number of deaths in the first year of life / Number of live births) × 1000

Maternal mortality ratio (MMR)
= (Number of maternal deaths during a given time period / Number of live births in the same period) × 100000

Maternal mortality rate
= (Number of maternal deaths / Number of women in the reproductive age group) × 100000

Under-5 mortality rate
= (Number of deaths under 5 years of age in a year / Total live births in that year) × 1000

Age-specific death rate
= (Number of deaths in specific age groups / Mid-year population of that age group) × 1000

Sex-specific death rate
= (Number of deaths per year of that sex / Mid-year population of that sex) × 1000

Case fatality rate
= (Total number of deaths due to a particular disease / Total number of cases due to the same disease) × 100

Proportional mortality from a particular disease
= (Number of deaths from a specific disease in a year / Total deaths from all causes in a year) × 100

Survival rate
= (Total number of patients alive after 5 years / Total number of patients diagnosed or treated) × 100

Morbidity indicators

- Attendance rates at out-patient departments, health centres
- Admission, readmission and discharge rates
- Duration of
- Spells of sickness or absence from work or school.

HALE (Health Adjusted Life Expectancy)

- Based on life expectancy at birth but includes an adjustment for ___________ in poor health.

- Equivalent number of years in full health that a newborn can expect to live based on current rates of ill-health and mortality.

QALY (Quality Adjusted Life years)

- Measure of disease burden including both _______ and __________ of life lived.
- Based on number of years of life that would be _______ by intervention.
- Each year in **perfect health** is assigned a value of ____ down to a value of _____ for **death**.

DFLE (Disability Free Life Expectancy)

Average number of years an individual is expected to live _______ of disability.

DALY (Disability adjusted life years)

- It is a **measure of ___________ burden**, expressed as a number of years lost due to ill-health, disability or early death.
- Used in field of public health and health impact assessment.
- **One DALY,** therefore, is **equal to one year of _____________.**

Formula: DALY =

Components:

- ________________ **:** Calculated from the number of deaths at each age multiplied by the expected remaining years of life according to a global standard life expectancy.
- ______________________ **:** where the number of incident cases due to injury and illness is multiplied by the average duration of the disease and a weighting factor reflecting the severity of the disease on a scale from 0 (perfect health) to 1 (dead).

Health care delivery indicators

- **Doctor-**
- Doctor-
- **Population-**
- Population per
- Population per

Socio-Economic indicators of Health

H	
-	
F	
L	
A	
G	
G	
E	
D	

Utilization rates

- Proportion of infants who are fully immunized against the __________ diseases.
- Proportion of pregnant women who receive antenatal care, or have delivery supervised by trained attendant.
- Percentage of population using various methods of _________________
- **Bed ______________**
- **Average _______________**
- **Bed ________________**

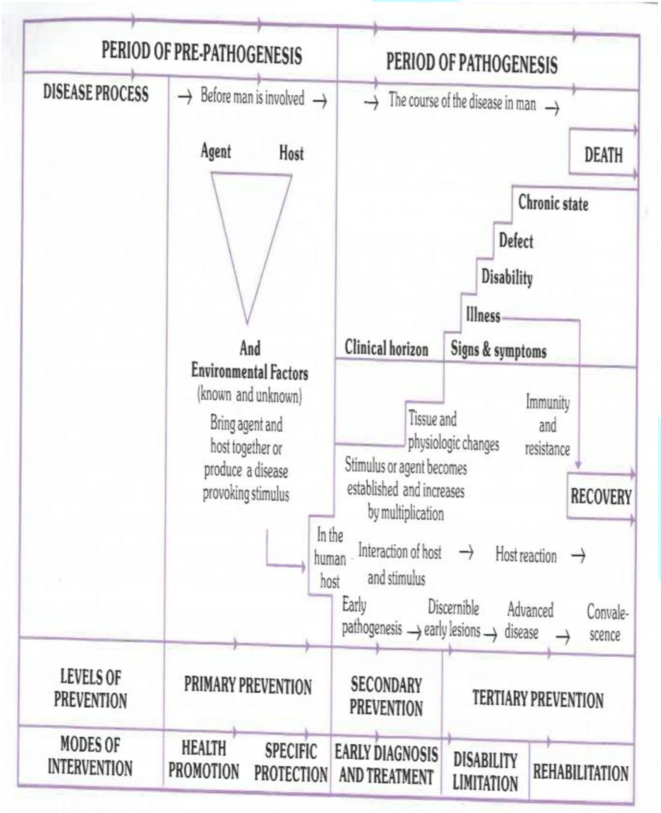
Characteristics of health care	
• A • A • A • A • A	• C • F

Ottawa charter

The Ottawa charter incorporates five key action areas in health promotion. **They are :**

-
-
-
-
-

Natural history of disease



__________**:** Ability of an infectious agent to invade and multiply (produce infection) in a host.

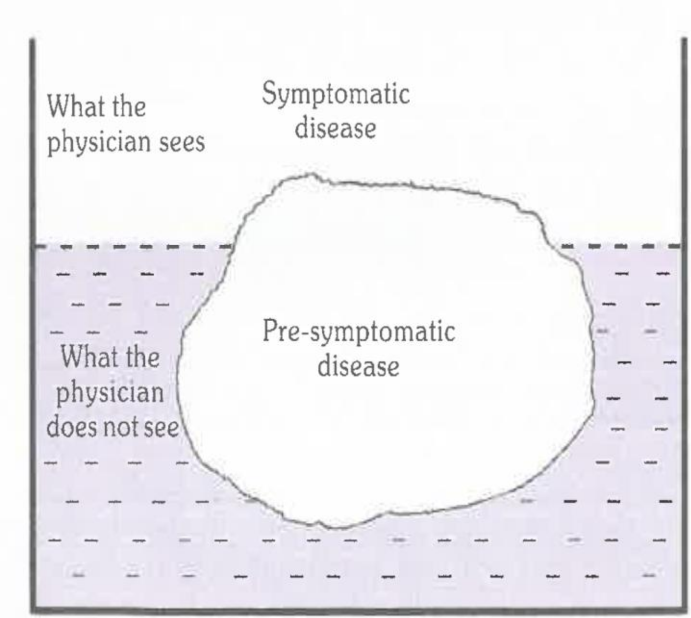
_____________: Ability to induce clinically apparent illness.

____________**:** Proportion of clinical cases resulting in severe clinical manifestations.

Iceberg of disease

- **Floating tip of iceberg** represents what the physician sees in the community, i.e., ____________
- The vast ___________ **portion** of iceberg represents hidden mass of disease, i.e., **latent, inapparent, presymptomatic** and **undiagnosed cases** and **carr**iers in community.

- The **waterline** represents **demarcation between ___________ and ______________ disease.**



Levels of Prevention

Level	Primordial	Primary	Secondary	Tertiary
Phase of disease	Underlying socio-economic & environmental conditions leading to causation			
Aim	Establish and maintain conditions that minimiz	Reduce the incidence of disease	Reduce prevalence by shortening	Reduce number an impact of complications

	e hazards to health		duration	
Actions	Inhibition of emergence of socio-economic, environmental and behavioural conditions	Protection of personal and community health by eliminating risks	Early detection and intervention to minimize disability	Soften the impact of long-term disease and disability, minimize suffering

Mode of intervention				
Targets	Total population or select groups	Total population or high risk individual/ groups	Individuals/ patients	Patients

Concept of Disability

The sequence of events leading to disability and handicap :

__________ → ____________ → ____________ →
____

_____________: "Any loss or abnormality of psychological, physiological or anatomical structure or function", e.g., loss of foot, defective vision or mental retardation.

__________: "Any restriction or lack of ability to perform an activity in the manner or within range considered normal for a human being."

___________: "A disadvantage for a given individual, resulting from an impairment or a disability, that limits or prevents the fulfilment of a role that is normal for that individual."

Accident............	Disease (or disorder)
Loss of foot	Impairment (extrinsic or intrinsic)
Cannot walk	Disability (objectified)
Unemployed	Handicap (socialized)

FIG. 11
Concept of disability

Notes

Chapter 3: Principles of Epidemiology and Epidemiological methods

Epidemiology

Defined as the study of the ____________________ of health-related events, states, and processes in _________ populations, including the study of the ___________ influencing such processes and the application of this knowledge to control relevant health problems.

Three components common to most of epidemiologist are:_______________:

- The basic measure of disease frequency is a ____________
- This is a ______ step in the development of strategies for prevention or control of health problems ________________:
- An important outcome of this study is **formulation of aetiological hypothesis.**
- This aspect of epidemiology is known as "___________ **epidemiology"**__________________:
- Test **aetiological hypothesis** and identify the underlying causes or risk factors of disease.
- This aspect of epidemiology is known as "___________ **epidemiology".**

Uses of Epidemiology

1. To study historically the __________ of disease in the population

2. ___________ diagnosis

3. _______ and _________

4. Evaluation of __________ risks and chances

5. _________ identification

6. Completing the ___________ of disease

7. ___________ for causes and risk factors

Epidemiology and clinical medicine:

- In **epidemiology** the **unit of study is a ______________ or population** at-risk; in **clinical medicine**, the unit of study is a **"_____" or "______"**.
- In **clinical medicine**, the physician is concerned with **disease in the __________ patient**, whereas the

epidemiologist is concerned with disease patterns in the entire __________, thus concerned with both the sick and healthy.

- In **clinical medicine,** the **patient comes to the ________**; in **epidemiology**, the investigation goes out into the ____________.
- **Clinical medicine** is based on ____________ **concepts**. In contrast, the subject matter of epidemiology is "___________".
- Clinical medicine and epidemiology are _____ **antagonistic.**

Measurements in Epidemiology:

- **_________:** Any piece of information referring to the patient or his disease is called a _______. This can be discrete, that is it can be present or absent, e.g., cancer lung, broken leg, or rash in measles or it can be continuously distributed, e.g., blood pressure, serum cholesterol, height, etc.
- **____________:** Any factor in the environment that might be suspected of causing a disease, e.g., air pollution, polluted water, etc

Tools of measurement:

1) ______

- It measures the occurrence of some particular event (development of disease or the occurrence of death) in a defined population during a given time period.

- The time dimension is usually a __________ year.

$$\textbf{DEATH RATE} = \frac{\textbf{Number of deaths in one year}}{\textbf{Mid-year population}} \textbf{ X 1000}$$

2) __________

- Expresses a relation in size between two random quantities.
- The numerator is not a component of the denominator.
- It is the result of dividing one quantity by another.
- It is expressed in the form of

$$\mathbf{X : Y \text{ or } \frac{\mathit{X}}{\mathit{Y}}}$$

3) ____________

- It is a ratio which indicates the relation in magnitude of a part of the whole.
- The numerator is always included in the denominator.

- A proportion is usually expressed as percentage.

Example :

$$\frac{\textbf{The number of childern with scabies at a certain time}}{\textbf{The total number of children in the village at the same time}} \textbf{ X 100}$$

Mortality

Limitation of mortality data:

- __________ reporting of deaths
- Lack of _________
- Lack of __________
- Choosing a _______ cause of death
- Changing coding system and changing fashion
- Disease with _____ fatality

Mortality Rates and Ratios

1) Crude death rate

- The simplest measure of ________ is the crude death rate.
- Defined as the number of deaths (from all causes) per 1000 estimates mid-year population in one year, in a given place.

$$\frac{\textit{Number of death during the year}}{\textit{Mid-year population}} \textbf{ X 100}$$

2) Specific death rate due to tuberculosis

$$= \frac{Number\ of\ deaths\ from\ tuberculosis\ during\ a\ calender\ year}{Mid-year\ population} X1000$$

3) Case fatality rate (Ratio)

- Represents the ______________ of a disease.
- It is simply the ratio of _______________
- The time interval is ________________
- Typically used in ________ infectious disease (e.g., food poisoning, cholera, measles)
- Case fatality is closely related to ___________

$$\frac{Total\ number\ of\ deaths\ due\ to\ a\ particular\ disease}{Total\ number\ of\ cases\ due\ to\ the\ same\ disease}\ \text{X}\ 1000$$

4) Proportional mortality rate (Ratio)

- Number of deaths due to a particular cause (or in a specific age group) per 100 (or 1000) total deaths.

Proportional mortality from a specific disease

$$= \frac{Number\ of\ deaths\ from\ the\ specific\ disease\ in\ a\ year}{Total\ deaths\ from\ all\ causes\ in\ that\ year}\ \text{X}\ 100$$

Under-5 proportionate mortality rate

$$= \frac{Number\ of\ deaths\ under\ 5\ years\ of\ age\ in\ the\ given\ year}{Total\ number\ of\ deaths\ during\ the\ same\ period}\ \text{X}\ 100$$

5) Survival rate

- It is the proportion of survivors in a group.
- Survival experience can be used as a yardstick for the assessment of ____________________

$$\text{Survival rate} = \frac{\textit{Total number of patients alive after 5 years}}{\textit{Total number of patients diagnosed or treated}} \times 100$$

Standardized mortality ratio (SMR)

- The **simplest and most useful form of __________ standardization** is the Standardized Mortality Ratio (SMR).
- It is a ________ (usually expressed as a percentage) of the total number of deaths that occur in the study group to the number of deaths that would have been expected to occur if that study group had experienced the death rates of a standard population.

$$\textbf{SMR} = \frac{\textbf{\textit{Observed deaths}}}{\textbf{\textit{Expected deaths}}} \textbf{ X 100}$$

- If ratio had value > 100, then the occupation would appear to carry a __________ mortality risk than that of the whole population.
- If the ratio had value < 100, then it would be proportionately ______.

INCIDENCE

- Defined as the number of ________ cases occurring in a defined population during a specified period of time.
- Incidence rate must include the ___________ used in the final expression.
- It is not influenced by the _____________ of the disease.
- The use of incidence is generally restricted to _________ conditions.

$$\textbf{Incidence} = \frac{\textit{Number of new cases of specific disease during a given time period}}{\textit{Population at-risk during that period}} \textbf{ X } \textbf{1000}$$

For **example**, if there had been 500 new cases of an illness in a population of 30,000 in a year, the incidence rate would be:

= 500/30,000 x 1000

= 16.7 per 1000 per year

Incidence rate refers to:

- Only to ___________
- During a given period (usually ___________)
- In a specified population or "population at risk", unless other denominators are chosen.

- It can also refer to new spells or episodes of disease arising in a given period of time, per 1000 population.

Special incidence rates:

A) Attack rate

It is an incidence rate (usually expressed as a percent), used only when the population is exposed to risk for a limited period of time such as during an epidemic.

$$\textbf{Attack rate} = \frac{\text{Number of new cases of a specified disease during a specified time interval}}{\text{Total population at risk during the same interval}} \times 100$$

B) Secondary attack rate

Defined as the number of __________ persons developing the disease within the range of the ______________ period following exposure to a __________ case.

Uses Of Incidence Rate

- Useful for taking action to ___________ disease and for ___________.
- If the incidence rate is increasing, it might indicate _________ or __________________ of the current control programmes.

- Rising incidence rates might suggest the need for a new disease control or preventive programme, or that reporting practices had improved.

- A change or fluctuation in the incidence of disease may also mean a change in the ____________ of disease.

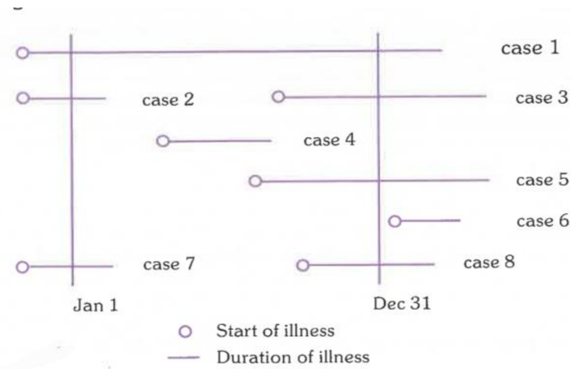
PREVALENCE

It refers specifically to all _________ cases (___________) existing at a given point in time, or over a period of time in a given population.

A) Point prevalence

Defined as the number of all _________ cases (___________) of a disease at ______________ of time, in relation to a defined population.

$$= \frac{\text{Number of all current cases (old and new) of a specified disease existing at a given point in time}}{\text{Estimated population at the same point in time}} \times 100$$



Incidence would include cases – 3,4,5, and 8
Point prevalence (Jan 1) cases – 1,2, and 7
Point prevalence (Dec.31) cases – 1,3,5 and 8
Period prevalence (Jan–Dec) cases – 1,2,3,4,5,7, and 8

Number of cases of a disease beginning, developing and ending during a period of time

B) Period prevalence

It measures the ________ of all current cases (old and new) existing during a defined period of time (e.g., annual prevalence) expressed in relation to a defined population =

$$\frac{\text{Number of.existing cases (old and new)of a specified disease during a given period of time interval}}{\text{Estimated mid–interval population at–risk}}$$

X100

Relationship between prevalence and incidence:

P = __________

= ___________ x _____________

Uses of prevalence:

- Prevalence helps to estimate the __________ of health/ disease problems in the community, and identify potential high-risk populations
- Prevalence rates are especially useful for ____________ and ___________ purposes, e.g., hospital beds, manpower needs, rehabilitation facilities, etc

Epidemiologic Methods

Observational studies
• **Ecological or Correlational:** ____________ as unit of study • **Cross-sectional or Prevalence:** _________ as unit of study • **Case-control or Case-reference:** _________ as unit of study • **Cohort or Follow-up:** ___________ as unit of study

Experimental studies Intervention studies	
Randomized controlled trials or Clinical trials	_________ as unit of study
Field trials	______________ as unit of study
Community trials or Community intervention studies	___________ as unit of study

Short-term fluctuations

- The best-known short-term fluctuation in the occurrence of a disease is an ____________.
- It is defined as "the occurrence in a community or of cases of an illness or other health-related events region clearly in excess of ____________________

Main features of a "point-source" epidemic are:

- The epidemic curve _________________, with ____ secondary waves
- The epidemic tends to be __________, there is clustering of cases within a _________ interval of time.
- All the cases develop within _____ incubation period of disease.

Main features of a Continuous or repeated exposure are:

- _____ evidence of secondary cases among persons who had contact with ill persons.

- Epidemic reaches a _________ but tails off ________ over a longer period of time.

<table>
<tr><th colspan="3">Types of Epidemic</th></tr>
<tr><td rowspan="2">Common-source epidemics</td><td></td><td>The exposure to the disease agent is brief and essentially</td></tr>
<tr><td>Continuous or Multiple exposure

epidemics</td><td>Exposure from the same source may be prolonged - continuous, repeated or intermittent – not necessarily at the same time or place.
E.g.,</td></tr>
<tr><td rowspan="3">sPropagated epidemics</td><td>Person-to-person</td><td></td></tr>
<tr><td>Arthropod vector</td><td rowspan="2"></td></tr>
<tr><td>Animal reservoir</td></tr>
<tr><td colspan="3">Slow (modern) epidemics</td></tr>
</table>

Main features of a Propagated epidemics are:

- __________ rise and tails off over a much ______ period of time.
- The speed of spread depends upon _____________, opportunities for __________ and ___________________

Cross-sectional study

- It is the simplest form of an _____________ study, also known as "____________ study".
- Based on a ________ examination of a cross-section of population at one point in time.
- They are more useful for _________ than short-lived diseases.

Case Control Study

- It is often called ______________ studies.
- The unit is the ___________ rather than the ________.
- The focus is on a disease or some other health problem that has __________ developed.
- Some investigators select cases from one source and controls from more than one source to avoid the influence of "________ _____".

Its distinct features are:

a) Both exposure and outcome (disease) have occurred _______ the start of the study

b) The study proceeds _____________ from effect to cause; and

c) It uses a control or comparison group to support or refute an inference.

Matching

- It is defined as the process by which we select controls in such a way that they are ________ to cases with regard to certain pertinent selected variables.

- A ___________________ is defined as one which is associated both with exposure and disease and is distributed unequally in study and control groups.

Relative risk or risk ratio

It is defined as the ratio between the incidence of disease among ____________ persons and incidence among ______________

A case control study of smoking and lung cancer

	Cases (with lung cancer)	Controls (without lung cancer)	Total
Smokers (less than 5 cigarettes a day)	33 (a)	55 (b)	88 (a+b)
Non-smokers	2 (c)	27 (d)	29 (c+d)
Total	35 (a+c)	82 (b+d)	n = a+b +c+d

$$\textbf{Relative risk} = \frac{\text{Incidence among exposed}}{\text{Incidence among non-exposed}}$$

$$\textbf{Relative risk} = \frac{a}{(a+b)} + \frac{c}{(c+d)}$$

Odds Ratio (Cross-product ratio)

	Diseases: Yes	Diseases: No
Exposed	a	b
Not exposed	c	d

Odds ratio = ad/bc

$$\textbf{Odds ratio} = \left(\frac{a}{b}\right) / \left(\frac{c}{d}\right) = \frac{ad}{bc}$$

$$= \frac{33\ X\ 27}{55\ X\ 2} = 8.1$$

Bias in case control studies

Bias is any systematic error in the determination of the ____________ between the exposure and disease.

Types of biases are:

- Bias due to ______________: Removed by matching.
- __________ or _________ bias
- ____________ bias
- _______________ bias
- ________________ bias: Eliminated by double-blinding

Examples of case control studies

- Adenocarcinoma of vagina
- Oral contraceptives and thromboembolic disease
- Thalidomide tragedy

COHORT STUDY

- Known as: _____________ study, ___________ study, __________ study, and _________________ study.
- In a cohort study the exposure has __________ but the disease _________.

The distinguishing features of cohort studies are:

- Cohorts are identified _________ to the appearance of the disease under investigation
- Study groups, so defined, are observed over a period of time to determine the frequency of disease among them.
- The study proceeds __________ from cause to effect.

Indications for cohort studies

- There is good evidence of an association between exposure and disease
- When exposure is ___, but the incidence of disease ___ among exposed
- When attrition of study population can be _______________
- When ample _________ are available

Types of cohort studies

- Prospective cohort studies
- Retrospective cohort studies, and
- A combination of retrospective and prospective cohort studies.

Examples of cohort studies

- Smoking and lung cancer
- The Framingham heart study
- Oral contraceptives and health

RELATIVE RISK

- It is a direct measure (or index) of the "_________" of the association between suspected cause and effect.
- A relative risk of _____ indicates no association; _____ suggests "positive" association between exposure and the disease under study.

- A relative risk of ___ indicates that the incidence rate of disease is 2 times higher in the exposed group as compared with the unexposed.

$$RR = \frac{\text{Incidence of disease (or death) among exposed}}{\text{Incidence of disease (or death) among non-exposed}}$$

ATTRIBUTABLE RISK

Cigarette smoking	Developed lung cancer	Did not devel[op] lung cancer
Yes	70 (a)	6930 (b)
No	3 (c)	2997 (d)

Incidence rates

(a) among smokers = 70/7000 =

(b) among non-smokers = 3/3000 = 1

Statistical significance : $P < 0.00$

$$= \frac{\text{Incidence of disease rate among exposed} - \text{incidence of disease rate among nonexposed}}{\text{Incidence rate among exposed}} \times 100$$

$$= \frac{10 - 1}{10} \times 100 = 90 \text{ per cent}$$

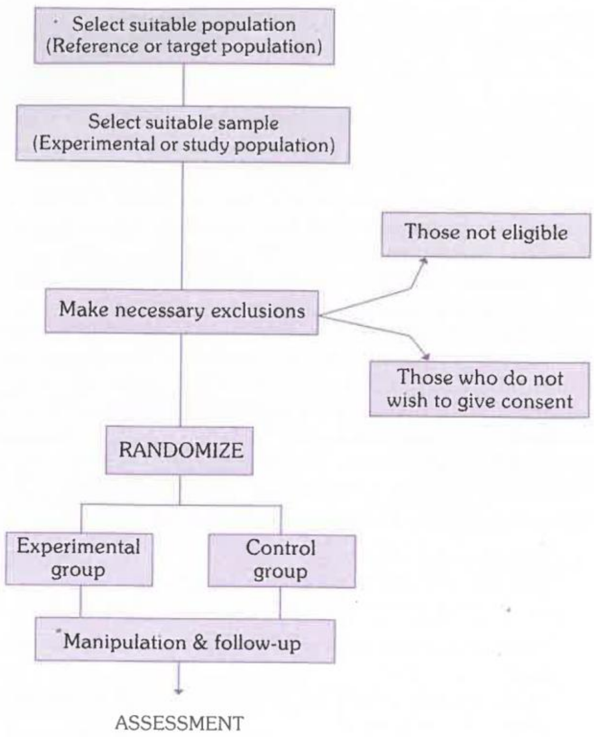
Case control study	Cohort study
	Proceeds from "cause to effect"
	Starts with people exposed to risk factor or suspected cause.
	Tests whether disease occurs more frequently in those exposed, than in those not similarly exposed.
	Reserved for testing of precisely formulated hypothesis.
	Involves larger number of subjects.
	Long follow-up period often needed. involving delayed results.
	Inappropriate when the disease or exposure under investigation is rare.
	Yields **incidence rates**, **RR** as well as **AR**
	Can yield information about more than one disease outcome
	Expensive

Randomized Controlled Trials

- Randomization is the "_________" of a control trial.
- Randomizations eliminate what is known as "_______________".
- By random allocation, every individual gets an _____ chance of being allocated into either group or any of the trial groups.
- It is best done by using a table of ___________ numbers

The basic steps in conducting a RCT include the following:

1.
2.
3.
4.
5.
6.



Study Designs of controlled trials

Concurrent parallel study designs

- Comparisons are made **between** ________________ **assigned groups**.
- One group __________ to specific treatment, and the other group **not exposed.**
- Patients remain in the study group or the control group **for the** _________ of the investigation.

Cross-over type of study designs

- **Each patient serves as his own ________** and are randomly assigned to a study and control group.

- **_________ group** receives **treatment** under consideration while **______ group** receives some alternate form of active treatment or **placebo**.

- The two groups are observed over time and then each group are taken off their medication or placebo to allow for elimination of medication from body and for possibility of any "carry over" effects.

- After this period of medication, the two groups are __________.

Advantages:

- All patients can be assured that sometime during the course of investigation, they will receive the ____________

- Such studies generally economize on the total number of patients required at the expense of the time necessary to complete the study.

Study not suitable if:

- If the drug of interest ________ the disease

- If the drug is ____________ only during a ________ stage of the disease

- If the disease changes _________ during the period of time required for the study.

Randomized Controlled Trials	Non- Randomized Trials
Types	

Association and Causation

- Association may be defined as the concurrence of ___ variables more often than would be expected by chance.
- Association does not necessarily imply a _______ relationship.
- Correlation indicates the degree of association between ______ characteristics.
- The correlation coefficients range from _____ to ______
- The correlation coefficients of 1.0 means that the two variables exhibit a perfect ________ relationship.

Causal relationship is increased by the presence of the following criteria:

- __________ association
- ________ of association
- ___________ of the association
- __________ of the association
- Biological ___________
- ___________ of the association

Infectious Disease Epidemiology

________: The entry and development or multiplication of an infectious agent in an organism, including the body of man or animals.

____________: The presence of an infectious agent on a body surface; also, on or in clothes, beddings, toys, surgical instruments or dressings, or other inanimate articles or substances including water, milk and food.

__________: For persons or animals, the lodgment, development and reproduction of arthropods on the surface of the body or in the clothing, e.g., lice, itch, mite.

Host:

- _________ **host:** Means the only host, e.g., man in measles and typhoid fever.
- _________ **or** _________ **hosts:** in which the parasite attains maturity or passes its sexual stage
- _________ **or** ____________ **hosts:** Those in which the parasite is in a larval or asexual state.
- __________ **host:** Is a carrier in which the organism remains alive but does not undergo development.
- __________ **Disease:** A disease due to an infectious agent.
- __________ **Disease:** A disease that is transmitted through contact. Examples include scabies, trachoma, STD and leprosy.

- __________: The occurrence in a community or region of cases of an illness, specific health- related behaviour, or other health-related events clearly in excess of normal expectancy.

- ___________: Refers to **constant presence** of a disease or infectious agent within a given geographic area or population group, **without importation from outside.** Also defined as **"Usual"** or expected frequency of the disease within such an area or population group.

- **"_______________":** Expresses that the **disease is constantly present at a high incidence and/or prevalence** rate and affects all age groups equally.

- **"______________":** High level of infection beginning early in life and affecting most of the child population, leading to a state of equilibrium such that the adult population shows evidence of the disease much less commonly than do the children, as in the case of malaria.

- ___________: It means scattered about. The cases occur irregularly, haphazardly from time to time, and generally infrequently. The cases are so few and separated widely in space and time that they show little or no connection with each other. E.g., tetanus, herpes-zoster and meningococcal meningitis.

- **__________:** An epidemic occurring over a very wide area, crossing international boundaries, and usually affecting a large number of people. Examples are influenza pandemics and cholera pandemics.

- **__________:** Diseases which are imported into a country in which they do not otherwise occur.

- **___________:** An infection or infectious disease transmissible under natural conditions from vertebrate animals to man.

- ____________ **Infection:** Nosocomial (hospital acquired) infection is an infection originating in a patient while in a hospital or other health care facility.

- ____________ **Infection:** Infection with organism(s) that are normally innocuous but become pathogenic when the body's immunological defenses are compromised, as in AIDS.

- **______________:** Termination of all transmission of infection by extermination of the infectious agent. It is "all or none" phenomenon, restricted to termination of an infection from the whole world. To-date only one disease has been eradicated, that is smallpox. Diseases which are amenable to eradication are measles, diphtheria, polio and guinea worm.

Dynamics of disease transmission

Source and reservoir

- ________________ is defined as "the person, animal, object or substance from which an infectious agent passes or is disseminated to the host".

- _____________ is defined as "any person, animal, arthropod, plant, soil or substance (or combination of these) in which an infectious agent lives and multiplies on which it depends primarily for survival, and where it reproduces itself in such manner that it can be transmitted to a susceptible host.

- In **______________ infection**, the reservoir is man, but 'the source of infection is the soil contaminated with infective larvae.

- In __________ the reservoir and source are the same, that is soil.

Cases

- __________ **case** refers to the first case of a communicable disease introduced into the population unit being studied.
- ________ **case** refers to the first case to come to the attention of the investigator; it is not always the primary case.
- ___________ **cases** are those developing from contact with primary case.
- __________ **case** is an individual (or a group of individuals) who has all the signs and symptoms of a disease or condition yet has not been diagnosed as having the disease or had the cause of the symptoms connected to the suspected pathogen.

Carriers

- A carrier is defined as "an _________ person or animal that harbours a specific infectious agent in the _________ of discernible clinical disease and serves as a potential source of infection for others".
- "________________" is a classic example of a carrier.

Types of Carriers

Pseudo-carriers: Carriers of _________ organisms are called pseudo-carriers.

Incubatory Carriers:

- Shed the infectious agent during the __________ period of disease.

- Capable of infecting others before the onset of illness.
- E.g.,

Convalescent Carriers:

- Continue to shed the disease agent during the period of _______________.
- E.g.,

Healthy Carriers :

- Healthy carriers emerge from ____________ cases.
- They are victims of subclinical infection who have developed carrier state without suffering from overt disease, but are nevertheless shedding the disease agent.
- E.g.,

Epidemiological classification of vector-borne diseases

By Transmission chain:

- **Man and a non-vertebrate host**
  - Man-arthropod-man (e.g., __________)
  - Man-snail-man (e.g., ________________).
- **Man, another vertebrate host, and a non-vertebrate host**
  - Mammal-arthropod-man (__________)
  - Bird-arthropod- man (____________).
- **Man and 2 intermediate hosts**
  - Man-cyclops-fish- man (______________)
  - Man- snail-fish- man (_________________)
  - Man-snail-crab-man (________________-).

Biological transmission:

The infectious agent undergoing replication or development or both in vector. **Its types are:**

- _______________: The agent multiplies in vector, but no change in form, e.g., _____________________

- ___________________: The agent changes in form and number, e.g., ______________________________

- ______________________: The disease agent undergoes only development but no multiplication, e.g., _____________________

- _________________________: When the infectious agent is transmitted vertically from the infected female to her progeny in the vector.

- _________________________: Transmission of the disease agent from one stage of the life cycle to another as for example nymph to adult.

Airborne Transmission

- Airborne transmission refers to the spread of infectious agents via droplet nuclei or dust particles that remain suspended in the air and can be inhaled by others.

- Diseases spread by droplet nuclei include ___________________________ ____ ______

- Diseases carried by infected dust include ________________ and __________________ infection, ____________, ______________, _________ and _______________.

Susceptible Host

Incubation period:

- The time interval between ________ by an infectious agent and __________ of the first sign or symptom of the disease in question.

- **_________ incubation period**, defined as the time required for 50 per cent of the cases to occur following exposure.

Serial interval: The gap in time between the onset of the ________ case and the ____________ case is called the "serial interval".

Generation time: It is defined as "the interval of time between _________ of infection by a host and maximal ____________ of that host".

Communicable period: Defined as "the time during which an infectious agent may be ___________ directly or indirectly from an __________ person to another person, from an __________ animal to man, or from an _________ person to an animal, including ___________.

Secondary attack rate: It is defined as "the number of __________ persons developing the disease within the range of the __________ period, following exposure to the __________ case". The primary case is __________ from both numerator and denominator.

$$\text{SAR} = \frac{\text{Number of exposed persons de}\ldots\ \text{the disease within the range of}\ldots\ \text{incubation period}}{\text{Total number of exposed/“susce}\ldots\ \text{contacts}}$$

Host Defenses

- Maternal antibody transmitted to infant is gradually lost over a period of ___________
- This protective “_____________” is due to the presence of ____ levels of immunoglobulins _____ and especially _____ in the cord blood and plasma of infants born of immune mothers.

The Immune Response

Primary Response:

- Occurs when an **antigen is administered for the ______ time** to an individual who has never been exposed to it before.
- There is a **latent period of _______ days** before antibodies appear in the blood.
- The first antibody elicited is of **_____ type.** The IgM antibody titre rises steadily during the next 2-3 days or more, reaches a peak level and then declines almost as fast as it developed.

- Meanwhile, if the antigenic stimulus was sufficient, ____ **antibody appears within 7-10** days and then gradually falls over a period of weeks or months.

- The antigenic dose required for the induction of IgG is about ____ times that which is required to induce IgM antibody.

- There is production of what are known as **"_________ cells" or "_______ cells"** by both B and T lymphocytes.

Secondary (Booster) Response:

Differs from the primary response:

- _________ latent period.
- Production of antibody more _________.
- Antibody _______ abundant.
- Antibody response maintained at _____ levels for a _____ period of time
- The antibody elicited tends to have a greater avidity or capacity to bind to the antigen.

Vaccines

- Live vaccines should not be administered to persons with immune deficiency diseases or to persons whose immune response may be suppressed because of leukaemia, lymphoma or malignancy or because of therapy with corticosteroids, alkylating agents, antimetabolic agents, or radiation.

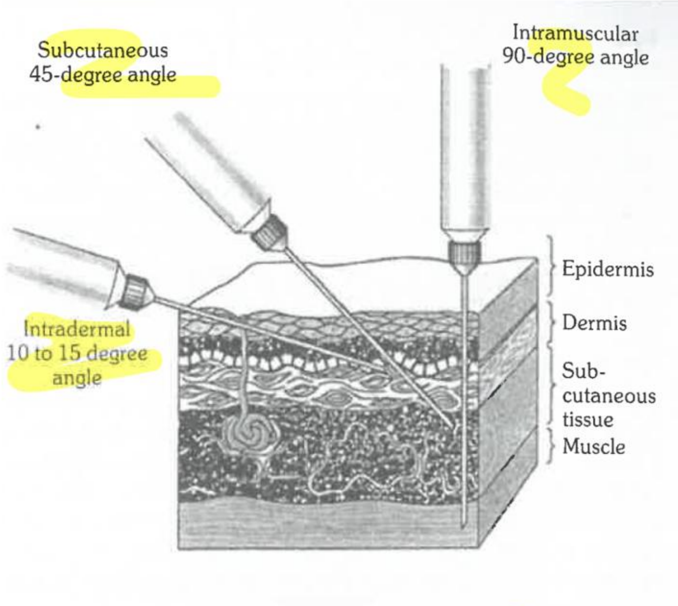
- Pregnancy is another ________________ unless the risk of infection exceeds the risk of harm to the foetus of some live vaccines.
- When two live vaccines are required, they should be given either simultaneously at ___________ sites or with an interval of atleast _ weeks.
- In the case of live vaccines, protection is generally achieved with a _____ dose of vaccine.

Type	Vaccines
Live attenuated	

Killed whole organism	
Toxoid/Protein	
Polysaccharide	
Glycoconjugate	
Recombinant	

Characteristic	Killed Vaccine	Live Vaccine
Number of doses		Single
Need for adjuvant		No
Duration of immunity		Longer
Effectiveness of protection		Greater
Immunoglobulins produced		IgA and IgG
Mucosal immunity produced		Yes
Cell-mediated immunity produced		Yes
Residual virulent virus in vaccine		No
Reversion to virulence		Possible
Excretion of vaccine virus and Transmission to non-immune contacts		Possible
Interference by other viruses in host		Possible
Stability at room temperature		Low

Recommended route of administration of vaccines	
Oral	
Intramuscular (IM)	
Subcutaneous (SC)	
Intradermal (ID)	
Intranasal spray	



Technique of giving vaccine injection

Sensitivity to light and heat

- Vaccines sensitive to light are BCG, measles, measles-rubella, measles-mumps-rubella and rubella.

- These vaccines are supplied in dark glass vials that give them some protection.

Group A	**Most sensitive to heat**	•
Group B		•
Group C		• • JE (freeze-dried) • Measles or MR or MMR (freeze-dried)

Group D		• • DTaP-Hep. B-Hib-IPV (hexavalent) • DTwP or DTwP-Hep. B-Hib (pentavalent) • Hib (liquid) • Measles (freeze-dried) • Rotavirus (liquid and freeze-dried) • Rubella (freeze-dried) • Yellow fever (freeze-dried)

Group E	**Least sensitive to heat**	• • • •
Group F		• • • •

Cold chain equipment

1) ________________: It maintains a temperature between -15°C to -25°C. WIF are usually installed at national, state and regional vaccine stores. It is used for bulk storage of OPV vaccine.

2) ________________: Maintain a temperature of +2°C to +8°C. For storage of large quantities of all UIP vaccines like BCG, hepatitis B, DPT, pentavalent, IPV, measles. Installed at government medical store depots, state and regional vaccine stores.

3) ________________: DF has top opening lid to prevent loss of cold air during door opening. Maintained temperature between -15° to -25°C. This is used for storing of OPV vaccine for 3 months (district level and above only) and also for freezing of ice packs (at sub-district level only).

4) Ice lined refrigerator (ILR):

- Maintain a cabinet temperature between ___________ and are used to store vaccines at district and sub-district level.

- These types of refrigerators are ____________

- It can keep vaccine safe with minimum ___ hours continuous electricity supply in a 24-hour period.

- Based on the temperature zone, inside of the ILR can be divided into ___ parts, ___________ and ____________-

- The _______ part is cooler compared to the ______ part. Hence _____ part is preferred location for storing the freeze sensitive vaccines.

- Vaccines like OPV, BCG, measles, and JE can be kept at ________ of the basket while DPT, TT, hep B, IPV and pentavalent vaccines and diluents are kept in ___________ of the basket.

- These vaccines should never be kept directly on the ________ of the refrigerator as they can freeze and get damaged.

Hold-over time of the equipment

- It is defined as the "______ taken by the equipment to ______ the inside cabinet temperature from its temperature at the time of _________, to maximum temperature limit of its recommended range."

- For example, in case of ILR if the cabinet temperature is +4°C at the time of power-cut, then the time taken to reach +8°C from +4°C will be hold-over time for that ILR.

Factors affecting Hold-over time:

- __________ temperature: More the ambient temperature less will be the hold-over time
- **_________- of opening of lid and use of basket**
- __________ of vaccines kept inside with adequate space between the containers (Equipment empty/ loaded)
- Condition of ice-pack ________ (Frozen/ partially frozen/melted).

5) ___________: Are supplied to all peripheral centers. These are used mainly for transportation of the vaccines.

6) _____________: Are used to carry small quantities of vaccines (16-20 vials) for the out of reach sessions. 4 fully frozen icepacks are used.

7) ____________: are used to carry small quantities of vaccines (6-8 vials) to a nearby session. Two fully frozen packs are to be used.

Open Vial Policy

Applicability

- It applies only for ______, ____, ___________, ___________ Vaccine (OPV), Liquid _____________, ________ and injectable _______.

(Mnemonic: Three letters: DPT, OPV, IPV, PCV, HBV for hepatitis B except BCG)

- **Does not apply to** Measles/MR, ______, _______________________ vaccines, ______________ vaccine.

Conditions for Use of Open Vials

Vials opened in session-site used in **>1 immunization session up to ___ weeks** if:-

- ___________ date not passed.
- Vaccines stored under appropriate temperature range both during transportation & storage at a cold chain point.
- The vaccine vial septum has not been _____________ in water or contaminated in any way.

- __________ technique has been used to withdraw all doses.
- The vaccine vial monitor (VVM) has not reached the ________ point.

Recording and Return

- Record date and time of opening for all open vials.
- Return open vials to the Cold Chain Point after the session.

Segregation and Disposal

- At Cold Chain Point, open vials are segregated into Reusable (DPT, TT, Hep B, Pentavalent) and Non-reusable (Measles/MR, BCG &JE) vials.

- **Non-reusable open vials of BCG, Measles and JE** should be **destroyed after 48 hours or before the next session,** whichever is earlier.

Reporting of adverse events

- Store all open vials properly until the investigation is complete, if any adverse event following immunisation (AEFI) is reported.

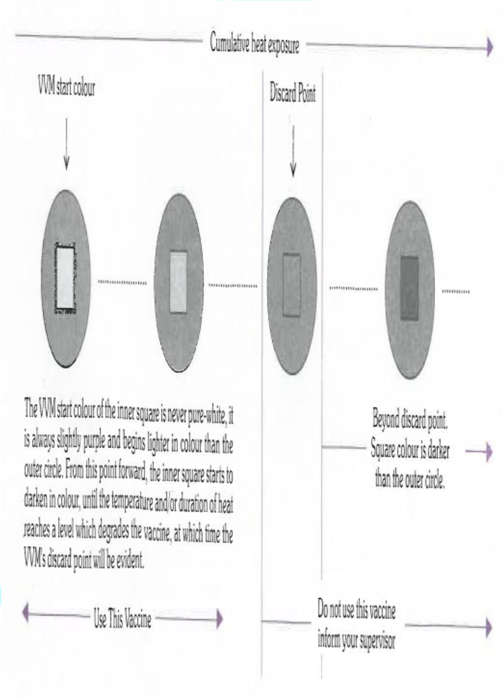
Transportation

- Transport all vials (open or unopened) in a ________ bag in the vaccine carrier and record in the stock register.

Monitoring heat exposure using vaccine vial monitors

- If the colour of the inner square is the same colour or darker than the outer circle, the vaccine has been exposed to too much heat and should be discarded.

- Four types of VVM (VVM2, VVM7, VVM14 and VVM30) are chosen to match the heat sensitivity of the vaccine.



Rare Vaccine Reactions	
Vaccine	**Reaction**
BCG	
Hepatitis B	
Influenza (inactivated)	
Influenza (live-attenuated)	
Japanese encephalitis (inactivated)	
Measles/MMR/MR	
Oral poliomyelitis	
Pertussis (DTwP)	
Tetanus toxoid, DT	
Yellow fever	
Varicella	

International Health Regulations (IHR)

Under the International Health Regulations (IHR), certain prescribed diseases are notified by the national health authority to WHO. These can be divided into:

a) **Those diseases subject to** International Health Regulations (1969), Third Annotated Edition, 1983 –

-
-
-

b) **Diseases under surveillance by WHO:**

- Louse-borne typhus fever
- Relapsing fever
- Paralytic polio
- Malaria
- Viral influenza-A
- SARS
- Smallpox

______________**:** It is defined as "separation, for the period of communicability of infected persons or animals from others in such places and under such conditions, as to prevent or limit the direct or indirect transmission of the infectious agent from those infected to those who are susceptible, or who may spread the agent to others".

_______________**:** It is defined as "the limitation of freedom of movement of such well persons or domestic animals exposed to communicable disease for a period of time not longer than the longest usual incubation period of the disease, in such manner as to prevent effective contact with those not so exposed”.

National Immunization Schedule (NIS)				
Vaccine	**When to give**	**Dose**	**Route**	**Site**
For Pregnant Women				
Tetanus Toxoid (TT)/ & adult Diphtheria (Td)-1	Early in pregnancy	0.5 ml	IM	Upper Arm
TT/Td-2	4 weeks after TT-1			
TT/Td-Booster	If received 2 TT doses in a pregnancy within the last 3 years*			
For Infants				
BCG	At birth or as	0.1ml (0.05 ml until	ID	

	early as possible till one year of age	1 month age)		Left Upper Arm
Hepatitis B - Birth dose	At birth or as early as possible within 24 hours	0.5 ml	IM	Antero-lateral side of mid-thigh
Oral Polio Vaccine (OPV)-0	At birth or as early as possible within	2 drops	Oral	Oral

	the first 15 days			
OPV 1, 2 & 3	At 6 weeks, 10 weeks & 14 weeks (OPV can be given till 5 years of age)	2 drops	Oral	Oral
	At 6 weeks, 10 weeks		IM	Antero-lateral side

Pentavalent 1, 2 & 3	& 14 weeks (can be given till one year of age)	0.5 ml		of mid-thigh
Pneumococcal Conjugate Vaccine (PCV)	Two primary doses at 6 and 14 weeks followed by booster dose at 9-12 months	0.5 ml	IM	Antero-lateral side of mid-thigh

Rotavirus (RVV)	At 6 weeks, 10 weeks & 14 weeks (can be given till one year of age)	3 drops	Oral	Oral
Inactivated Polio Vaccine (IPV)	Two fractional dose at 6 and 14 weeks of age	0.1 ml ID	ID, two fractional dose	Right Upper Arm

Measles Rubella (MR) 1st dose	9 completed months -12 months. (Measles can be given till 5 years of age)	0.5 ml	SC	Right Upper Arm
Japanese Encephalitis (JE) -1	9 completed months -12 months.	0.5 ml	SC	Left Upper Arm
Vitamin A (1st dose)	At 9 completed months with Measles-Rubella	1 ml (1 lakh IU)	Oral	Oral
For Children				
Diphtheria, Pertussis & Tetanus (DPT) booster-1	16-24 months	0.5 ml	IM	Antero-lateral side of mid-thigh

MR 2nd dose	16-24 months	0.5 ml	SC	Right Upper Arm
OPV Booster	16-24 months	2 drops	Oral	Oral
JE-2	16-24 months	0.5 ml	SC	Left Upper Arm
Vitamin A(2nd to 9th dose)	16-18 months. Then one dose every 6 months upto the age of 5 years	2 ml (2 lakh IU)	Oral	Oral

DPT Booster-2	5-6 years	0.5 ml	IM	Upper Arm
TT/Td	10 years & 16 years	0.5 ml	IM	Upper Arm

Indications for chemoprophylaxis	
Disease	**Chemoprophylaxis**
Cholera	
Conjunctivitis, bacterial	
Diphtheria	
Influenza	
Meningitis, meningococcal	
Plague	

DISINFECTION

____________: Usually a chemical agent (but sometimes a physical agent) that destroys disease causing pathogens or other harmful microorganisms, but might not kill bacterial spores.

______________: Thermal or chemical destruction of pathogen and other types of microorganisms but not bacterial spores.

__________: Validated process used to render a product free of all forms of viable microorganisms including bacterial spores.

__________: Substance that prevents or arrests the growth or action of micro-organisms by inhibiting their activity or by destroying them.

_________: Prevention of contact with micro-organism.

___________: Agent that reduces the number of bacterial contaminants to safe levels as judged by public health requirements.

_________: State of being free from all living micro - organisms.

__________: Agent that destroys micro-organisms, especially pathogenic organisms.

__________: Surface cleaning agent that makes no antimicrobial claims on the label.

Properties of an ideal disinfectant

1. ________ spectrum: Should have a wide antimicrobial spectrum.

2. ______ **acting:** Should produce a rapid kill.

3. Not affected by __________ factors: Should be active in the presence of organic matter (e.g., blood, sputum, faeces) and compatible with soaps, detergents, and other chemicals encountered in use.

4. _________: Should not be harmful to the user or patient.

5. __________ compatibility: Should not corrode instruments and metallic surfaces, and should not cause the deterioration of cloth, rubber, plastics, and other materials.

6. _________ effect on treated surfaces: Should leave an antimicrobial film on the treated surface.

7. Easy to use with _______ label directions.

8. ___________: Should have a pleasant odour or no odour to facilitate its routine use.

9. ___________: Should not be prohibitively high in cost.

10. _________: Should be soluble in water.

11. __________: Should be stable in concentrate and use dilution.

12. _________: Should have good cleaning properties.

13. _______________ friendly: Should not damage the environment on disposal.

Investigation of an Epidemic

a. **____________ of diagnosis:** It is the first step in an epidemic investigation.

b. **____________ of the existence of an epidemic:** Compare the disease frequencies during the same period of previous years

c. **_________ the population at risk:** Obtaining a map of the area and counting the population.

d. **______ search for all cases and their characteristics:** Medical survey, Epidemiological case sheet, Searching for more cases.

e. **______ analysis:** Using the classical epidemiological parameters- time, place and person

f. ____________ **of hypothesis:** Possible source, causative agent, possible modes of spread and environmental factors

g. ___________ of hypothesis

h. ______________ of ecological factors

i. Further ____________ of population at risk

j. _________ the report

NOTE

Chapter-4: Screening for disease

Screening test	Diagnostic test
	Done on those with indications or sick
	Applied to single patients, all diseases are considered
	Diagnosis is not final but modified in light of new evidence, diagnosis is the sum of all evidence
	Based on evaluation of a number of symptoms, signs (e.g., diabetes) and laboratory findings
	More accurate
	More expensive
	Used as a basis for treatment
	The initiative comes from a patient with a complaint

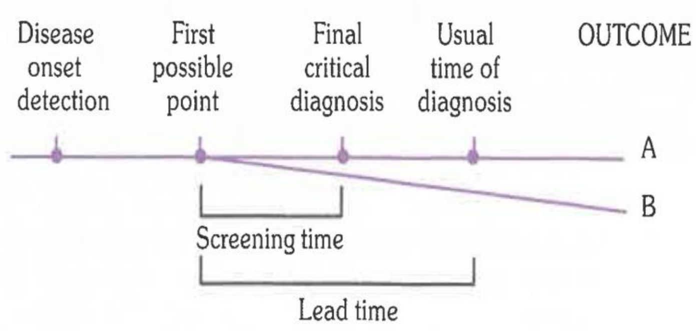


FIG.1
Model for early detection programmes

Lead time: is the advantage gained by screening that is the period between diagnosis by _____________ and diagnosis by ____________

Uses of screening:

- **Case detection:** Also known as _____________ screening. It is defined as the presumptive identification of unrecognized disease, which does not arise from patients request.
- **Control of disease:** Also known as _______________ screening. People are examined for the benefit of others.
-
-

Types of screening		
Screening of whole population or subgroup	Targets specific high risk groups based on epidemiological evidences	Combines two or more screening tests for detecting multiple disease simultaneously

Offered to everyone, regardless of individual risk	Focuses on screening of risk factors to prevent onset of disease	Include interviews, medical examination and lab test
E.g., TB	Cervical cancer and diabetes screening	Lack proper validation
Lacks preventive value so least useful	Offers better resource utilization and effectiveness	

Criteria For Screening

Disease to be screened:

- The condition sought should be an **important ______________** (in general, prevalence should be high)
- There should be ____________________________________ stage
- The __________ history of the condition, including development **from ________________ disease,** should be ____________ understood (so that we can know at what stage the process ceases to be reversible)
- There is a test that can detect the disease _______ to the onset of signs and symptoms
- **Facilities** should be available for **____________ of the diagnosis;**
- There is an effective ___________
- There should be an **agreed-on policy concerning _________ treat as patients**

- There is good evidence that **early detection and treatment reduces morbidity** and mortality

- The **expected benefits** (e.g., the number of lives saved) of early detection **exceed the risks and costs.**

Test to be applied:

The test must satisfy the criteria of ____________, ____________ and _________.

Screening test

Screening test result by diagnosis

Screening test results	Diagnosis		Total
	Diseased	Not diseased	
Positive	a (True-positive)	b (False-positive)	a+b
Negative	c (False-negative)	d (True-negative)	c + d
Total	a + c	b + d	a + b + c + d

est:

(a) Sensitivity = $a/(a + c) \times 100$

(b) Specificity = $d/(b + d) \times 100$

(c) Predictive value of a positive test = $a/(a + b) \times 100$

(d) Predictive value of a negative test = $d/(c + d) \times 100$

(e) Percentage of false-negatives = $c/(a + c) \times 100$

(f) Percentage of false-positive = $b/(b + d) \times 100$

Sensitivity: Defined as the ability of a test to identify correctly all those who have the disease, that is __________.

Specificity: Defined as the ability of a test to identify correctly all those who do not have the disease, that is ____ ___________.

Diagnosis of brain tumours by EEG

EEG results	Brain tumour	
	Present	Absent
Positive	36	54,000
Negative	4	306,000
	40	360,000

Sensitivity = $36/40 \times 100 = 90$ per cent

Specificity = $306{,}000/360{,}000 \times 100 = 85$ per cent

Predictive accuracy of a screening test

- Predictive value reflects the _______________ **of a test.**
- The predictive accuracy depends upon ____________, _________ and _________________
- The predictive value of a positive test indicates the probability that a patient with a positive test result has in fact the disease in question.
- The more prevalent a disease in a given population, the more accurate will be the predictive value of a positive screening test.
- The predictive value of a positive result ______ as disease prevalence ________.

Yield

- It is the **amount of _____________________ disease that is diagnosed as a result of the screening effort.**
- It **depends upon** many factors:
    -
    -
    -
- For example, by limiting a diabetes screening programme to persons over 40

years, we can increase the yield of the screening test.

- ___________ populations are usually selected for screening, thus increasing yield.

Cut-off point

In screening for disease, a prior decision is made about the cut-off point, on the basis of which individuals are classified as normal or diseased.

Following factors are taken into consideration:

- _________________: When the prevalence is _____ in community, screening level set at ______ level, which will ___________ sensitivity
- The __________: If the disease is very _______ and early detection markedly improves prognosis, a greater degree of sensitivity, even at expense of specificity is desired.
- Proportion of false-________ is tolerable but not false-_________.
- Specificity must be high but false positives should be limited.

Some Screening test	
Pregnancy	

Infancy	
Middle-aged men and women	
Elderly	

NOTE

Chapter-5: Epidemiology of Communicable disease

I. Respiratory Infections

Human Monkeypox

- Monkeypox virus is an ________ ____________ ______ **virus.**
- It belong to _______________ genus of Poxviridae family.
- Incubation period = _______ **days**

Chickenpox (Varicella)

- It is caused by **varicella-zoster (V-Z) virus** also known as ___________________________
- Period of communicability range from ______ days before the appearance of rash, and _____ days thereafter.
- Virus tends to die out before _________ stage
- Patient ceases to be infectious once the lesions have ________.
- Secondary attack ratio of approximately ______
- Incubation period is usually ______ **days.**
- Characteristic feature of rash in chickenpox is its ____________, that is,

all stages of rash (papules, vesicles and crust) may be seen simultaneously at one time in the same area.

Characteristics	Smallpox	Chickenpox
Incubation		
Prodromal symptoms	Severe	Mild
Distribution of rash	• • Palms and soles frequently involved	• • • •
	• Axilla usually free • Rash predominant on extensor surface and bony prominences	

Characteristic of rash	• • Vesicles multilocular and umbilicated • Only one stage of rash is seen at one time	• Superficial • Unilocular, dew drop like appearance • Pleomorphic rash • Area of inflammation seen around rash
Evolution of rash	Slow	Very rapid
Fever	Subsides with appearance of rash	Temperature rises with each crop of rash

Measles (Rubeola)

- Measles is caused by an ___________________
- The only source of infection is _____ **of measles.** ________ are not known to occur.
- It is highly infectious during the prodromal period and at time of eruption
- Period of communicability is ___ days before and ___ days after appearance of rash
- One attack of measles confers ___________________
- Incubation period is 10 days from exposure to onset of fever, and 14 days to appearance of rash.

- A day or two before the appearance of rash __________ like table salt crystals appear on buccal mucosa opposite _____ **and** _________________

- Koplik spot are small, bluish white spots on red base. Their presence is ______________ of measles.

Treatment and prevention:

- All cases of severe measles, and all cases of measles in areas with high case-fatality rates should be treated with _________

- In countries with ongoing transmission in which the risk of measles mortality remains high, MCV1 should be given at age __ months. MCV2 should be given between _____ months.

- Minimum interval between MCV1 and MCV2 is __ weeks.

- In countries with low levels of measles transmission, MCV1 may be administered at ____ months of age

Rubella (German measles)

- Rubella is caused by an ______ **virus of ________ family.**

- Congenital infection is confirmed if infant has _____ **rubella antibodies shortly after birth** or if ______ **antibodies persist for more than 6 months.**

- The classical triad of congenital defects are ________, ______ malformations and _________.

- Rubella infection **inhibits __________**, and this is probably **the reason for congenital malformations and low birth weight.**

Mumps

- The causative agent, **Myxovirus parotiditis** is a **_____ virus** of Myxovirus family.

- Period of maximum infectivity is just before and at onset of parotitis.

Complications:

- ____________________ and ___________ denote _________, which is the most common extra salivary gland manifestation of mumps in adults.

Vaccination:

Widely-used live attenuated mumps vaccine strains include:

- Jeryl- Lynn
- RIT 4385
- Leningrad-3
- L-Zagreb
- Urabe strains

WHO recommends that **__________** mumps vaccine should **not** be **used** in national immunization programmes because it demonstrates **low effectiveness.**

Influenza

- It is classified within the family _______________.
- There are ______ **viral subtypes**, namely influenza type A, B, C and D.
- These four viruses are **antigenically _________. There is ____ cross immunity** between them.
- When there is a sudden complete or major change, it is called a ________.
- When the antigenic change is gradual over a period of time, it is called a ______.
- ______ **syndrome** (fatty liver with encephalopathy) is a rare and severe complication of influenza, usually A and B type, particularly in young children between 2 and 16 yrs of age.
- Avian influenza (_______) is ______ while Pandemic Influenza A (_______) is ______

Treatment:

- _________ and _________ is indicated for treatment of influenza.
- In adults the recommended oral dose is 75 mg twice daily for 5 days.
- Zanamivir is indicated in adults and children (>5 years)
- ______________ is the drug of choice for chemoprophylaxis given till 10 days after last exposure.

Diphtheria

- Caused by C. diphtheriae (**gram ________, _________, _________** producing)
- It is one of the diseases protected from by the Pentavalent (diphtheria, tetanus, pertussis, hepatitis B and Hib) vaccine given in National Program
- Each dose of these antigens generally contains ___ **Loeffler (Lf)** units of diphtheria toxoid.

Incubation period: _____ days

Treatment:

- **Cases:** _____________ should be given without delay, IM or IV, days
- **Carriers:** ________________ for 10 days

Whooping cough (Pertussis)

- It is caused by B. pertussis
- The Chinese call it a _________________
- Incubation period is usually _______ days.

Pneumococcal Pneumonia vaccine

PPV23:

- It is available for children over ____ years of age.
- It is administered as single ______________ dose in a _______ muscle or as subcutaneous dose.

PCV:

- Two conjugate vaccines are available ______ and _______
- National immunization schedule recommends PCV for infants upto 1 year of age in three doses at ___ weeks, ___ weeks and __ months.

The Integrated Global Action Plan for the Prevention and Control of Pneumonia and Diarrhoea (GAPPD)
Specific Goals For 2025
Reduce mortality from **pneumonia** in children < 5 years of age to ____ per 1000 live births
Reduce mortality from **diarrhoea** in children < 5 years of age to ___ per 1000 live births

Reduce the incidence of **severe pneumonia** by ____ **%** in children < 5 years of age as compared to 2010 levels
Reduce the incidence of **severe diarrhoea** by _____ **%** in children < 5 years of age as compared to 2010 levels
Reduce by ____ **%** the global number of children < 5 years of age who are **stunted** as compared to 2010 levels.

Mucormycosis or Black fungus

- It is a rare fungal infection caused by group of fungi called **mucormycetes**.
- Risk factors are uncontrolled pre-existing diabetes, rampant overuse of steroids in management of COVID-19.
- Blackish discoloration of skin over nasolabial groove/ alae nasi.
- __________________ is the treatment of choice.

COVID-19

Covaxin vaccine

- The Bharat Biotech's vaccine, developed in collaboration with ICMR and NIV Pune, is a __________________ coronavirus vaccine.

- It is given ______________ in __________ muscle in two doses 28 days apart, to persons above 18 years of age.

- The vaccine should be stored at 2-8°C.

Covishield vaccine

- It is a recombinant ___________________________vaccine, administered intramuscularly in deltoid muscle.

Sputnik V vaccine

- It is a ________________ vaccine administered by intramuscular injection in two doses to people 18 years and above.

COVAX Initiative- Key Agencies

Major partners

-
-
-

Purpose of COVAX initiative

- The COVAX initiative was launched to ensure equitable access to covid-19 vaccines across all countries, especially low and middle income nations.

Tuberculosis

- M. tuberculosis is a ________________ **parasite**, it is readily ingested by phagocytes and is resistant to intracellular killing.

- **Sputum smear examination by direct microscopy** is now considered the method of choice.
- Sputum smear microscopy for tubercle bacilli is positive when there at least ______ of sputum.

Tuberculin Test

- It determine ______________ **infection by M. tuberculosis.**
- It is the only means of **estimating ____________ of infection in a population.**
- Only **two tuberculins** are accepted as standard tuberculin by **WHO**:
    - **Purified protein derivative-S (PPD-S)**
    - **PPD-RT 23**

Mantoux Test:

- Is carried out by injecting ___ TU of PPD in ____ ml _______________ on the ________ surface of the left forearm, mid-way between elbow and wrist.
- The injection should be made with a __________ syringe, with the needle bevel facing ________.
- When placed correctly, injection should produce a _____ wheal of the skin, _________ mm in diameter.
- Result of test is **read after 48-96 hours but _____ hours (3rd day) is the ideal.**
- Tuberculin Reactions:

- **10 mm** =
- **< 6 mm** =
- **6 - 9 mm** =
- **No induration** =

Anti-tuberculosis drugs

Rifampicin

- It is a powerful ___________ **drug.**
- Effective against ___________ as well as __________ bacilli.
- Only bactericidal drug active against ________ or ________ bacilli which are found in solid caseous lesions
- It is used only as ________ **drug** and its absorption is _______ by food.

Isoniazid (INH)

- It is the **most __________ drug** in treatment of tuberculosis
- Effective against intracellular as well as extracellular bacilli.
- Its action is most marked on ________________ **bacilli**

Bedaquiline

- It is a new **class of drug _____________** that specifically **targets mycobacterial _____________**
- It is a __________ **drug** with ___ volume of tissue distribution, highly bound to plasma proteins and hepatically metabolized.

- Drug has ________ **half-life**, present in plasma up to ____ **months** post stopping BDQ.

- Improve the time to **culture convertion in MDR-TB patients.**

Contraindication:

Adverse reaction	**Drug responsible**
Severe rash, agranulocytosis	
Hearing loss or disturbed balance	
Visual disturbance (poor vision and colour perception)	
Renal failure, shock or thrombocytopenia	
Hepatitis	

BCG VACCINATION

- It is derived from _____________ strain of tubercle bacilli.

- ____________ strain is recommended by WHO for production.

- **Dosage:** Usual strength is ___ mg in 0.1 ml. For newborns below 4 weeks, a reduced dose of 0.05 ml is used due to thinner skin.

- Vaccine injected intradermally using a tuberculin syringe.

Complications:

- Severe ulceration
- Suppurative lymphadenitis
- Osteomyelitis
- Disseminated BCG infection

Contraindications:

-
-
-
- **Deficient Immunity:** Symptomatic HIV infection, congenital immunodeficiency, leukemia and lymphoma
- **On immunosuppressive treatment:** Corticosteroids, alkylating agents, antimetabolites and radiation therapy
-

END TB STRATEGY

- The strategy envisions universal access to high-quality TB care and goes ___________ to promote TB prevention.
- The End TB Strategy identifies ______ barriers to achieving progress in the fight against TB. **They are:**
    - Weak health system

- Underlying determinants of TB such as poverty, undernutrition, migration and ageing population and risk factors such as diabetes, silicosis and smoking
- Lack of effective tools
- Continuous unmet-funding needs.

VISION

- A world free of tuberculosis: _______ **deaths, disease and suffering** due to TB.

GOAL

- End the global tuberculosis ___________.

INDICATORS	MILESTONES	
	2020	2025
Reduction in number of TB deaths compared with 2015 (%)		
Reduction in TB incidence rate compared with 2015 (%)		
TB-affected families facing catastrophic costs due to TB (%)		

II. Intestinal Infections

Poliomyelitis

- tOPV is now replaced with bOPV containing only type ______ poliovirus and withdrawing OPV type 2 from all immunization activities called the _______.

	IPV	Fractional IPV (fIPV)
Volume per dose		
Schedule		

Administration		
Site of administration		
Syringe		
Are two fractional doses as effective as a single standard dose ?	Two fractional doses of IPV (ID) given at 6 and 14 weeks produces better immunogenicity than a single standard dose (IM) given at 14 weeks	

Vaccine derived Poliovirus

- Vaccine associated paralytic poliomyelitis (VAPP) is the most important of these rare events.

- VAPP occurs in both OPV recipients and their unimmunized contacts.

- It is most frequently associated with ________ followed by Sabin 2 and Sabin 1.

- VDPVs are divided into three categories:
    - **__VDPV:** Evidence of person-to-person transmission in community exists
    - **__VDPVs:** ________________ associated VDPV
    - **__VDPV:** ________ VDPV

Sequential administration of IPV and OPV

- **Schedule:** _____ doses of IPV followed by ____ doses of OPV.

- Combined schedules of IPV and OPV appear to ______ **or prevent VAPP** while **maintaining** the _____ levels of intestinal mucosal immunity conferred by OPV.

- In addition, such schedules economize on limited resources by reducing the number of doses of IPV, and may optimize both the humoral and mucosal immunogenicity of polio vaccination.

- The effectiveness of this approach in preventing polio caused by WPV as well as VAPP has been documented by 2 large studies.

IPV (Salk type)	OPV (Sabin type)
Induces circulating antibody, but no local (intestinal) immunity	
Prevents paralysis, but does not prevent reinfection by wild polio viruses	Prevents not only paralysis, but also intestinal reinfection
Not useful in controlling epidemics	Can be effectively used in controlling epidemics. Even a single dose elicits substantial immunity (except in tropical countries)
More difficult to manufacture	
The virus content is 10,000 times more than OPV. Hence costlier	
Does not require stringent conditions during storage and transportation. Has a longer shelf-life	Requires to be stored and transported at sub-zero. temperatures, unless stabilized

Mopping Up:

- Involves ___________ immunization in high risk districts, where wild polio virus is known or suspected to be still circulating.
- Mopping up activities are usually the _____ stage in polio eradication

Pulse Polio Immunization:

- The term "pulse" has been used to describe this sudden, simultaneous, mass administration of OPV on a single day to all children 0-5 years of age, regardless to previous immunization.
- PPIs occur as two rounds about 4 to 6 weeks apart during low transmission season of polio, i.e. between **___________ to ___________.**
- In India, the peak transmission is from **_______ to ___________.**
- The dose of OPV during PPIs are extra doses which supplement, and do not replace the doses received during routine immunization services.

Viral Hepatitis B

- HBV is a complex, **42 nm, double shelled _______ virus,** originally known as **_______ particle.**
- It replicates in the liver cells. HBV occurs in **3 morphological** forms in the serum of patient.

- Of the three morphological forms, only the DANE particle is considered infectious, he other circulating morphological forms are not infectious.

- Man is the only reservoir of infection which can be spread either from carriers or from cases.

- Incubation period is of ______ **days.**

Recombinant Hepatitis B Vaccine

- **Active substance:** ________

- When immunizing against HBV at birth, only ___________ hepatitis B vaccine should be used.

- Dose for adults is ________ µg at 0, 1 and 6 months on deltoid.

- Children under 10 years of age should be given half of adult dose at same time intervals.

- For greatest reliability of absorption, the _______ muscle is preferred for injection as ________ injection often results in deposition of vaccine in fat rather than muscle, with fewer serologic conversion.

- For infants and children under 2 years, anterolateral aspect of ________ is used as vaccination site.

- ___________ administration is not recommended because the immune response is less reliable particularly in children.

- **Contraindicated in** those allergic to vaccine components.

- **Neither** pregnancy nor lactation is a contraindication for use of this vaccine.

The WHO "5Cs"

These are principles that apply to all models of hepatitis testing and in all settings:

- C
- C
- C
- C
- C

Acute Diarrhoeal Disease

- It is most common in children especially those between 6 months and 2 years.
- Incidence is **highest in the age group ____________**, when weaning occurs.
- It reflects the combined effects of _________ levels of maternally acquired antibodies, the lack of active immunity in the infant, the human of contaminated food, and direct contact with also or animal faeces when the infant starts to crawl.

Composition of reduced osmolarity ORS		
Reduced osmolarity ORS	**grams/ litre**	**mmol / litre**
Sodium chloride	2.6	Na = Cl =
Glucose, anhydrous	13.5	
Potassium chloride	1.5	

Trisodium citrate, dihydrate	2.9	
Total weight	20.5	
Total osmolarity		

Stool	> 3 loose stools per day, watery like rice water	> 3 stools per day, with blood or pus
Drug of choice	Doxycycline, tetracycline, TMP-SMX and erythromycin	

Symptoms	**Cholera**	**Shigella**
Diarrhoea		
Fever		
Abdominal cramps		
Vomiting		
Rectal pain		

Zinc Supplementation

WHO and UNICEF therefore recommend:

- Daily ___ mg of zinc for infants under ___ months of age
- **____ mg for children older than ___ months for ______ days.**

Cholera

Types of Oral Cholera Vaccine

Dukoral (WC-rBS)	• Monovalent vaccine contains formalin and heat-killed V. cholerae 01 with recombinant cholera toxin B subunit. • Vaccine is provided in 3 ml single-dose vials together with bicarbonate buffer (effervescent granules in sachets to protect the toxin B subunit from being destroyed by gastric acid). • Shelf life of ___ years at 2-8°C and remains stable for 1 month at 37°C. • Primary immunization consists of ___ oral

	doses given ≥7 days apart. • Protection expected ___ week after last scheduled dose. • Not licensed for children aged ____ years

Sanchol and mORCVAX	• Bivalent contains serogroups ___ and _________. • Do not contain the bacterial toxin B subunit therefore it does not require buffer. • Administered orally in 2 liquid doses 14 days apart for individuals aged 21 year. • A booster dose is recommended after 2 years.
Euvichol	It has same characteristics as Sanchol.

Typhoid fever

ANTI-TYPHOID VACCINES

The Vi polysaccharide vaccine

- It is composed of purified **Vi capsular polysaccharide from the Ty2 S. Typhi strain.**
- Induces a ______________________ **response**
- Vaccine is licensed for **individuals aged _____ years.**
- Administered ______________________________
- Single human dose: ____ **µg of antigen.**
- Storage temperature: **2-8°C**
- Vaccine stability: For **6 months at 37°C**, and for **2 years at 22°C.**

Schedule:

- __________ dose is required, and vaccine confers **protection 7 days after injection.**
- **Re-vaccination** is recommended **every ____years.**
- Can be **co-administered with** yellow fever, hepatitis A, and with vaccines of the routine childhood immunization programmes.

The Ty21a vaccine

It is an _______ administered, _____**-attenuated Ty2 strain** of S. Typhi available as **enteric-coated capsules**.

Schedule:

- Licensed for individuals **aged ____ years.**
- Dosage: ______________ **day (3-dose regimen),** protection achieved **7 days after** the last dose.
- **Revaccination schedule every 3 years** for people in endemic areas, and **annually** for travellers to endemic regions.
- Can be **co-administered with live vaccines** against polio, cholera, and yellow fever, or the measles, mumps and rubella (MMR) combination.

Safety

- **Proguanil and antibacterial** should be **stopped 3 days** before and after the administration.
- Not effective if taken during diarrhoea.
- Safety during pregnancy is not well established.
- Safe for **HIV-positive, asymptomatic individuals** with CD4 count **>200/mm³.**

B. cereus food poisoning

- Bacillus cereus is an _______**, spore-bearing,** ______**, gram** ______________
- Produce atleast _____ **distinct enterotoxins**; Diarrhoeal and emetic form.

- Diagnosis can be confirmed by isolation of _____ **or more** B. cereus organisms per gram of epidemiologically incriminated food.

Soil-Transmitted Helminthiasis

Include:

-
-
-

Secondary prevention:

Effective drug for treatment of human reservoir

- These are piperazine, mebendazole, levamisole and pyrantel; last two drugs are effective in single dose.
- **Albendazole** : The usual dose for adults and children over 2 years is 400 mg as a ______ dose. Contraindicated in children below 2 years and in pregnancy.
- **Mebendazole :** The usual dose is 100 mg ______ daily for 3 days for all ages above 2 years.
- **Levamisole :** It is the drug of choice. A single oral dose of 2.5 mg/kg of body weight (maximum 150 mg) has been recommended.
- **Pyrantel:** Effective in single dose of 10mg/kg of body weight with maximum of 1g.

III. Arthropod- borne Infections

Dengue

- It belongs to the genus ___________.
- There are ______ virus serotypes: **DENV-1, DENV-2, DENV-3 and DENV-4**.
- Infection with any _____ serotype confers lifelong immunity to that virus serotype.
- All four serotypes are **antigenically ________**, they are different enough to elicit cross-protection for only a few months after infection by any one of them.
- Secondary infection with dengue serotype ____or multiple infection with different serotypes lead to severe form of dengue DHF/DSS.
- **1st infection ________ the patient**, while **2nd infection** with different serotype produces **immunological _____________.**

▪ Most important vectors are **Aedes ________** and **Aedes ___________.**

Aedes aegypti	**Aedes albopictus**
Highly domesticated and strongly anthropophilic	Invades peripheral areas of urban cities.
Nervous feeder (i.e., it bites more than one host to complete one blood meal)	**Aggressive feeder** (i.e., the species can complete its blood meal in one go on one person)
Discordant species (i.e., it needs more than one feed for the completion of the gonotropic cycle).	**Concordant species** (i.e., does not require a second blood meal for the completion of the gonotropic cycle.)

Transmission of disease:

- After an extrinsic incubation period of _______ days, the mosquito becomes ________ and can transmit the infection.
- Once the mosquito becomes infective, it remains so for _____.
- The genital tract of the mosquito gets ________.
- ____________ transmission of dengue virus occurs when virus enters fully developed eggs at the time of oviposition.

Tourniquet test:

- Is considered positive when ____ or more petechiae per 2.5 x 2.5 cm are observed.
- In DHF, the test usually gives a definite positive with ____ petechiae or more.

Criteria for discharge of patients:

- Absence of fever for atleast 24 hours without the use of anti-pyretic drugs.
- Return of appetite.
- Visible clinical improvement.
- Good urine output.
- Minimum of 2-3 days after recovery from shock.
- No respiratory distress from pleural effusion or ascites
- Platelet count ___________

Vaccines:

- ___________ is a prophylactic, tetravalent, live attenuated viral vaccine developed by Sanofi Pasteur in December 2015.

- The vaccination schedule consist of 3 injections of 0.5 ml administered at 6 months intervals.

Malaria

- Malaria is caused by **four distinct species** of malaria parasite- P. vivax, P. falciparum, P. malariae and P. ovale.
- Parasite undergoes **_________ cycle** (human cycle) and _______ cycle (mosquito cycle).
- _____ is the intermediate host and _______ is the definitive host.
- The vector mosquito must live for at least **______ days** after an infective blood meal to become infective.

Measurements of malaria	
	It is widely used for measuring the endemicity of malaria in a community
	It is a useful malariometric index
	It indicates average degree of parasitaemia in a sample of well-defined group of the population.
	Most sensitive index of recent transmission of malaria in a locality
	This is a crude index because the cases are not related to their time/ space distribution

Annual parasite incidence (API)	Areas with API ______ per 1000 population per year have been classified as high risk areas in India

Treatment of malaria		
	Drug of choice	**Primaquine dose**
P. vivax malaria	Chloroquine 25 mg/kg divided as: 10 mg/kg on day 1 10 mg/kg on day 2 5 mg/kg on day 3	0.25 mg/kg for 14 days
P. falciparum malaria North east In other states	ACT-AL for 3 days ACT-SP for 3 days	0.75 mg/ kg on day 2
Mixed Infection North east In other states	ACT-AL for 3 days ACT-SP for 3 days	0.25 mg/ kg for 14 days
Resistance to ACT/ Severe malaria		
Pregnancy: P. vivax		
Pregnancy: P. Falciparum	**1st trimester:** **2nd and 3rd trimester:**	

Goals	Milestones		Target
	2020	2025	2030
Reduce malaria mortality rates globally compared with 2015	At least _____%	At least _____%	At least ____%
Reduce malaria case incidence globally compared with 2015	At least 40%	At least 75%	At least _____%
Eliminate malaria from countries in which malaria was transmitted in 2015	At least 10 countries	At least 20 countries	At least _____ countries
Prevent re-establishment of malaria in all	Reestablishment prevented	Reestablishment prevented	Reestablishment prevented

countries that are malaria-free			

Malaria vaccine:

- The ____________ vaccine is the first and currently the only malaria vaccine to be recommended for use by WHO.
- The shelf life is of ____ years.

Lymphatic Filariasis

- The Mf of **W. bancrofti** and **B. malayi** occurring in India display a _________ **periodicity.**
- The maximum density of Mf in blood is reported between ___ **pm and** ___ **am.**
- _____ is the definitive host and ________ the intermediate host of Bancroftian and Brugian filariasis.
- The males are about ____ mm long and the females ________ mm long.
- The females are _________.
- In occult filariasis classical clinical manifestations are not present and Mf are not found in the blood. The example is _______________________
- ____________________________ **methods** is the most sensitive method

currently available for detecting low density microfilaraemia.

Human filarial infections		
Organism	**Vectors**	**Disease produced**
Wuchereria bancrofti		Lymphatic filariasis
Brugia malayi		
Brugia timori		
Onchocerca volvulus		
Loa loa		
T. perstans		
T. streptocerca		
Mansonella ozzardi		

Characteristics	**Mf. W. bancrofti**	**Mf. B. malayi**
General appearance		Crinkled, secondary curves
Length	244 to 296 μ	177 to 230 μ
Fee cephalic space		Nearly twice as long as broad
Excretory pore		Prominent

Caudal end	Uniformly tapering to a delicate point No terminal nuclei present	
Nuclear column		

IV. Zoonoses

Rabies

- It is caused by _____________, a _______ **shaped** __________ ______ containing virus from family Rhabdoviridae.
- The virus is excreted in the _______ of the affected animals.
- The virus recovered from naturally occurring cases of rabies is called "______ **virus**". It is pathogenic for all mammals and shows a long variable incubation period (20-60 days in dogs).
- A "**______" strain** of virus may be defined as one that has a short, fixed and reproducible incubation period (4-6 days) when injected intracerebrally into suitable animals. It **does not form _______ bodies.**
- The _______ **virus** is used in the **preparation of antirabies vaccine.**
- All warm blooded animals including man are susceptible to rabies.
- Rabies in man is a dead end infection and has no survival value for the virus.

Category	Description	Post exposure prophylaxis measures
Category I		
Category II		
Category III		

Rabies immunoglobulin for passive immunization

- It is given only ______, preferable at or soon after the ______ of _______ exposure vaccination.
- After the ______ day following the first vaccine dose, RIG is not recommended because an active antibody response is expected by then.
- The dose of human RIG is _____ IU/kg body weight, while for equine RIG it is _____ IU/kg body weight.
- All RIG as much as possible should be injected _____________ wound site without causing compartment syndrome.
- There is no scientific basis for performing _______ test before giving Equine RIG as test cannot predict reactions.

Yellow fever

- The causative agent **______________** classified as group B arbovirus, member of togavirus family.
- One attack of yellow fever gives **__________ immunity.**

Yellow Fever Vaccine

- **_____ Vaccine** is a _____ attenuated vaccine prepared from non-virulent strain, grown in chick embryo.
- **Stored at:** __________ degrees Celsius, preferably below zero.

- Administered __________ at **insertion of deltoid.**
- **Single dose of _____ ml**, irrespective of age.
- Immunity begins to appear on 7th day.
- Reconstituted vaccine is discarded if not used within half an hour.

Interference with other vaccines

- ________ and _______ fever vaccines should not be administered together or within ____weeks of each other due to potential interference.

Requirement for International travellers

- In **May 2014**, the **World Health Assembly** adopted an amendment of **International Health Regulations (2005)**, concerning yellow fever vaccination.
- **The amendment changed** the which period of protection afforded by yellow fever vaccination, and the term of **validity of the certificate** from ___ **years to** _______ of person vaccinated.
- On **11th July 2016**, the amendment entered into force and is legally binding upon all IHR states.
- Lifetime validity applies automatically to all _______ and ____ certificates.
- **Validity of certificate begins** ____ days after the date of vaccination.
- Accordingly, as of **11th July 2016**, revaccination or a booster dose of yellow fever vaccine will ______ required for international travellers as a condition of

entry into a State Party, **regardless of the date that their international certificate** of vaccination was initially issued.

- All travellers including infants exposed to the risk of yellow fever or passing through endemic zones of yellow fever must possess a **valid international certificate of vaccination against yellow fever** before they are allowed to enter yellow fever "receptive" areas.

- If no such certificate is available, the traveller is placed on quarantine, in a mosquito-proof ward, for ________ from the date of leaving an infected area.

- If the traveller arrives before the certificate becomes "valid", he is **isolated till the certificate becomes valid.**

Kyasanur forest disease

- It is caused by an _______________ and transmitted to man by bite of infective ________.

- It was first recognized in Shimoga district of _________ State.

- Local inhabitants called the disease "_________ disease" because of its association with ____________

- The disease was later named after the locality - Kyasanur Forest - from where the virus was first isolated.

Natural Hosts and Reservoirs

- **Small mammals (rats and squirrels):**

- **Monkeys**: _______________ for the virus.
- **Cattle:** Maintain Haemaphysalis ticks' population by provide plentiful source of blood meals,
- **Man:** Incidental or dead-end host and plays no part in virus transmission.

Incubation Period: ________ days.

Chikungunya fever

- It is caused by a __________ **virus,** the chikungunya virus and transmitted by __________ mosquitoes.

- Chikungunya is a local word meaning "___________" owing to excruciating joint pains.
- The **maximum cases** were reported from ___________.
- The incubation period is _______ days
- Disease has a _______ **onset with fever, chills, cephalalgia, anorexia, lumbago, conjunctivitis and adenopathy.**
- 60-80 % patients have ____________ **rash**, occasionally with purpura, on the trunk and limbs.
- The cutaneous eruption may recur every 3 to 7 days. Other symptoms are _____________ vomiting, epistaxis and petechiae.

- A prominent symptom, seen especially in adult patients is ____________, from which the disease gets its name.

- There is **no specific treatment of chikungunya infection** and it is usually ______________

- **Drugs like _______ and __________ should be avoided.**

Plague

- Y. pestis is a **gram-_________, __________, ____________ that exhibits _______ staining with special stains (_________ stain).**

- The commonest and the most efficient vector of plague is the _____________________

- _____________ is the specific flea index. It is more significant than total flea index.

- The drug of choice is ______________.

- For chemoprophylaxis the drug of choice is _____________.

Rickettsial zoonoses

- _____________ is the drug of choice for all rickettsial diseases.

Diseases	Rickettsial agent	Insect vectors	Mammalian reservoirs
1. Typhus group			
a. Epidemic typhus			Humans
b. Murine typhus			Rodents
c. Scrub typhus			Rodents
2. Spotted fever group			
a. Indian tick typhus			
b. Rocky mountain spotted fever			Rodents, dogs
c. Rickettsial pox			Rodents, dogs
			Mice
3. Others			
a. Q fever			Cattle, sheep, goats
b. Trench fever			Humans

Leishmaniasis

- **Leishmania donovani** is the causative agent of kala-azar.
- **L. tropica** is the causative agent of cutaneous leishmaniasis.
- **L. braziliensis** is the causative agent of mucocutaneous leishmaniasis
- Kala-azar can occur in all age groups including infants below the age of one year.
- **Males** are affected twice as often as females.

Dosage guide for children (2-11 years)		
Body weight	**Daily dosage** (after meal)	**Number of capsules of Miltefosine**
9-11 kg		capsules of 10 mg
12-16 kg		capsules of 10 mg
17-20 kg		capsules of 10 mg
21-25 kg		capsule of 50 mg
26-31 kg		1 capsule of 50 mg & 1 capsule of 10 mg
32-39 kg		1 capsule of 50 mg & 3 capsule of 10 mg
40 kg and above		capsules of 50 mg

Miltefosine dosage guide for persons (> 12 years)		
Weight	**Morning dose (after meal)**	**Evening dose (after meal)**
More than 25 kg	1 capsule of Miltefosine 50 mg	1 capsule of Miltefosine 50 mg
Less than 25 kg	1 capsule of Miltefosine 50 mg	Drug not to be given at evening

V. Surface Infections

Trachoma

Trachoma is caused by ___________________

Diagnosis:

Cases must have atleast 2 of the following diagnostic criteria:

- Follicles on the _____________ conjunctiva
- Limbal follicles or their sequelae, ________________
- Typical conjunctival ___________ (________________)
- Vascular ___________, most marked at ________ limbus

Tetanus

- C. tetani is a **gram ________, ____________, ________ bearing organism**.
- The spores are ____________ and give a ___________ **appearance.**
- Spores are highly ___________ to number of injurious agents.
- They germinate under anaerobic conditions and produce potent ______________ **(tetanospasmin).**
- Incubation period is usually __________ **days.**

Prevention of neonatal tetanus

- Training of birth attendants alone can reduce death due to neonatal tetanus by ______ %.
- Tetanus toxoid will protect both the _______ and her ______.
- In **unimmunized pregnant women**, two doses of tetanus toxoid should be given, the first as early as possible during pregnancy and the second at least a month later and at least 3 weeks before delivery.
- According to the National Immunization Schedule, these doses may be given between 16-36 weeks of pregnancy, allowing an interval of 1-2 months between the 2 doses.
- In **previously immunized** pregnant women, a ________ **dose** is considered sufficient.
- The golden rule is that no pregnant mother should be denied even one dose of tetanus toxoid if she is late in pregnancy.

- The infants born to the mothers who have not previously received 2-doses of tetanus toxoid can be protected by injection of antitoxin (heterologous serum, ______ IU), if it is administered within ____ hours of birth.

Leprosy

- It is caused by **M. leprae.**
- They are **acid-fast** and occur in human host both intracellularly and extracellularly.
- They occur in

  ____

- They have an affinity for Schwann cells and cells of reticulo-endothelial system.

Characteristics	PB (Pauci-bacillary)	MB (Multi-bacillary)
Skin lesions		
Peripheral nerve		
Skin smear		
Treatment		

Bacterial Index:

- The bacteriological or bacterial index indicates the ________ of leprosy bacilli in smears and includes both _________ (solid-staining) and _______ (fragmented or granular) bacilli.
- It ranges from ___________
- Based on number of bacilli seen in an average microscopic field of the smear using an oil immersion objective.

Morphological Index:

The criteria for calling the bacilli solid rods are:

- ○
- ○
- ○
- ○

It has been widely believed that only solid-staining-organisms are viable.

Lepromin test:

Early reaction	**Late reaction**
Also known as	
• Inflammatory response develops within 24 to 48 hrs and disappear after 3-4 days. • Diameter of red area is >10 mm	• Reaction develops late, become apparent in 7-10 days and reach maximum in 3 or 4 weeks.

at end of 48 hrs, test is considered as positive	• Test is read at 21 days. • If nodule is >5 mm in diameter then it is positive

Type I (Reversal Reaction)	Type II (ENL)
Occurs in both PB and MB cases in unstable types like BT.BB.BL.	

Skin lesions suddenly become reddish, swollen, warm, painful, and tender. New lesions may appear	Red, painful, tender, sub-cutaneous nodules - ENL may appear, commonly on face, arms, legs, bilaterally symmetrical. They appear in crops and subside within few days even without treatment (Evanescent skin nodules). Nodules are better felt than seen and these are recurrent (episodic)

Nerves close to skin may be enlarged, tender and painful (neuritis) with loss of its functions (loss of sensory, motor or autonomic) which may appear suddenly.	
Other organs - not affected.	
General symptoms - not common	Fever, joints pain, red eyes with watering may be associated
Eye-Lagophthalmos and corneal anaesthesia due to neuritis	

.

Sexually transmitted diseases

- A group of communicable diseases that are transmitted predominantly by ___________________________

Pathogen	Disease or syndrome
	Gonorrhoea. urethritis, cervicitis. epididymitis, salpingitis. PID, neonatal conjunctivitis
	Syphilis
	Chancroid
	LGV, urethritis, cervicitis, proctitis. epididymitis, infant pneumonia, Reiter's syndrome, PID, neonatal conjunctivitis

	Donovanosis (granuloma inguinale)
	Genital herpes
	Acute and chronic hepatitis
	Genital and anal warts
	AIDS
	Genital molluscum contagiosum
	Vaginitis
	Vaginitis

AIDS

- HIV virus is a protein capsule containing ___________ strands of genetic material (RNA) and enzymes.
- It can pass through the blood brain barrier and can destroy some brain cells.
- The virus is killed by _______.
- Once the person is infected the virus remains in the body ___________.

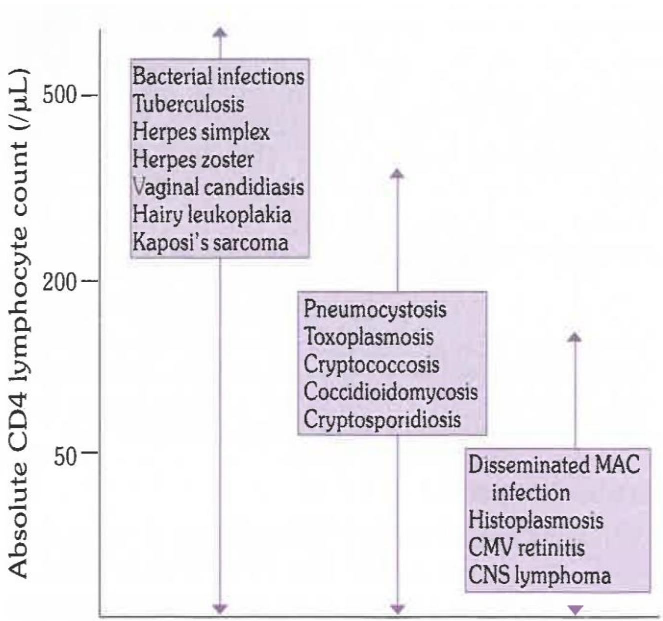


FIG. 4

Relationship of CD4 count to development of opportunistic infection

Classification of drugs used for ART	
	• Abacavir (ABC) • Didanosine (ddl) • Emtricitabine (FTC) • Lamivudine (3TC) • Stavudine (d4T) • Zidovudine (AZT)
	Tenofovir (TDF)
	• Efavirenz (EFV) • Etravirine (ETV) • Nevirapine (NVP)

	• Atazanavir + ritonavir (ATV/r) • Darunavir + ritonavir (DRV)r) • Fos-amprenavir + ritonavir (FPV/r) • lndinavir + ritonavir (IDV/r) • Lopinavir'ritonavir (LPV/r) • Saquinavir + ritonavir (SQVtr)
	Raltegravir (RAL)

Particulars	**Recommended Regimen**
1. Age less than 6 years and body weight less than 20 kg	
2. Age between 6 and 10 years and body weight between 20 kg and 30 kg	
3. Age more than 10 years and body weight more than 30 kg	

Tomato fever (Tomato flu)

- It is a virus infection emerged in state of ________ in children younger than _____ yrs of age.
- First identified in ___________ district on ____________
- It is so called because of the eruptions of _____ and __________ blisters ____________ the body.

NOTES

Chapter-6: Epidemiology of chronic non-communicable disease

Chronic Diseases

It is **defined** as "comprising all impairments or deviations from normal, which have **one or more** of the following characteristics :

- Are _____________
- Leave __________________
- Are caused by ______________ pathological alteration
- Require special training of the patient for _____________
- May be expected to require a long period of supervision, observation or care."

Cancer Prevention

- **Vaccination** against ____________, major cause of **liver cancer.**
- Vaccination against _______________________ main cause of **cervical cancer.**

WHO Global Action Plan for the Prevention and

Control of NCDs (2013–2020)

The voluntary global targets are:

- _____% relative reduction in risk of premature mortality from **cardiovascular diseases**, cancer, diabetes and chronic respiratory disease;
- At least ____% relative reduction in the harmful use of **alcohol**
- A ____% relative reduction in **insufficient physical activity**
- A ____% relative reduction in mean population intake of **salt/ sodium**
- A ____ % relative reduction in prevalence of current **tobacco** use in persons aged 15+ years;
- A ____ % relative reduction in prevalence of **raised blood pressure**
- _____ **the rise of diabetes and obesity**
- At least ____% of eligible people receive drug therapy and counselling (including glycaemic control) to prevent heart attacks and strokes
- An ____ % availability of the affordable basic technology and essential medicines including generics, required to treat major NCDs in both public and private facilities.

Coronary Heart Disease

- CHD is our **modern** _________, i.e., a disease that affects populations, not an unavoidable attribute of ageing.
- The WHO has completed a project known as __________ (multinational monitoring of trends and determinants in

cardiovascular diseases) to elucidate changing trends in CHD.

- _______________ is single most useful test for identifying individuals at high risk of developing CHD. __________ **BP** is better predictor of CHD than is the __________.
- ___________ is identified as a major CHD risk factor with several **possible mechanisms:**
  - _________________ induced atherogenesis
  - Nicotine stimulation of __________ drive ________ both blood pressure and myocardial oxygen demand
  - Lipid metabolism with _____ in "protective" high-density lipoproteins.

Risk factors for CHD	
Not modifiable	**Modifiable**
• • • • •	• • • • • • •

Prevention of CHD

WHO considered following dietary modification:

- Reduction of fat intake to _______ % of total energy intake

- Consume saturated fats to _____% of total energy intake.

- Reduction of dietary cholesterol to _____ mg per 1000 kcal/day

- __________ in complex carbohydrate consumption

- ___________ of alcohol consumption

- Reduction of salt intake to _____ daily or less

Risk Factor Intervention Trials

- Since 1951, one of the best known large prospective studies, the ____________ Study, has played a **major role in establishing the nature of CHD risk factors and their relative importance.**

Hypertension

Classification of blood pressure measurements		
Category	**Systolic (mm of Hg)**	**Diastolic (mm of Hg)**
Optimal		
Normal		
High normal		
Grade 1 Hypertension		

Grade 2 Hypertension		
Hypertensive crisis		
Isolated systolic hypertension		

Rule of halves:

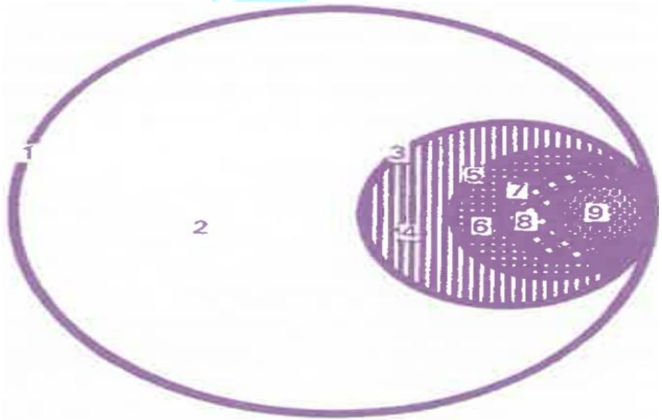


FIG. 1
Hypertension in the community

Modifiable risk factors:

- **Central obesity** indicated by an _________ waist to hip ratio, has been positively correlated with high blood pressure in several populations.

- **High salt intake** ________ blood pressure proportionately
- Potassium antagonizes biological effects of sodium and thereby ________ blood pressure
- Calcium, cadmium and magnesium also ________ blood pressure levels.
- **Saturated fat** _____ blood pressure as well as serum cholesterol
- CHD and hypertension is _________ related to dietary fibre
- **High alcohol intake** is associated with _______ risk of high blood pressure

Lifestyle modification to manage hypertension

- Weight reduction
- Adopt DASH eating plan
- Dietary sodium reduction
- Physical activity
- Moderation of alcohol consumption

______ **diet:** Diet rich in fruits, vegetables and low fat dairy products with a reduced content of saturated fat and total fat

Rheumatic heart disease

_______________ **Mode** project on Community Control of RF /RHD in India is being carried out with **four main components:**

- To study the epidemiology of streptococcal sore throats
- Establish registries for RF and RHD
- Vaccine development for streptococcal infection
- Conducting advanced studies on pathological aspects of RF and RHD

Prevention:

Primary prevention:

- Single IM injection of 1.2 million units of ______________________________ for adults and 600,000 units for children is adequate, or oral penicillin should be given for 10 days.
- If allergic to penicillin then first generation _____________ i.e., cefadroxil or cefalexin is used.

Secondary prevention:

- Single IM injection of __________________________ (1.2 million units in adults and 600,000 units in children) at intervals of 3 weeks.
- This must be continued for at least 5 years or until the child reaches 21 years whichever is later.
- For patients with carditis the treatment should continue for 10 years after the last attack or atleast until 40 years of age whichever is longer.

Cancer

Causes of cancer	
Causative agent	**Cancer**
Smoked fish	
Dietary fibre	
Beef consumption	
High fat diet	
Hepatitis B and C virus	
HIV infection	
AIDS	
EBV	
Cytomegalovirus (CMV)	
Human papilloma virus (HPV)	
Schistosomiasis	

Early warning signs (danger signals) of cancer:

- _______________ area in the breast
- Change in a ___________
- ___________ change in digestive and bowel habits
- __________ cough or hoarseness
- Excessive loss of blood at the monthly period or loss of blood outside the usual dates

- Blood loss from any natural orifice
- Swelling or sore that does not get better
- Unexplained loss of weight.

Screening of cancer cervix:

Methods of screening:

- PAP smear
- **VIA (Visual inspection with 5% acetic acid)**
- VIAM (VIA with magnification)
- VILI (Visual inspection post application of Lugol's iodine)
- HPV DNA and HPV mRNA tests

Time of screening:

- Screening should start at age of _______ with PAP smear and then every _____ years thereafter.
- Then add HPV test every ________ years
- If women living with HIV then start screening at age of _____ years.

Diabetes Mellitus

- Type 1 diabetes is associated with HLA-B8 and B15, more powerfully with HLA-_______________
- Type 2 diabetes is ______ HLA-associated.
- ____________________________________ are more powerful determinants of subsequent risk of type 2 diabetes than BMI.

- ______________ is also an important determinant of insulin resistance, the underlying abnormality in most cases of type 2 diabetes.

Diagnostic criteria for diabetes:

- Fasting plasma glucose _________ mmol/L
- 2-hour post-load plasma glucose _________ mmol/L
- Hba1c _________ mmol/mol

Diagnostic criteria for gestational diabetes:

- Fasting plasma glucose _________ mmol/L
- 1-hour post-load plasma glucose _________ mmol/L
- 2-hour post-load plasma glucose ___________ mmol/L

Insulin resistance syndrome (Syndrome X)

- Seen in _______ patients with _________ diabetes.
- Also known as __________ syndrome.

Association of:

-
-
-

-

Screening for diabetes

Target population:

- Age group ___________
- Family history of ___________
- ___________
- Women having baby weighing

  ____
- Women show excess ____________ during pregnancy
- Patient with premature ______________

Obesity

Use of **certain drugs,** e.g., ______________, ____________, _______, ____________________ etc can promote **weight gain**.

Body mass index (BMI)

- It is defined as weight in __________ divided by the square of the ______ in metres (_________).

- For example, an adult who weighs 70 kg and whose height is 1.75 m will have a BMI of 22.9:

  BMI = 70 (kg)/1.75^2 (m^2) = 22.9

Classification of adults according to BMI		
Classification	**BMI**	**Risk of comorbidities**
Underweight		Low (but risk of other clinical problems increased)
Normal range		Average
Overweight		
Pre-obese		Increased
Obese class I		Moderate
Obese class II		Severe
Obese class III		Very severe

Visual Impairment and Blindness

- Blindness is defined as visual acuity of ______________ or its equivalent.
- Near vision impairment defined as presenting near vision acuity **worse than ___ or _______ at _____ cm** with existing correction.
- The nomenclature of National Programme for control of blindness is changed to National Programme for control of **blindness and _______________.**
- _______________ are a major cause of preventable blindness. Mass campaigns with topical _____________ markedly reduced the severity of trachoma.
- Under the vitamin A distribution scheme in India, 200,000 IU of vitamin A are

given orally at 6 monthly intervals between the ages 1-6 years.

Drowning

- The victim loses consciousness after approximately ___ **minutes** of immersion, and irreversible brain damage can take place after _____ **minutes.**
- Children **under____ years** of age have the highest drowning mortality rates worldwide.
- ______ are especially at risk
- _________ access to water is another risk factor for drowning.

Early clues that a patient has severe envenoming:

- Snake identified as a very dangerous one
- Rapid early extension of local swelling from the site of the bite;
- Early tender enlargement of local lymph nodes, indicating spread of venom in the lymphatic system
- Early systemic symptoms: collapse (hypotension, shock), nausea, vomiting, diarrhoea, severe headache, "heaviness" of the eyelids, inappropriate (pathological) drowsiness or early ptosis/ ophthalmoplegia
- Early spontaneous systemic bleeding
- Passage of dark brown/black urin

NOTES

Chapter 7: Health Programmes In India

National vector borne disease control programme: For prevention and control of vector borne diseases namely _______, _________, __________, _________________, _________ and _____________.

Malaria

National Framework for Malaria Elimination in India (2016-2030)		
Goals	• Eliminate malaria (zero indigenous cases) throughout the entire country by 2030 • Maintain malaria-free status in areas with interrupted transmission and prevent reintroduction of malaria	
Objective	**By 2022:**	
	By 2024:	
	By 2027:	
	By 2030:	

Target	**By 2027:**	
	By 2030:	

Category 1: Elimination phase	
Category 2: Pre-elimination phase	
Category 3: Intensified control phase	

Classification of States/UTs for malaria elimination in India (2014)	
Category	**Definition**
Category 0: Prevention of re-establishment phase	

Integrated vector management (IVM)

- The measures of vector control and protection include:

- **Measures to control adult mosquitoes:**
- **Antilarval measures** chemical, biological and environmental.
- **Personal protection:** use of bed-nets, including insecticide-treated nets.

- The national malaria control program is currently using ____ **as the primary method of vector control in rural settings**, and _____________ **measures in urban areas.**
- The population living in areas with **API** ___ is to be covered by **LLINs**
- Population living in endemic areas with **API ≥ 2** is covered with **conventional net treated with insecticides and IRS.**
- **IRS is the preferred method in areas with very hot summers** and where ITNs are not acceptable to population.
- ______ is the insecticide of choice; and alternatives are malathion and synthetic pyrethroids.
- **_____ rounds of spraying are done for DDT and synthetic pyrethroids** while **______ rounds of spraying for malathion.**
- The real coverage by IRS is, however, limited by the low community acceptance due to the ______ **marks left on plastered surface**, acrid smell associated with malathion.

Anti-malaria month campaign

- **Timing:** Observed every _____ in the month of ___ throughout the country.

- **Purpose**: Enhance __________ and encourage ___________ participation

- **Methods**: Through ___________ campaign and ___________ communication

- **Goals**: Consolidate inter-sectoral collaborative efforts with other government departments, corporate and voluntary agencies at national, state and district levels.

Elimination of Lymphatic Filariasis

- The elimination is defined as "lymphatic filariasis ceases be public health problem, when the number of microfilaria carriers is less than___ per cent and the children born after initiation of ELF are _____ from circulating antigenaemia (presence of adult filaria worm in human body).

- The microfilaria survey in all the implementation units is being done through ______ blood survey before MDA.

- The survey is done in ____ sentinel and ___ random sites collecting total _______ slides (500 from each site).

Kala-Azar

- The frequency of case searches has been increased from a ______ annual case search to _______ case searches.

- The active case searches are carried out during a fortnight designated as the "Kala-azar __________", during which the peripheral health workers and volunteers are engaged to make _____________ search.

- An incentive amount of Rs. _____ is provided to ASHA for identifying each case of kala-azar and Rs. ______ for ensuring one round and Rs. ___ for two rounds of insecticide spraying.

- Even the patient being treated in the hospital will be given Rs. _____ as compensation of daily wage for the time he spends in the hospital during the treatment for kala-azar and Rs. _______ for PKDL.

- The new strategy also includes introduction of Rapid Diagnostic Kit developed by ICMR into the programme and single dose treatment with Liposomal Amphotericin B, which is given intravenously in 10 mg/kgbw dose.

- It is to _______ the human reservoir of infection.

- WHO will supply the drug _______ of cost.

National Leprosy Eradication Programme

- More focus has now been given to **new _____ detection** as they are main indicator for programme monitoring.

- More emphasis is given on **providing disability prevention** and medical rehabilitation (**DPMR**) services to leprosy affected persons. The aid provided is as follows :

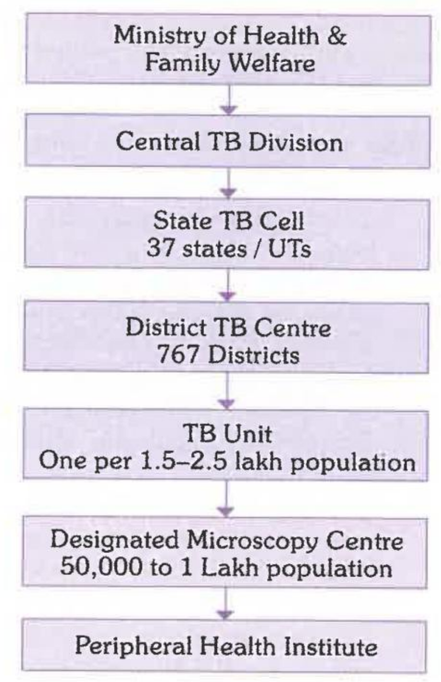
- Dressing materials, supportive medicines and ulcer kits
- **Micro-cellular rubber footwear** is provided for protection of insensitive feet.
- An amount of Rs. ______/- is provided as incentive to **each leprosy affected person** from BPL family undergoing reconstructive surgery
- Support is also provided **to government institutions/ PMR centres** in the form of Rs _______/- per reconstructive surgery conducted.

Activities to be performed by ASHA are as follows :

- _____________ suspected leprosy cases from their villages for _______ and _________ at PHC
- __________ of confirmed cases for their treatment ___________.
- To facilitate the involvement of ASHA in the programme, they are being paid incentive money as below :

ASHA Incentives	
On confirmed diagnosis of case brought by them	Rs. 250/-
On completion of full course of treatment of the case within specified time - PB leprosy case	Rs. 400/-
MB leprosy case	Rs. 600/-
An early case before onset of any visible deformity	Rs. 250/-
A new case with visible deformity in hands, feet or eye	Rs. 200/-

Organization structure of NTEP



National TB Elimination Programme

NTEP endorsed TB diagnostics

1. Smear microscopy for acid fast bacilli:

- Sputum smear stained with Ziehl-Neelsen staining; or
- Fluorescence stains and examined under direct or indirect microscopy with or without LED.

2. Culture

- Solid (____________________) media; or
- Liquid media (______________) using manual semi-automatic or automatic machines, e.g., Bactec, MGIT etc.

3. Rapid diagnostic molecular test:

- Conventional PCR based Line Probe Assay for MTB complex; or
- Real-time PCR based Nucleic Acid Amplification Test NAAT for MTB complex, e.g. GeneXpert.

4. Radiography where available.

5. Tuberculin skin test.

_____________: TB surveillance using case based, web based IT system, meaning eradication of TB.

Sputum microscopic examination

- A senior TB laboratory supervisor is appointed for every __ microscopy centres.

- The senior TB laboratory supervisor rechecks all the ______ slides and ____ per cent of the negative slides of these five microscopy centres.

- It is essential to examine __ sputum specimens of each patient before a conclusive diagnosis can be made.

- One sputum sample is not sufficient for diagnosis as the chance of detecting smear positive case is only ____ per cent.

- Sputum microscopy not only confirms the diagnosis, but also indicates the degree of _________ and _________ to treatment.

Newer initiative in treatment

- For the intensive phase, each blister pack contains ____ day's medication.

- For the continuation phase, each blister pack contains ____ weeks supply of medication.

- The boxes are coloured according to the category of the regimen, ___ for category I patients, _____ for category II patients.

National Strategic Plan for TB elimination

The aim of the National Strategic Plan is to achieve elimination of TB by 2025. During plan period, **targets for TB are:**

- 80% reduction in TB _________

- 90% reduction in TB __________

- 0% patient having ______________ expenditure due to TB

National AIDS Control Programme

Prevention of Parent-To-Child Transmission of HIV

Aim: To offer HIV testing to every pregnant woman in the country.

The essential package of PPTCT services in India are as follows :

- HIV counselling and testing to all pregnant women enrolled into antenatal care, with an '**_________ option.**
- Involvement of spouse and other family members, and move from 'ANC-Centric' to a "**________________ approach.**
- Provision of life-long-ART (_____________) to all pregnant and breast-feeding HIV infected women, regardless of CD4 count and clinical stage of HIV progression.
- Promotion of _________ deliveries of all HIV infected pregnant women.
- Provision of care for associated conditions (STI/RTI, TB and other opportunistic infections).

- Provision of nutrition, counselling and psychosocial support for HIV infected pregnant women.

- Provision of counselling and support for initiation of exclusive breast-feeds within an hour of delivery as the preferred option and continued for 6 months.

- Provision of ARV prophylaxis to infants from birth upto a minimum of 6 months.

- Integrating follow-up of HIV-exposed infants into routine healthcare services including immunization.
- Ensuring initiation of Co-trimoxazole Prophylactic Therapy (CPT) and Early Infant Diagnosis (EID) using HIV-DNA PCR at 6 weeks of age onwards, as per the EID guidelines.

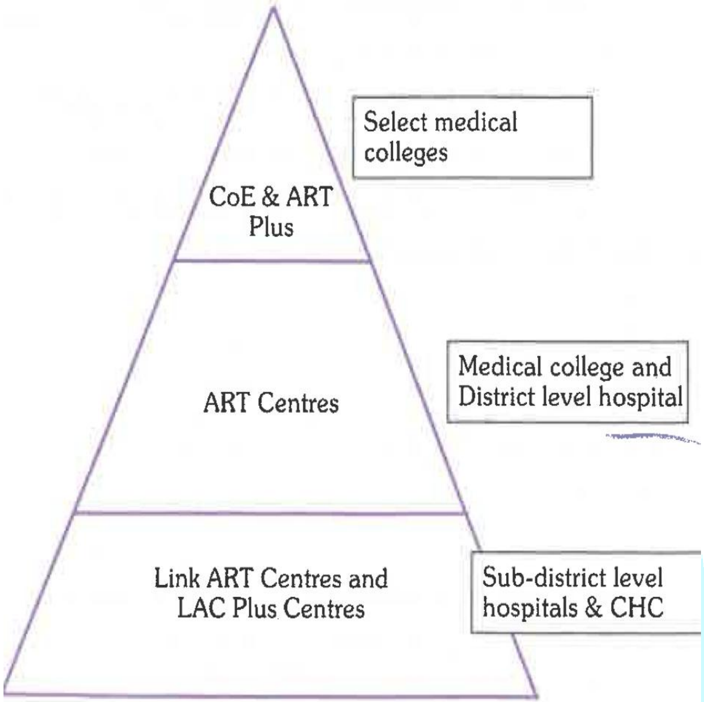
- Strengthening community follow-up and outreach through local community networks to support HIV- positive pregnant women and their families.

Country has adopted 95-95-95 target:

- 95 per cent of PLHIV ____________________, of which

- 95 per cent of PLHIV are on _______, of which

- 95 per cent of PLHIV have ________________



Model of HIV treatment services

Note: NACO has branded the STI/RTI services as "____________________

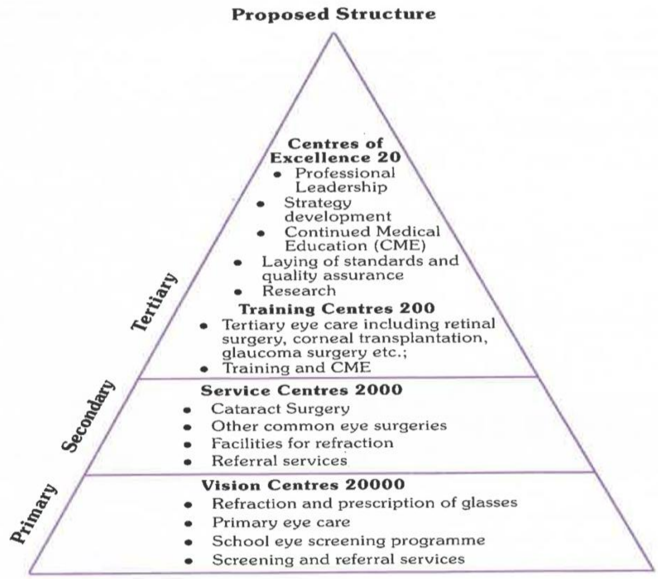
Pre-Packed STI/RTI colour coded kits are as follows:

- **Kit 1:**
- **Kit 2:**
- **Kit 3:**
- **Kit 4:**
- **Kit 5:**
- **Kit 6:**
- **Kit 7:**

Vision 2020 : The Right to Sight

- Target diseases are _________, _______________, ________________, corneal blindness, ____________, _________________.

- The proposed four tier structure includes Centres of Excellence (20), Training Centres (200), Service Centres (2000), and Vision Centres (20,000).



Proposed structure for Vision 2020 : The Right to Sight

Universal Immunization Programme

- WHO launched its "Expanded Programme on Immunization" (EPI) against six, most common, preventable childhood diseases, viz. _________, _________ (whooping cough), _______, ______, __________ and __________.
- The first vaccine to be introduced in India was _____ in 1962 as part of the National Tuberculosis Programme.

Pulse Polio Immunization Programme

Under this programme children under _______ years of age are given additional oral polio drops in ___________ and __________ every year on fixed days.

Inactivated Polio Vaccine (IPV)

Fractional IPV of ___ ml dose as __________ injection at __ and __weeks of life

Introduction of Pentavalent vaccine

- The pentavalent vaccine contains five antigens i.e., ____________, __________, ________, _____ and _____________________ (____) vaccine.

- Pentavalent vaccine is given at _th, __th and __th weeks as primary dose.
- Initially, India introduced pentavalent vaccine in two states — _____ and ___________. Presently it covers the whole country.

Questions about all vaccines

Q. If the mother/caregiver permits administration of only one injection during an infant's first visit at 9 months of age, which vaccine should be given ?

Ans. At 9 months of age, the priority is to give measles vaccine with OPV and Vitamin-A.

Q. Which vaccines can be given to a child between of 1-5 years of age, who has never been vaccinated ? .

Ans. The child should be given DPT1, OPV-1, measles and 2 ml of vitamin A solution. It should then be given the second and third doses of DPT and OPV at one month intervals. Measles second dose is also to be given as per the schedule. The booster dose of OPV/DPT can be given at a minimum of 6 months after administering OPV3/DPT3.

Q. Which vaccines can be given to a child between 5-7 years of age, who has never been vaccinated ?

Ans. The child should be given first, second and third doses of DPT at one month intervals. The booster dose of DPT can be given at a minimum of 6 months after administering DPT3 upto 7 years of age.

Q. Should one re-start with the first dose of a vaccine if a child is brought late for a dose ?

Ans. Do not start the schedule all over again even if the child is brought late for a dose. Pick up where the schedule was left off. For example: If a child who has received BCG, HepB-1, DPT-1 and OPV-1 at 5 months of age, returns at 11 months of age, vaccinate the child with DPT-2, HepB-2, OPV-2 and measles and do not start from DPT-1, HepB-1 again.

Q. Why it is not advisable to clean the injection site with a spirit swab before vaccination ?

Ans. This is because some of the live components of the vaccine are killed if they come in contact with spirit.

DPT vaccine

Q. If a child could not receive DPT 1, 2, 3 and OPV 1, 2,3 according to the schedule, upto what age can the vaccine be given ?

Ans. The DPT vaccine can be given upto 7 years of age and OPV can be given upto 5 years of age. If a child has received previous doses but not completed the schedule, do not restart the schedule and instead administer the remaining doses needed to complete the series.

Q. Why should there be a minimum gap of 4 weeks between two doses of DPT?

Ans. This is because decreasing the interval between two doses may not obtain optimal antibody production for protection.

Q. Why give the DPT vaccine in the antero-lateral mid-thigh and not the gluteal region ?

Ans. To prevent damage to the sciatic nerve. Moreover, the vaccine deposited in the fat of gluteal region does not invoke the appropriate immune response.

Q. What should one do if the child is found allergic to DPT or develops encephalopathy after DPT ?

Ans. They should be given the DTaP/DT vaccine instead of DPT for the remaining doses, as it is usually the P (whole cell Pertussis) component of the vaccine which causes the allergy/encephalopathy. It may be purchased with locally available resources.

Q. Why DT is replaced by DPT vaccine for children above 2 years of age ?

Ans. As pertussis cases were reported in higher age group children and the risk of AEFIs were not found to be more after DPT vaccine as compared to DT vaccine.

Measles vaccine

Q. Why give the measles vaccine only on the right upper arm ?

Ans. To maintain uniformity and to help surveyors in verifying the receipt of the vaccine.

Q. If a child has received the measles vaccine before 9 months of age, is it necessary to repeat the vaccine later ?

Ans. Yes, the measles vaccine needs to be administered, according 'to the National Immunization Schedule i.e. after the completion of 9 months until 12 months of age and at 16-24 months. If not administered in the ideal age for measles vaccine, it can be administered upto 5 years of age.

Q. What is measles catch-up campaign ?

Ans. A measles catch-up campaign is a special campaign to vaccinate all children in a wide age group in a state or a district with one dose of measles vaccine. The catch-up campaign dose is given to all children, both immunized and un-immunized, who belong to the target age group of 9 months to 10 years. The goal of a catch-up campaign is to quickly make the population immune from measles and reduce deaths from measles. A catch- up campaign must immunize nearly 100% of target age group children. .

Q. Why 2nd dose of measles vaccine is introduced in the National Immunization Programme ?

Ans. Measles is highly infectious disease causing illness and death due to complications such as diarrhoea, pneumonia or brain infection. One dose of measles vaccine at 9 months of age protects 85% of infants. With 2nd dose the aim is to protect all the children who remain unprotected after first dose.

Q. If a child comes late for the first dose, then can it get the second dose ?

Ans. All efforts should be made to immunize the children at the right age i.e. first dose at completed 9 months to 12 months and second dose at 16-24 months. However if a child comes late then give two doses of measles vaccine at one month interval until 5 years of age.

Q. If a child received one dose of measles vaccine during an SIA campaign, should it receive the routine dose of measles vaccine ?

Ans. Yes, the child should receive routine doses of measles vaccine according to the Immunization Schedule irrespective of the measles SIA dose.

BCG vaccine

Q. Why give BCG vaccine only on the left upper arm ?

Ans. BCG is given on the left upper arm to maintain uniformity and for helping surveyors in verifying the receipt of the vaccine.

Q. Why do we give 0.05 ml dose of BCG to newborns (below 1 month of age)

Ans. This is because the skin of newborns is thin and an intra- dermal injection of 0.1 ml may break the skin or penetrate into the deeper tissue and cause local abscess and enlarged axillary lymph nodes.

Q. Why is BCG given only upto one year of age ?

Ans. Most children acquire natural clinical/sub-clinical tuberculosis infection. by the age of one year. This too protects against severe forms of childhood tuberculosis e.g. TB meningitis and miliary disease.

Q. If no scar appears after administering BCG, should one re-vaccinate the child?

Ans. There is no need to re-vaccinate the child, even if there is no scar.

OPV

Q. Till what age can a child be given OPV ?

Ans. OPV can be given to children till 5 years of age.

Q. Can OPV and vitamin A be given together with DPT- booster dose ?

Ans. Yes

Q. Can an infant be breast-fed immediately after OPV ?

Ans. Yes

TT vaccine

Q. If a girl has received all doses of DPT and TT as per the NIS till 16 years of age and she gets pregnant at 20 years, should she get one dose of TT during pregnancy ?

Ans. Give 2 doses of TT during the pregnancy as per the schedule.

Q. Is TT at 10 years and 16 years meant only for girls ?

Ans. No, it is to be given to both boys and girls.

Q. Can TT be given in the first trimester of pregnancy ?

Ans. Yes, it should be given as soon as pregnancy is diagnosed.

Hepatitis B vaccine

Q. Up to what age can hepatitis B vaccine be given ?

Ans. According to the National Immunization Schedule, Hepatitis B vaccine should be given with the first, second and third doses of DPT till one year of age.

Q. Why give the birth dose of hepatitis B vaccine only within 24 hours of birth ?

Ans. The birth dose of Hepatitis B vaccine is effective in preventing perinatal transmission of Hepatitis B if given within the first 24 hours.

JE vaccine

Q. If a child 16~24 months of age has been immunized with JE vaccine during an SIA, can it receive the JE vaccine again, as part of routine immunization ?

Ans. No, currently this is a single dose vaccine and should not be repeated.

Q. If a child above 2 years (24 months) of age has not received the JE vaccine through either RI or an SIA, should she/he be given the JE vaccine ?

Ans. Yes, the child is eligible to receive a dose of the JE vaccine, through RI, till the age of 15 years

National Rural Health Mission

Kilkari

- Kilkari is an Interactive _______ Response (IVR) based mobile service that delivers time- sensitive audio messages (voice call) about pregnancy and child health directly to the mobile phones of pregnant women, mothers of young children and their families.

- The service covers the critical time period where the most maternal infant deaths occur from the ___th month of pregnancy until the child is _____ year old.

- Families which subscribe to the service receive one pre-recorded system generated call per _______.

- Each call will be ___ minutes in length and serve as reminders for what the family should be doing that week depending on woman's stage of pregnancy or the child's age.

- Kilkari services will be available to states in regional dialect too.

ANMOL

ANMOL is a tablet-based application which helps ______ in carrying out their day to day work efficiently and effectively by entering and updating service records of _____________ on real time basis.

Reproductive and child health Programme

Essential obstetric care

- _________ registration of pregnancy (within 12-16 weeks)

- Provision of minimum ______ antenatal check-ups by ANM or medical officer to monitor progress of the pregnancy

- Provision of _______ delivery at home or in an institution

- Provision of _________ postnatal check-ups to monitor the postnatal recovery and to detect complications.

Emergency obstetric care

- Complications associated with pregnancy are not always predictable, hence, emergency obstetric care is an important intervention to prevent maternal morbidity and mortality.

Essential newborn care

The primary goal of essential newborn care is to ______ perinatal and neonatal mortality. The main components are _________ of newborn with asphyxia, prevention of _________, prevention of __________, exclusive ___________ and referral of ____________

Prevention and control of anaemia in children

- Infants from the age of 6 months onwards upto the age of 5 years are to receive iron supplements in liquid formulation in doses of ___ mg elemental iron and _____ mcg folic acid per day for ____ days in a year.

- Children 6 to 10 years of age will receive iron in the dose of ___ mg elemental iron and ____ mcg folic acid for ____ days in a year.

First Referral Unit (FRU):

There are three critical determinants of a facility being 'declare" as a FRU. They are:

-
-
-

Janani Suraksha Yojana

- The National Maternity Benefit scheme has been modified into a new scheme called Janani Suraksha Yojana (JSY).
- It was launched on 12th April, 2005.
- The objectives of the scheme are — _______ maternal mortality and neonatal mortality through encouraging delivery at health institutions, and focusing at institutional care among women in below poverty line families.
- It is a 100 per cent ___________ sponsored scheme
- This benefit will be given to ____ women, both _____ and ______, belonging to _______ poverty line household.

Vandemataram scheme

- This is a _________ scheme wherein any obstetric and gynaec specialist, maternity home, nursing home, lady doctor/MBBS doctor can ________ themselves for providing safe motherhood services.
- The enrolled doctors will display ____________________ at their clinic.
- Iron and Folic Acid tablets, oral pills, TT injections etc. will be provided by the respective ________________ Officers to

the 'Vandemataram doctors/ clinics' for ______ distribution to beneficiaries.

- The cases needing special care and treatment can be referred to the government hospitals, who have been advised to take due care of the patients coming with Vandemataram cards.

Safe abortion services

A. Medical method of abortion

- _______________ (RU 486) followed by _______________.
- In India is recommended upto____ weeks (49 days) of amenorrhoea in a facility

B. Manual Vacuum Aspiration (MVA)

- Manual Vacuum Aspiration is safe and simple technique for termination of early pregnancy, makes it feasible to be used in primary health centres or comparable facilities, thereby increasing access to safe abortion services.
- The project of introducing the MVA technique has been piloted in coordination with FOGSI, WHO.

Janani-Shishu Suraksha Karyakram (JSSK)

- Government of India launched this on 1st June 2011, to make available better health facilities for _________ and _________.
- All pregnant women delivering in public health institutions to have absolutely _______ and ____ expense delivery, including caesarean section.
- The entitlements include ____ drugs and consumables, ___ diet upto ___ days during normal delivery and upto ___ days for C-section, free diagnostics, and free blood wherever required.
- This initiative would also provide for free transport from home to institution, between facilities in case of a referral and drop back home.

Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA)

- The Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA) has been launched by the Ministry in _____________
- Under PMSMA, all ________ women in the country are provided _____ day, _____ of cost assured and quality antenatal care.
- As part of the campaign, a minimum package of antenatal care services (including investigations and drugs) is being provided to the beneficiaries on the ___th day of every month.

- The Abhiyan also involves private sector's health care providers as volunteers to provide Specialist care in government facilities.

SUMAN (Surakshit Matritva Aashwasan)

- Launched by **Ministry of ______________________ on 10th October 2019.**
- With an aim to provide **assured, dignified, respectful, and quality healthcare at ____ cost** and zero tolerance for denial of services for every women and newborn visiting the public health facility.
- In order to end all preventable maternal and newborn deaths and morbidities and provide positive birthing experience.
- The **expected outcomes** of this new initiative is **______ preventable maternal and newborn deaths and high quality of maternity care delivered with dignity and respect.**

LAQSHYA PROGRAMME

- __________ launched "LaQshya program" to improve quality of care in labour room and maternity OTs in public health facilities in 2017.

- The LaQshya programme is evidence based approach to improve quality of maternal and newborn care and provide respectful care, particularly during the intrapartum and postpartum periods, which are the most vulnerable periods for a woman and contribute to a significant proportion of maternal deaths.

ANAEMIA MUKT BHARAT PROGRAMME (AMB)

- To achieve the envisaged target of ____ reduction in anaemia prevalence every year under the POSHAN Abhiyaan.

- The 6 x 6 x 6 strategy to reduce anaemia prevalence (nutritional and non-nutritional) in six age groups namely pre-school children (6-59 months), children (5-9 years), adolescent girls and boys (10-19 years), pregnant women, lactating women and in women of reproductive age group (15-49 years) in programme mode.

- The **six interventions** are prophylactic ______________ supplementation, periodic ______________, intensified year-round behaviour change communication campaign including _____________ clamping, testing and treatment of ___________ using digital methods and point of care treatment, mandatory provision of iron folic acid fortified foods in public health programmes and addressing non-nutritional causes of anaemia in endemic pockets, with special focus on malaria, haemoglobinopathies and fluorosis.

- ____________ IFA supplements are provided to children aged _______ months through ASHAs and Weekly IFA supplements to children of 5-10 years and adolescents 10-19 years of age.

- ______ doses of **IFA** supplements are also being provided to pregnant and lactating women during ANC and PNC period, respectively.

- ____________ **Vitamin-A** supplementation is being done for all children **below five years of age.**

SAANS:

Management of SAM

Other micronutrients should be given daily for at least 2 weeks:

Multivitamin supplement:

Folic acid:

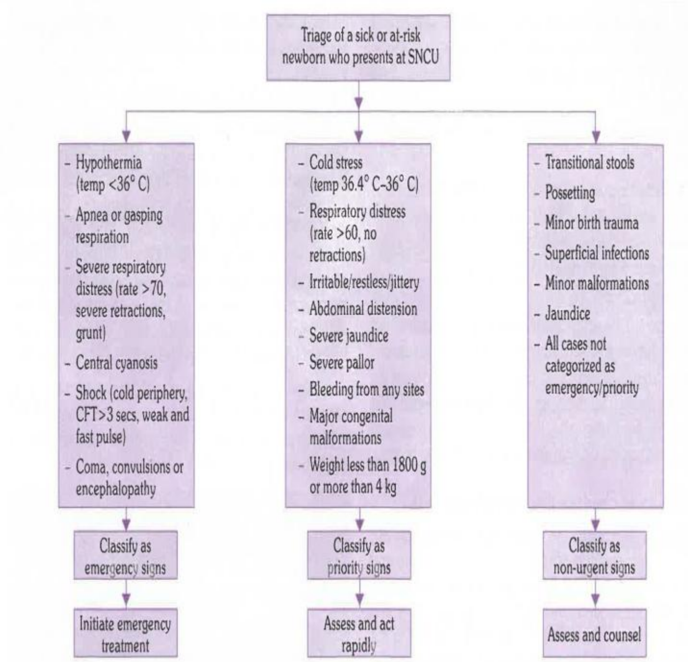
Elemental Zinc:

Copper:

Iron:

Follow-up of children discharged from NRC:

The AWWs should prioritize these children for home visits, ______ week in the first ___weeks and then once in ___ weeks till the child is discharged from the programme.



Newborns classified as "Emergency" require urgent intervention and emergency measures. All such newborns will be admitted to SNCU a

Home Based Newborn Care (HBNC)

ASHA will make visits to all newborns according to specified schedule upto 42 days of life. **The schedule of visit is as follows :**

- _____ visits in the case of institutional delivery -Day ____________, and ____
- _____ visits in the case of home delivery (Day ___________________).
- In cases of Caesarean section delivery, where the mother returns home after 5-6 days, ASHAs are entitled to full incentive of Rs. ____ if she completes all five visits starting from Day __ to Day ____.

- In cases when a newborn is discharged from SNCU, ASHAs are eligible to full incentive amount of Rs. ______ for completing the remaining visits. In addition, ASHAs are also eligible for an incentive of Rs. ___ for monthly follow-up of low birth weight babies and newborns discharged from SNCU

- In cases of twin or triples the incentive amount for ASHA would be two time of the regular HBNC incentive of Rs. 250/- (i.e., Rs. 500/-) or three times of Rs. 250/- (i.e., Rs. 750/-) respectively.

The incentive money is paid to ASHA on 45th day subject to the following:

- Record of birth weight in the mother and child protection card;

- Immunization of newborn with BCG, first dose of OPV, hep B and DPT /pentavalent vaccine and entry into the mother and child protection card;

- Registration of birth; and

- Both mother and newborn are safe until 42nd day of delivery

Home-Based Care for Young Child (HBYC)

ASHA will provide home visits on ____________________________ months to promote early initiation of breast feeding, exclusive breast feeding till 6 months and continued breast feeding till ____nd year of life.

Navjat Shishu Suraksha Karyakram (NSSK)

- It is a programme aimed to _____ health personnel in basic newborn care and resuscitation.
- It has been launched to address care at birth issue i.e. prevention of ___________, prevention of ___________, early initiation of breast-feeding and basic newborn resuscitation.
- The objective of the new initiative is to have a trained health person in basic newborn care and resuscitation unit at every delivery point.

Weekly Iron and Folic Acid Supplementation (WIFS)

Ministry of _______________________ has launched the Weekly Iron and Folic Acid Supplementation (WIFS) Programme to meet the challenge of high prevalence and incidence of anaemia amongst adolescent girls and boys.

The key interventions under this programme are as follows :

- Administration of supervised weekly iron-folic acid supplements of ____ mg elemental iron and ______ µg folic acid using a fixed day approach.
- Screening of target groups for moderate/severe anaemia and referring these cases to an appropriate health facility.

- ___________ de-worming (Albendazole _____ mg), six months apart, for control of helminths infestation.

- Information and counselling for improving dietary intake and for taking actions for prevention of intestinal worm infestation.

Dashboard indicators for INAP	
Impact level indicators	• • • • • • Survival rate of newborns discharged from SNCU/NICU at one year of age

	• Cause-specific neonatal mortality

Framework element	Targets for NCD prevention and control		
	Outcome	2020	2025
Premature mortality from NCDs	Relative reduction in overall mortality from cardiovascular disease, cancer, diabetes, or chronic respiratory disease	10%	
Alcohol use	Relative reduction in alcohol use	5%	
Obesity and diabetes	Halt the rise of obesity and diabetes prevalence	No target set	
Physical inactivity	Relative reduction in prevalence of insufficient physical activity	5%	
Raised blood pressure	Relative reduction in prevalence of raised blood pressure	10%	
Salt/sodium intake	Relative reduction in mean population intake of salt, with aim of achieving recommended level of less	20%	

	than 5 gms per day		
Tobacco use	Relative reduction in prevalence of current tobacco use	15%	
Drug therapy to prevent heart attacks and strokes	Eligible people receiving drug therapy and counselling (including glycemic control) to prevent heart attacks and strokes	30%	

Essential NCD medicines and basic technologies to treat major NCDs	Availability and affordability of quality, safe and efficacious essential NCD medicines including generics, and basic technologies in both public and private facilities	60%	
Household indoor air pollution	Relative reduction in household use of solid fuels as a primary source of energy for cooking	25%	

Ayushman Bharat- PMJAY

- Ayushman Bharat- PMJAY is jointly funded by the _______ and _____ governments.
- **Health Coverage:** Rs ____ lakh per family per year
- **Target beneficiaries:** 10.74 crore poor and vulnerable families (Socio-Economic Caste Census data)

Expected outcomes:

- **Increased ______** in public healthcare facilities
- **Reduction in ______________** expenditure
- Enhanced awareness of preventive & promotive healthcare
- Encouraging healthy lifestyles: **Yoga & proper nutrition**

Swajaldhara

- Swajaldhara is a community led participatory programme, which aims at providing ______ drinking water in _______ areas, with full ownership of the community, building awareness among the village community on the management of drinking water projects, including better hygiene practices and encouraging water conservation practices along with rainwater harvesting.
- Swajaldhara has **two components** : **Swajaldhara I (First Dhara**) is for a ________________ or a group of panchayats (at block / tahsil level) and **Swajaldhara II (Second Dhara)** has _______ as the project area.

Pradhan Mantri Awas Yojna (URBAN)

- ______________________________ launched on 25th June, 2015, as a flagship mission of Government of India.
- It addresses ______ housing shortage among the Economically Weaker Section (EWS)/Low Income Group (LIG) category including slum dwellers by ensuring a pucca house to eligible urban households.
- An eligible beneficiary family will comprise of a _________, ____, ________ sons and /or _________ daughters.
- The beneficiary family should not own a _______ House (an all whether dwelling unit) either in his/ her name or in the name of any member of his/her family in any part of India.
- An adult earning member (irrespective of marital status) can be treated as separate household.

Chapter 8: Demography and family planning

Demography

It focuses on three readily observable human phenomena :

a. changes in ________________ (growth or decline);
b. the ____________ of the population; and
c. the ____________ of population in space.

It deals with five "demographic processes", namely:

a.
b.
c.
d.
e.

Stages of demographic cycle	
Stage 1	• High birth rates and high death rates render the population _________ • _________ demographic gap

Stage 2	• The death rate _________ and the birth rate remains ______________ • The demographic dap starts __________ and becomes __________
Stage 3	• Death rate _________ further and birth rate ________ • ________ is in this stage currently • The demographic gap starts __________
Stage 4	• Low birth rate and low death rate render the population ___________ • __________ demographic gap

Stage 5	• The population begins to ________ as the birth rate is _______ than the death rate • The demographic gap is _________.

World population trends

- Three countries of **SEAR**, i.e. ________, ____________ and _______________ are among the most populous ten countries of the world.
- When the crude death rate is substracted from the crude birth rate, the net residual is the **current** __________________ **rate**, exclusive of ____________.

Demographic trends in India

- **Population statistics** include indicators that measure the population _____, _____ ratio, ________ and __________ ratio.

- **Vital statistics** include indicators such as _____ rate, ____ rate, natural growth rate, _____ expectancy at birth, ________ and _______ rates.

- The year 1921 is called the "__________" because the absolute number of people added to the population during each decade has been on the increase since 1921.

- _____ **ratio** is defined as the number of females per 1000 males.

Population Pyramid:

- It is also known as ______________ **pyramid.**
- It is a ___________ representation of the age and sex of a population.

- It shows the distribution of ______ across a population divided down the centre between male and female members of the population.

- The graphic starts from the youngest at the ________ to oldest at the _____.

Dependency ratio

- It represents the proportion of persons above ___ years of age, and children below ___ years of age who are dependent on the economically productive age group (________ years).

- Also referred to as the ___________ **dependency ratio,** which reflects the need for a society to provide for their younger and older population groups.

- It can be subdivided into **young age dependency ratio** (_____ years); and **old age dependency ratio** (____ years and more).

Total Dependency Ratio =

$$\frac{\text{Children (0-14 years)} + \text{Elderly (65+ years)} \times 100}{\text{Working – age population (15-64 years)}}$$

Given:

- Population aged 0–14 = **391,558,367**
- Population aged 15–64 = **856,076,200**
- Population aged 65+ = **71,943,390**

Step-by-step Calculation:

$$\text{Numerator} = 391,558,367 + 71,943,390 = 463,501,757$$

$$\text{Dependency Ratio} = \left(\frac{463,501,757}{856,076,200}\right) \times 100$$

$$\text{Dependency Ratio} \approx 0.5414 \times 100 = 54.14\%$$

- The young age dependency ratio is **38.3 per cent**, and old age dependency ratio is **9.8 per cent** for the year **mid 2020.**

- There is shift from child dependency to old age dependency, as fertility _______ and life expectancy _________.

- "**Demographic _______**" connotes the period when the dependency ratio in a population declines because of decline in fertility, until it starts to rise again because of increasing longevity.

- "**Demographic ________**" is used to connote the increase in the total dependency ratio during any period of time, mostly caused by increased old age dependency ratio.

Literacy and Education

- **______ literacy rate:** Literacy rate taking in account the total population in the __________.

- **________ literacy rate:** the literacy rate calculated taking into account the ___ years and above population in the denominator.

$$\text{Crude literacy rate} = \frac{\text{Number of literate persons} \times 100}{\text{Total population in a given year}}$$

$$\text{Effective literacy rate} = \frac{\text{Number of literate persons aged 7 and above} \times 100}{\text{Population aged 7 and above in a given year}}$$

Fertility

- There is ______ **association** between **fertility and educational** status.

- **Economic status** also bears _______ **relationship** with fertility.

Fertility Related Statistics	
	• Simplest indicator of fertility • Defined as "the **number of live births** per 1000 estimated **mid-year population**, in a given year".
	• It is the "number of live births per 1000 women in the reproductive age-group (15-44 or 49 years) in a given year". • Better measure of fertility than crude birth rate. • Major weakness of this rate is that not all women in the denominator are exposed to the risk of child birth.
	It is the "number of live births per 1000 married women in the reproductive age group (15-44 or 49) in a given year".
	Defined as the "number of live births in a year to 1000 women in any specified age-group".

	It is the number of live births in a year to 1000 married women in any specified age group.
	Represents the average number of children a woman would have if she were to pass through her reproductive years bearing children at the same rates as the women now in each age group
	Average number of **children** that would be born to a married woman if she experiences the current fertility pattern throughout her reproductive span.
	Average number of **girls** that would be born to a woman if she experiences the current fertility pattern throughout her reproductive span (15-44 or 49 years), assuming no mortality.
	• Defined as the **number of daughters a newborn girl**

	will bear during her lifetime assuming fixed age-specific fertility and mortality rates. • NRR of 1 is equivalent to attaining approximately the 2 child norm.

Causes of	
High birth rate	**Declining death rate**
• ____________ of marriage • ______ marriage • ______ puberty • _____ standard of living • _____ levels of literacy • Traditional customs and habits • ________ of family planning habit	• ___________ of natural checks • Mass control of diseases • Advances in medical science • Better health facilities • Impact of national health programmes • Improvements in food supply • International aids in severe directions • Development of social consciousness among the masses.

Eligible couples: Refers to a currently married couple wherein the wife is in the ____________ age, which is generally assumed to lie between the ages of ____ and ____.

Couple protection rate (CPR): Defined as the percent of eligible couples effectively protected against childbirth by one or the other approved methods of family planning, viz. __________, _____, _______ or _____ pills.

Contraceptive methods

Barrier methods

Condom:

- Known by its trade name ___________

- It protects both men and women from _______.

- Pregnancy rates vary from _____ per 100 women-years to more than ____ in typical users.

Diaphragm:

- Also known as _________

- It is inserted ______ sexual intercourse and must remain in place for **not less than ____ hours** after sexual intercourse.

- Failure rate vary from ______ per 100 women-years.

Vaginal sponge:

- Known by its trade name ________.

- It is a small __________ foam sponge , saturated with spermicide _____________

- Failure rate in parous women is between ________ per 100 women-years and in nulliparous women about ______.

Intra-Uterine Devices

Intra-uterine contraceptive devices	
First Generation IUDs	

Second Generation IUDs	**Earlier devices:** • • **Newer devices :** • • •
Third Generation IUDs	• •

Lippes Loop:

- It is __________ **shaped** device made of _______________.
- It contains a small amount of _____________ to allow X-ray observation.
- The Lippes Loop exists in _____ **sizes** A, B, C and D, the latter being the __________-.
- The larger loops (C and D) are more suitable for ___________ women.

Progestasert:

- It is a ___ shaped device filled with ____ **mg** of progesterone, the _________ hormone.

LNG-20 (Mirena)

- It is a ____-shaped IUD releasing ____ mcg of levonorgestrel (a potent __________ steroid)

- It has a _____ uterine pregnancy rate (____ per 100 women) and _____ number of ectopic pregnancies.
- Long-term exposure is associated with _______ menstrual blood loss and _______ days of bleeding
- It has an effective life of _____ years.
- It is valuable for women in developing countries, where excess blood loss caused by inert devices has been shown to result in significant ________.
- But these devices are more _________ to be introduced on a wide scale.

Change of IUD

- Inert IUDs such as Lippes Loop may be left in place as ______ required, if there are no side-effects.
- Copper device ______ be used indefinitely because they have to be replaced periodically. The same applies to the hormone-releasing devices.
- Cu-T 380A is approved for use for ____ years.
- Cu-T-200 is approved for ____ years
- Nova T for ____ years
- Progesterone-releasing IUD must be replaced ______ year
- The levonorgestrel IUD can be used for atleast ___ years, and probably ____ years.

Contraindication of IUD	
Absolute	**Relative**
• • • • •	• • • • • Distortions of uterine cavity d/t congenital malformations, fibroids •

The ideal IUD candidate:

- Who has borne atleast ____ child
- Has no history of ________ disease
- Has ________ menstrual periods
- Is willing to check the ______ ______
- Has access to ___________ and treatment of potential problems
- Is in a ___________ relationship

Side effect of IUD:

- _________: Commonest complaint
- _______: Second major side-effect
-
-
- Pregnancy
- Ectopic pregnancy
- Expulsion
- Fertility after removal
- Cancer and teratogenesis
- Mortality

Hormonal Contraceptives

Oral pills

Combined Pill:

- Combined pill contain _____ **mcg of a _________ oestrogen**, and _________ **mg of a _______________.**

- Pill is given orally for ____ **consecutive days beginning on ___ day** of menstrual cycle, followed by **break of ___ days** during which period menstruation occurs.

- This is actually __________ **bleeding** rather than menstruation.

- **Loss of blood** which occurs is about _____ that occurring in a women having ovulatory cycles.

- Pill should be taken ________ **day at a ______ time,** preferably before going to _____ at night.

- **Mechanism of action:** Prevent _____________ from the ovary. Achieved by blocking pituitary secretion of ______________.

Type of pills:

- Government of India has made available two types of ____-dose oral pills under brand name of _________ **and** ___________.

- It contains **Levonorgestrel _____ mg and ethinyl estradiol _____ mg.**

- MALA-D is a package of ____ pills (____ of oral contraceptive pills and ___ brown film coated ____ mg ________________ tablets) made available at a price of **Rs ____ per packet**.

- MALA-N is supplied ______________ through all PHCs.

Progesterone-only pill (POP):

- Commonly referred as _________ **or** ___________**.**

- It contains only progesterone like ______________ **and** _______________**.**

- They are prescribed to older women for whom the combined pill is contraindicated because of ___________________ risks.

- Also considered in young women with risk factors for _______.

- **Mechanism of action:** Render the cervical mucus ______ and ________ and thereby inhibit sperm penetration. Also inhibit tubal motility and delay transport of sperm and ovum to uterine cavity.

Post-coital contraception:

It is recommended within _____ hours of unprotected intercourse.

Methods:

Simplest technique is to insert an IUD, especially copper device within ____ day

or

Levonorgestrel used as one tablet of 0.75 mg within ____ hours of unprotected sex and the 2nd tablet after ____ hours of 1st dose.

or

_____ oral contraceptive pills containing 50mcg of ethinyl estradiol within 72 hours after intercourse, and the same dose after 12 hours.

or

______ oral contraceptive pills containing 30 or 35 mcg of ethinyl estradiol within 72 hours and 4 tablets after 12 hours.

or

_____________ 10 mg once within 72 hours.

Male Pill:

- It is made of __________ a derivative of __________ **oil.**

- Effective in producing azoospermia or severe oligospermia.
- If taken for ____ months 10% of men may be permanently azoospermic.
- Gossypol could be ________.

Note: Oral contraceptive effectiveness affected by ___________, _____________ and ___________.

Oral contraceptive pills		
Beneficial effects	**Contraindications**	
	Absolute	**Relative**
- -	-	• Age > 40 yrs • Smoking and age > 35 yrs

- - - -	- - - - -	• Mild hypertension • Chronic renal ds • Epilepsy • Migraine • Nursing mothers in first 6 months • Diabetes mellitus • Gall bladder ds • H/O infrequent bleeding • Amenorrhoea

Depot formulations

DMPA:

- Depot-medroxyprogesterone acetate given as intramuscular injection of _____ mg every ___ months.
- **MOA:**
- _________ affect lactation
- **Side effects:** Weight ________, _________ menstrual bleeding and prolonged __________.
- Used in multiparae of age over _____ yrs who have already completed their families.

NET-EN:

- Norethisterone enantate given intramuscularly in dose of _____ every _____ days
- **MOA:** Inhibition of ovulation, progestogenic effects on cervical mucus

Subdermal implants:

- Known as _________ consist of ___ silastic capsules containing 35 mg of levonorgestrel

Centchroman non-hormonal pill (Chhaya):

- Known as _______ pill as for the first ____ months taken biweekly and from ____ month onwards, once weekly.

Oral abortifacient

- **______________ in combination with _________** terminate pregnancies of upto **____ week's duration** with minimum complications.

- Regimen is _____________ mg orally on day 1, followed by ________________ mcg vaginally either immediately or within 6-8 hours.

- MTP kit have combipack tablets of _______________ mg one tablet and _______________ mcg 4 tablets (800 mcg).

Contraindication:

- H/O allergy or known hypersensitivity to mifepristone, misoprostol or other prostaglandin;
- Confirmed or suspected ectopic pregnancy or undiagnosed adnexal mass;
- ________ in place
- Chronic __________ failure;
- Haemorrhagic disorder or concurrent anticoagulant therapy;
- Inherited ___________;

- If a patient does not have adequate access to medical facilities equipped to provide emergency treatment of incomplete abortion and blood transfusion.

Abortion

- Defined as termination of pregnancy before the foetus becomes viable (**____ weeks with approx. weight ________).**

- MTP bill allow abortion upto ____ **weeks of pregnancy.**

The Medical Termination of Pregnancy Act

Conditions under which pregnancy can be terminated under MTP Act:

a. **__________:** Pregnancy can endanger mother's life

b. **_________:** Substantial risk of child being born with serious handicaps

c. **____________:** Pregnancy is result of rape

d. _______________

e. __________ **of contraceptive device**

Person who can perform abortion:

- Registered medical practitioner having experience in Gynecology and obstetrics to perform abortion where length of pregnancy does not exceed _____ weeks.
- When pregnancy exceeds ____ weeks and not more than ____ weeks, the opinion of _____ registered medical practitioners is necessary to terminate the pregnancy.
- _________________ of district certify that a doctor has necessary training in gynaecology and obstetrics to do abortions.
- Registered medical practitioner should qualify if he has assisted a RMP in a performance of ____ cases of medical termination of pregnancy in an approved institution. Or he should have one or more of following qualifications:
    - ___ months housemanship in obstetrics and gynaecology
    - _____________ qualification in OBG

- ____ yrs of practice in OBG for those doctors registered before 1971 MTP Act was passed.
- ___ yr of practice in OBG for those doctors registered on or after date of commencement of Act.

Terminal methods (Sterilization)

Guidelines for sterilization:

- The age of the husband should not be <___years nor > __ years.
- The age of the wife should not be < ___ years or > ___ years.
- The motivated couple must have ___ living children at the time of operation.
- If couple-has ___ or more living children, the lower limit of age of the husband or wife may be relaxed at the discretion of the operating surgeon, and
- Consent of his/her _______ to undergo sterilization operation

Male sterilization

- Male sterilization or _________ is a simple operation performed even in ______ by trained doctors under ______ anaesthesia.
- Acceptor is ____ immediately sterile after the operation, usually until approximately _____ ejaculations have taken place.

- During this intermediate period, another method of _______________ must be used.
- If properly performed vasectomy are almost _____% effective.
- Following vasectomy, sperm production and hormone output are _______________
- _______________ **vasectomy** is new technique that is safe, convenient and acceptable to males, funded by __________.

Complications:

- Pain, scrotal haematoma and local infection
- Sperm ___________
- ____________ recanalization
- ____________ response

Female sterilization

- Laparoscopy is not advisable for postpartum patients for ___ weeks following delivery.
- Haemoglobin per cent should not be less than ____.
- There should be no associated medical disorders such as _____ disease, ____________ disease, _________ and ___________.
- It is recommended that the patient be kept in hospital for a minimum of ____ hours after the operation.

- The cases are required to be followed-up by health workers once between ______ days after the operation, and once again between ____ and _____ months after the operation.

Evaluation of contraceptive methods

Pearl index:

- Defined as number of ________ per 100 woman-years of exposure (HWY).

$$\text{Failure rate per HWY} = \frac{\text{Total accidental pregnancies}}{\text{Total months of exposure}} \times 1200$$

- The factor _______ is the number of months in _____ years.

- Pearl index is usually based on ________ exposure and therefore fails to accurately compare methods at various __________ of exposure.

Life table analysis

- Calculates a ________ rate for each _______ of use.

- A ___________ failure rate can then _______ methods for any specific _______ of exposure.

New initiatives of delivery system

Home delivery of contraceptives:

______ is charging a nominal amount from beneficiaries for her effort to deliver contraceptives at doorstep i.e.,

- Rs. ___ for a pack of 3 condoms.
- Rs.___ for a cycle of OCPs and
- Rs. ____ for a pack of one tablet of ECP

Ensuring spacing at birth:

- Rs. _____/- to ASHA for delaying first child birth by 2 years after marriage.
- Rs. _____/- to ASHA for ensuring spacing of 3 years after the birth of 1st child.
- Rs.______/- in case the couple opts for a permanent limiting method upto 2 children only.

Pregnancy Testing Kits:

- ________-home based pregnancy test kits

Evaluation of impact in family planning

WHO proposed the following indices for evaluating the impact:

-
-
-
-
-
-

NOTES

Chapter 9: Preventive Medicine in Obstetrics, Paediatrics and Geriatrics

Currently, the main health problems affecting the health of the mother and the child in India, as in other developing countries, revolve round the **triad of:**

-
-
-

Antenatal Care

Antenatal visits:

- Ideally the mother should attend the antenatal clinic once a month during the first ___months; _______ a month, during the next month; and thereafter, _____ a week, if everything is normal.

- Registration of pregnancy within ____ weeks is the primary responsibility of the ANM.

- A minimum of ____ visits covering the entire period of pregnancy should be the target:

- **1st visit**- within 12 weeks, preferably as soon as the pregnancy is suspected, for registration of pregnancy and first antenatal check-up.
- **2nd visit**- between
- **3rd visit** - between
- **4th visit** - between

Estimation of number of pregnancies in a specified area and pregnancy tracking:

The number of expected pregnancies per year :

The ANM must know the population size and birth rate of the area under her jurisdiction.

The expected number of live births may be calculated as shown below :

Expected no. of live births (Y)/year = Birth rate (per 1000 population) x population of the area / 1000

As some pregnancies may not result in a live birth (i.e. abortions and stillbirths may occur), the expected number of live births would be an under-estimation of the total number of pregnancies.

Hence. a correction factor of 10% is required, i.e. add 10% to the figure obtained above.

So, the total number of expected pregnancies (z) = Y + 10% of Y

As a thumb rule, in any given month, approximately half the number of pregnancies estimated above should be in records.

Example of estimation of the number of pregnancies annually

Birth rate = 25/1000 population, Population under the sub-centre = 5000

Therefore, expected number of live births = (25 x 5000)/1000 = 125 births

Correction factor (pregnancy wastage) = 10% of 125 = say 13

Therefore, total no. of expected pregnancies in a year = 125 + 13 = 138

In any month, the ANM should have about 69 pregnancies registered with her.

Laboratory investigations:

A) At the sub-center:

-
-
-
-

B) At the PHC/CHC/FRU:

-
-
-
-
-
-

Identify "high risk" cases:

- Elderly primi (30 years and over).
- Short statured primi (140 cm and below).
- Malpresentations, viz breech, transverse lie, etc.
- Antepartum haemorrhage threatened abortion.
- Pre-eclampsia and eclampsia.
- Anaemia.
- Twins, hydramnios.
- Previous still-birth, intrauterine death, manual removal of placenta.
- Elderly grand multiparas.
- Prolonged pregnancy (14 days—after expected date of delivery).
- History of previous caesarean or instrumental delivery.
- Pregnancy associated with general diseases, viz. cardiovascular disease, kidney disease, diabetes, tuberculosis, liver disease, malaria, convulsions, asthma, HIV, RTI, STI, etc.

- Treatment for infertility.
- Three or more spontaneous consecutive abortions.

Drugs causing Foetal malformations:

- **Thalidomide:**
- **LSD:**
- **Streptomycin:**
- **Iodide-containing preparations:**
- **Corticosteroids:**
- **Sex hormones** may produce
- **Tetracyclines**:
- **Anesthetic agents** including **pethidine**: Depressant effect on the baby and delay the onset of effective respiration.

Health education during postnatal period:

Health education during the postnatal period should cover the following broad areas:

-
-
-
-
-

APGAR SCORE

- The Apgar score is taken at __ minute and again at __ minutes after birth.
- It requires immediate and careful observation of the heart rate, respiration, muscle tone, reflex response and colour of the infant.
- It provides an immediate estimate of the physical condition of the baby.

Apgar Score			
Sign	**Score**		
	0	1	2
Heart rate			
Respiratory effort			
Muscle tone			
Reflex response			
Colour			
Total score= 10	Severe depression	Mild depression	No depression

Breast-Feeding

- Breast-feeding should be initiated within an ____ of birth instead of waiting several hours as is often customary.

- The first milk which is called "_______" is the most suitable food for the baby during this early period because it contains a high concentration of ________ and other nutrient the body needs; it is also rich in anti-infective factors.
- The regular milk comes on the third to sixth day after birth.

Neonatal Screening

Routine Tests:

- Apgar score and routine clinical examinations on all newborn infants.

- 10 to 15 ml of cord blood should be collected at birth and saved in the refrigerator for _____ days for typing, coombs' testing.

Common disorders screened:

Phenylketonuria (PKU) :

- It is a rare disorder of ________________________

- This is an ____________________ trait in babies who are homozygous with deficiency in enzyme phenylalanine hydrozylase (PAH) which normally converts ____________ to ____________-.

- The deficiency results in raised serum phenylalanine concentrations causing mental retardation and tendency to seizures if the child does not receive low phenylalanine diet.

- Mass screening of blood phenylalanine in neonates is performed in many countries by the ___________ test.

- The treatment of PKU consists of a diet free of _______________.

Neonatal hypothyroidism :

- This is the most common disorder that is screened.
- The test involves measuring the radio-immuno-assays of the thyroid hormone T4 or the thyroid-stimulating hormone (TSH).

Coombs' test :

- This is done routinely on infants of all _____________ mothers.

Sickle cell or other haemoglobinopathies :

- Agar-gel electrophoresis is done on infants whose mothers have sickle cell or other haemoglobinopathies, e.g., thalassaemia, G6PD.

Congenital dislocation of hip :

- The test consists of manipulating the hip of the child from the _______ to _______ position while the thigh is flexed.
- A positive or abnormal result is indicated by a click or snap being produced during the test.
- The test is carried out on all babies at 6-14 days after birth, and at monthly intervals until the child is 4 months old.

Criteria to identify high risk Infants

- Birth weight less than _______
-

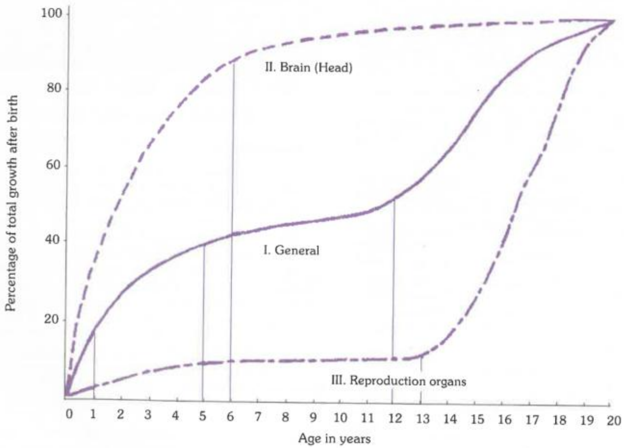
- Birth order ____________
- ___________ feeding
- Weight below _____ per cent of the expected weight
- Failure to gain weight during _______ successive months
- Children with PEM, Diarrhoea
- Working mother/one parent.

Small for Date (SFD) babies

- Also known as small for ______________ babies.
- These may be born at **term or preterm.**
- They weigh **less than the ____ percentile** for the gestational age.
- These babies are clearly the result of retarded intrauterine foetal growth.

$$\frac{\text{Live-born babies with birth weight less than 2.5 Kg}}{\text{Total number of live births}} \times 100$$

Growth and Development



Growth after birth

Reference or “standard” values of growth

- **Harvard (or Boston) standards**
- **WHO references values:** They are used for children upto 5 years of age since the influence of ethnic or genetic factors on young children at this age period is considered insignificant.
- **Indian standards**

Physical growth

1. Weight-for-age: Measurement of weight and rate of gain in weight are the best single parameters for assessing physical growth.

2. Height (length)-for-age:

- Height should be taken in a standing position without footwear.
- The length of a baby at birth is about 50 cm. It increases by about 25 cm during first year, and by another 12 cm during the second year.
- During growth spurt, boys add something like ___ cm in their height, and girls gain about ____ cm.
- Indian girls reach 98 per cent of their final height on an average by the age of _______ years, and boys reach the same stage by the age of _______ years.
- _________ is a stable measurement of growth as opposed to body weight.

Whereas weight reflects only the present health status of the child, height indicates the events in the past also.

- **Low height for age:** This is also known as nutritional __________ or ___________. It reflects _____ or __________ malnutrition.

3. Weight-for-height

- Weight in relation to height is now considered more important than weight alone. It helps to determine whether a child is within range of "normal" weight for his height.

- **Low weight for height:** This is also known as nutritional ________ or ____________________________

- A child who is less than ____ per cent of the expected weight- for-height is classed as severely wasted.

4. Head and chest circumference: At birth the head circumference is about 34 cm. It is about 2 cm more than the chest circumference. By ________ months, the two measurements become equal, after which the chest circumference overtakes the head circumference.

The Growth Chart

- ______ is the most sensitive measure of growth, and any deviation from "normal" can be detected easily by comparison with reference curves.

- A child can lose ______-, but not _______-.

- It is the _______ of growth that is more important than the _______ of dots on the line.

- The importance of the direction of growth is illustrated at the ____ hand, ______ corner of the chart.

- ________ or ________ of the child's weight curve signals growth _____, which is the earliest sign of the protein-energy malnutrition.

Uses of growth chart:

- For _______________

- __________ tool: for identifying "high-risk" children.
- _________ and policy making
- ____________ tool
- Tool for ________
- __________
- Tool for _________

Rights of the Child

- **Article** ____ prohibits employment of children below the age of 14 in factories.

- **Article** ____ prevents abuse of children of tender age; and

- **Article____** provides for free and compulsory education for all children until they complete the age of 14 years.

Integrated Child Protection Scheme (ICPS)

Ministry of ___________________- Development launched a new centrally sponsored scheme called "Integrated Child Protection Scheme" (ICPS).

The services provided under ICPS are as follows:

- Emergency outreach service through 'Child line', dedicated number is 1098. It is a 24-hour toll free telephone service available to all children in distress.
- Open shelters for children in need, in urban and semi-urban areas.
- Family based non-institutional care through sponsorship, foster-care, adoption, cradle baby centres and after-care.
- Institutional services through shelter homes, children homes, observation homes, special homes, and specialized services for children with special needs.
- Web-enabled child protection management system including website for missing children.
- General grant-in-aid for need based interventions.

Protection of Children from Sexual Offences (POCSO)

- Ministry of ________________ Development introduced the Protection of Children from Sexual Offences (POCSO) Act, 2012.

- The Act defines a child as any person below ___ years of age and regards the best interest and well-being of a child as being of paramount importance at every stage, to ensure a healthy physical, emotional, intellectual and social development of the child.

- An ordinance providing the death penalty for rapist of girls below ___ years of age "The Criminal Law Amendment Ordinance, 2018" was promulgated.

- A person is guilty of using a child for pornographic purposes if he uses a child in any form of media for the purpose of sexual gratification. The punishment is minimum 5 years in jail.

- Child offenders between __________ years of age should be covered under the ambit of the POCSO Act 2012 and be given the same punishment as is given to adults.

The salient features of the Ordinance are

a) Minimum punishment for rape made 10 years.
b) Minimum punishment of 20 years to a person committing rape on a girl aged below 16 years.
c) Minimum punishment of 20 years rigorous imprisonment and maximum death penalty

life imprisonment for committing rape on a girl aged below 12 years,

d) Police officer committing rape anywhere shall be awarded rigorous imprisonment of minimum 10 years.
e) Investigation of rape cases to be completed within 2 months.
f) Appeals in rape cases to be disposed within 6 months; and
g) No anticipatory bail can be granted to a person accused of rape of girl of age less than 16 years.

Maternal Mortality Ratio

Maternal death is defined as "the death of a woman while pregnant or within ____ days of termination of pregnancy, irrespective of the duration and site of pregnancy, from any cause related to or aggravated by the pregnancy or its management but not from unintentional or incidental causes"

$$= \frac{\text{Total no. of female deaths due to complications of pregnancy, childbirth or within 42 days of delivery from "puerperal causes" in an area during a given year}}{\text{Total no. of live births in the same area and year}} \times 1000 \text{ (or } 100{,}000\text{)}$$

Late maternal death: It is "the death of a woman from direct or indirect obstetric causes, after more than_____ days but less than _____ year after termination of pregnancy".

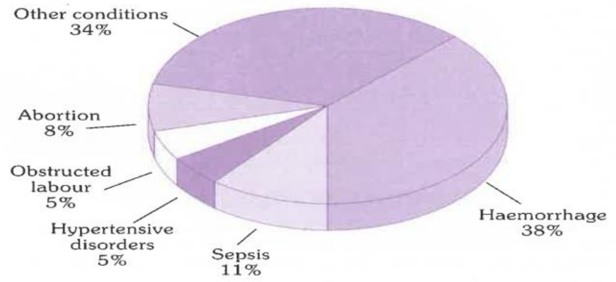
Pregnancy-related death (also known as death occurring during pregnancy, childbirth and puerperium): is defined as "The death of a woman while pregnant or within ____ days of ____________ of pregnancy, irrespective of the cause of death (obstetric and non-obstetric)"

Maternal mortality ratio (MMR): is defined as the number of maternal deaths during a given time period per 100,000 live births during the same time period.

Maternal mortality rate (MMRate): is defined and calculated as the number of maternal deaths divided by person-years lived by women of reproductive age in a population.

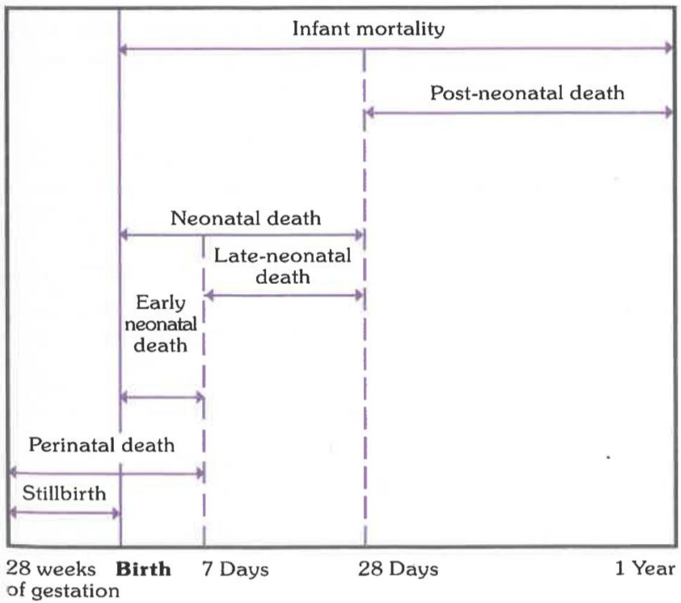
RHIME

SRS (Sample Registration System) included a new method called the "RHIME" or ____________, ____________, _______ __________ _________ of __________ with Medical __________. This is an enhanced form of "verbal autopsy" which is the key feature of a prospective study of 1 million deaths within the SRS.



Major causes of maternal deaths in India (2003)

Mortality in Infancy and Childhood



Mortality in and around infancy

Still Birth Rate

Death of a foetus weighing _______ (this is equivalent to 28 weeks of gestation) or more occurring during one year in every 1000 total births (live births plus stillbirth).

$$\text{Stillbirth rate} = \frac{\text{Foetal deaths weighing over 1000 g at birth during the year}}{\text{Total live + stillbirths weighing over 1000 g at birth during the year}} \times 1000$$

Neonatal Mortality Rate

- Deaths occurring during the neonatal period, commencing at ____ and ending _____ completed days after birth.

- Neonatal mortality rate is the number of neonatal deaths in a given year per 1000 live births in that year.

- The neonatal mortality rate is tabulated as :

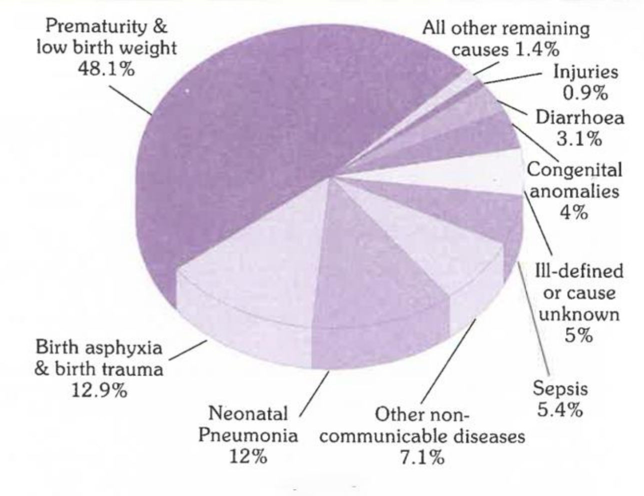
$$= \frac{\text{Number of deaths of children under 28 days of age in a year}}{\text{Total live births in the same year}} \times 1000$$

Causes of Neonatal Mortality:

- The neonatal mortality is _______ related to the **birth weight** and **gestational age.**

- Neonatal mortality rates of babies born to mothers with no **education** are nearly _____ as high as those of babies born to mothers with secondary education or higher.

- Ending **child marriage**, reducing **adolescent pregnancy** and extending **birth intervals** are crucial to _______ the risk of newborn mortality.



Causes of neonatal deaths in India, 2017

Source : (87)

Post-Neonatal Mortality Rate

- Deaths occurring from ____ days of life to under ____ year are called "post-neonatal deaths".
- The ratio of post-neonatal deaths in a given year to the total number of live births in the same year; usually expressed as a rate per 1000.
- The post-neonatal mortality rate is tabulated as:

$$= \frac{\text{Number of deaths of children between 28 days and one year of age in a given year}}{\text{Total live births in the same year}} \times 1000$$

Infant Mortality Rate

The ratio of infant deaths registered in a given year to the total number of live births registered in the same year; usually expressed as a rate per 1000 live births.

$$\text{IMR} = \frac{\text{Number of deaths of children less than 1 year of age in a year}}{\text{Number of live births in the same year}} \times 1000$$

The principal causes of infant mortality in India are: ____________ (57%), __________ infections (17%), ________ diseases (4%), congenital malformations (5%) and cord infection (2%), birth injury (3%) and unclassified about (18%).

Child Survival Index

$$\text{Child survival rate} = \frac{1000 - \text{under 5 mortality rate}}{10}$$

Integrated Management of Childhood Illness (IMCI)

The complete IMCI case management process involves the following elements:

- Assess a child by checking first for danger signs
- Classify using a **colour-coded triage system.**
  - urgent pre-referral treatment and referral (_______), or
  - specific medical treatment and advice (_______), or
  - simple advice on home management (_______).
- Identify specific treatments for the child.
- Provide practical treatment instructions,
- Counsel to solve any feeding problems found.
- Give follow-up care

Prenatal diagnosis of congenital anomalies

- **Alpha fetoprotein:** ___________ can be detected by measurement of a specific protein of foetal origin, called alpha fetoprotein in maternal blood and in amniotic fluid during pregnancy
- **Ultrasound:** Visualizes the foetus and detect abnormalities of the foetus.
- **Amniocentesis:** Performed after the completion of ___ **weeks** to detect **Down's syndrome** and **Neural tube defects.**
- **Chorionic villi sampling:** Performed at ______ **weeks of pregnancy** for early chromosomal status determination.

School health programme under Ayushman Bharat

Activities in school	
Weekly	
Fortnightly/Monthly	
Quarterly	
Bi-annual	

Age appropriate health promotion		
Primary school	*Middle school*	*High school*
• Health, growth and development • Personal safety • Nutrition and physical activity • Hygiene practices • Prevention of diseases like malaria, dengue, TB, worms infestation, diarrhoea and vaccine preventable diseases	• Puberty and related changes • Eye care, oral hygiene • Nutrition • Bullying prevention • Meditation and yoga • Internet safety and media literacy • Prevention of substance abuse • HIV/AIDS • Mental health	• Prevention of substance abuse • Sexual & reproductive health • Violence prevention • Unintentional injury • Road safety • Nutrition • Meditation and Yoga

Age groups	6-10 years	10-19 years
Intervention/ dose	Tablets of 45 mg elemental iron and 400 mcg of folic acid	Tablets of 100 mg elemental iron and 500 mcg of folic acid
Regime	Weekly, throughout the period 6-10 years of age	Weekly, throughout the period 10-19 years of age
Service delivery	Through teachers	Through teachers

Provision of services in 6-19 yr of age

Weekly iron folic acid supplementation through 6-19 years of age will follow the existing guidelines in the schools. These services will continue to be delivered through school teachers.

Deworming:

- Government of India has declared 10th ________ and 10th ______ as fixed days to provide Albendazole tablets for deworming school-age children.

- During NDD, Albendazole ______ chewable tablets will be administered to children at government, government-aided, and private schools.

Juvenile Delinquency

Defines delinquent as "a child who has committed an _______". Juvenile means a boy who has not attained the age of ____ years and a girl who has not attained the age of ____ years.

Battered baby syndrome

- Defined as "a clinical condition in young children, usually under __ years of age who have received non-accidental wholly inexcusable violence or injury, on one or more occasions, including minimal as well as severe fatal trauma, for what is often the most trivial provocation, by the hand of an adult in a position of trust, generally a parent, guardian or foster parent.
- In addition to physical injury, there may be deprivation of nutrition, care and affection in circumstances which indicate that such deprivation is not accidental".
- Found in all strata of society.
- Most worrying is the risk of mental and neurological complications.

Child abuse

All forms of physical and/or emotional ill-treatment, sexual abuse, neglect or negligent treatment or commercial or other exploitation resulting in actual or potential harm to the child's health, survival, development or dignity in the context of a relationship of responsibility, trust or power".

There are **four types of abuse:**

a)
b)
c)
d)

UJJAWALA

- A comprehensive scheme to combat _________ was launched in India by the Ministry of ________________ Development on 4th December 2007.

- The scheme has _____ components of prevention, rescue, rehabilitation, reintegration and repatriation of victims trafficked for commercial sexual exploitation

INSPIRE: Seven strategies for ending violence against children

INSPIRE is a set of seven evidence-based strategies for countries and communities working to eliminate violence against children.

I	
N	
S	
P	
I	
R	
E	

Child placement

______________: Children who have no home or who for some reason could not be cared for by their parents.

___________________:

- Fostering is an arrangement whereby a child lives, usually on a temporary basis, with an extended or unrelated family member.

- Such an arrangement ensures that the birth parents do not lose any of their parental rights or responsibilities.

BORSTALS:

- Boys over ____ years who are too difficult to be handled in a certified school or have misbehaved there, are sent to a Borstal.

- This institution falls in a category between a certified school and an adult prison.

- A Borstal sentence which is usually for ______ years, is regarded as a method of training and reformation.

- There are about ____ Borstals for boys in India but none for girls.

- Borstals do not come under the Children Act but are governed by the ________ Inspector General of Prisons.

______________: In the remand home, the child is placed under the care of doctors, psychiatrists and other trained personnel.

Integrated Child Development Services (ICDS)

Integrated Child Development Scheme (ICDS)	
Beneficiary	**Services**
Pregnant women	• • • •
Nursing mothers	• • •
Other women 15-45 years	
Children less than 3 years	• • • •
Children in age group 3-6 years	• • • • •
Adolescent girls 11-18 years	• •

Supplementary nutrition:

- Each child 6-72 months of age to get ____ calories and _____ grams of protein (financial norm of Rs 8.00 per child per day);

- Severely malnourished child 6-72 months to get ____ calories and _____ grams protein (financial norm of Rs 12.00 per child per day);

- Each pregnant and nursing woman to get ____ calories and ______ grams of protein (financial norm of Rs 9.50 per beneficiary per day)

Schemes for adolescent girls

- There are two schemes for adolescent girls viz. "_______________" and "Nutrition Programme for Adolescent Girls".

- Kishori Shakti Yojana is being implemented using the infrastructure of __________.

- The scheme targets adolescent girls in the age group of _________ years and addresses their needs of self-development, nutrition and health status, literacy and numerical skills, vocational skills etc.

Poshan Abhiyan

The goals are to achieve improvement in nutritional status of children from 0-6 years, adolescent girls, pregnant women and lactating mothers with fixed targets.

The objectives and targets are as follows	
Prevent and reduce stunting in children (0-6 years)	By
Prevent and reduce under nutrition and underweight prevalence in children (0-6 years)	By
Reduce the prevalence of anaemia among children (6-59 months)	By
Reduce the prevalence of anaemia among girls and women in the age group (15-49 years)	By
Reduce low birth weight (LBW)	By

Two more schemes are being implemented at the ICDS level. They are:

- **Rajiv Gandhi Scheme for Empowerment of Adolescent Girls** — "____________" for the age group 11 to 18 years to improve their nutritional and health status;

- **Indira Gandhi Matritva Sahyog Yojana (IGMSY)**, under which conditional cash transfer will be made to pregnant and lactating mothers in order to improve their nutritional and health status.

NOTES

Chapter-10: Health care of community

Primary Health Care

It is __________ **health care** made __________ **accessible** to individuals and acceptable to them, through their full participation and at a cost the community and country can afford.

Elements of primary health care	
E	__________ about prevailing health problems and the methods of preventing and controlling them
L	Prevention and control of _______ endemic diseases
E	Provision of __________ drugs
M	___________ and child health care, including family planning
E	___________ immunization against major infectious diseases
N	Promotion of food supply and proper ___________
T	Appropriate ___________ of common diseases and injuries
S	Adequate supply of safe water and basic __________

Principles of PHC

Suggested norms for health personnel	
Category of personnel	**Norms suggested 1 per**
Nurses	
Health workers female and male	
Trained Dai	
Health assistants (male and female)	
Pharmacists	
Lab. technicians	
ASHA	

ASHA

(Accredited Social Health Activist)

- ASHA must be resident of the _______- a ________ (married/ widow/divorced), aged _________ years
- Formal education **upto ________ class,** with communication skills and leadership qualities.
- The general norm of selection is one **ASHA for ________ population.**
- In tribal, hilly and desert areas the norm could be relaxed to **one ASHA per ______________.**

Responsibilities of ASHA

- Health **education** and awareness
- **Reproductive and child health** and family planning
- Facilitate the access to health service
- Promotion of __________ and household toilet construction
- Provision of primary medical care for **minor ailments**
- Provider of _________
- ______________ of essentials like ORS, iron and folic acid tablets, chloroquine, condoms, disposable delivery kits
- Dispersal of **health information** on her community to higher centers
- Inform about the ______________ in her village
- Develop a **comprehensive** _________________

Anganwadi worker

- Under the ICDS Scheme, there is an Anganwadi worker for a **population of** __________
- The Anganwadi worker is **selected from the** _________ she is **expected to serve.**
- She **undergoes training** in various aspects of health, nutrition, and child development for ____ **months.**

- She is a part-time worker and is paid an **honorarium** of **Rs. _______ per month** for the services rendered.

- Include health check-up including maintenance of growth chart, immunization, supplementary nutrition, health education, non-formal pre-school education and referral services.

- The **beneficiaries** are nursing mothers, pregnant women, other women (15-45 years), children below the age of 6 years and adolescent girls.
-

Indian Public Health Standards for Health and Wellness Centre- Sub Health Centre

The population norms are as follows:

- **HWC-SHC (rural):** In rural areas, one Sub Health Centre is established for every 5000 population in plain areas and 3000 population in hilly/tribal/desert areas.

- **UHWC (urban):** In urban areas one Urban-HWC per 15,000-20,000 population caters predominantly to poor and vulnerable populations, residing in slums or other such pockets.

Human Resource required	Required Numbers
Community Health Officer (CHO)	
Multipurpose Health Worker	

Population norms for HWC-PHC

Type of PHC facility	Plain (population)	Hilly and Tribal areas (population)
Rural PHC	30,000	20,000
Urban PHC	50,000	-
Polyclinic	2.5 lakh-3 lakh	-

Staffing Pattern at PHC		
Staff	**Essential**	**Desirable**
Medical Officer - MBBS	1	1
Medical Officer - AYUSH	-	1
Medical Officer - Dental	-	1
Specialist for medicine, obgy, pedia, ophthal, derma, psychi on rotation basis	-	-
Staff nurses	2	1
Pharmacist	1	-
Medical laboratory technologist/ Lab technician	1	-
Optometrist/ Ophthalmic assistant/ vision technician	-	1
Health worker (Female)/ ANM	1	-
Health worker/ health assistant (Male)	1	-
Health assistant (Female)/ Lady health visitor	1	-
Health educator/ Counsellor	-	1
Dental assistant	-	1

Cold chain & vaccine logistic assistant	-	1
Physiotherapist	-	-
Public health manager	-	-
Dresser	1	-
LDC-1/ Accountant	1	-
Data entry operator	1	-
Sanitation staff	1	-

Health Worker Female (ANM)

- Make **post-natal** home **visits** on ________________ **day** for deliveries at **home** and sub-centre and on ____________ **day** for **institutional delivery.**

- In case of low birth weight baby, a total of ____ post-natal visits are to be made on _________________ **day** to screen for congenital abnormalities, assess the neonate for danger signs of sickness etc

Responsibilities of Health Workers	
Female Health Worker	**Male Health Worker**

• Maternal and child health services • Family planning services and registers • Medical Termination of Pregnancy • Universal Immunization Program • Detection of non-communicable diseases • Dai training • Treatment of minor ailments • Participate in team activities	• National Health Programmes • Malaria slides and surveillance • Leprosy follow-up • Identification and follow-up cases of blindness • Reproductive and child health services • RNTCP • Expanded Immunization Program • Environmental sanitation including chlorination of water and proper waste disposal
Common Responsibilities	

- Nutrition
- Distribution of contraceptives
- Work on communicable diseases like DOTS activities
- Record and report vital events
- Record keeping and maintaining registers

Rashtriya Swasthya Bima Yojana (RSBY)

- It was a ___________ **sponsored scheme** to provide health insurance cover of Rupees ________**/- per family per annum** on family floater basis (maximum 5 members per family).

- To **Below Poverty Line (BPL) families** and **11 other categories of Unorganized Workers** i.e MGNREGA Workers, Construction Workers, Domestic workers, Sanitation Workers, Mine Workers, licensed

Railway Porters, Street Vendors, Beedi Workers, Rickshaw Pullers, Rag Pickers, and Auto/Taxi drivers.

- The Scheme was launched by **Ministry of Labour and employment in 2008** and transferred to _______________ ________ on "as is where is" basis with effect from 01.04.2015.
- Each family enrolled in the scheme was entitled to hospitalization benefits of up to INR 30,000 per annum in ____________ **as well as empanelled ________ hospitals.**
- ______ **treatment packages** were covered under RSBY.
- With the launch of **PMJAY** on 23.09.2018, RSBY was _________ in it.
-

Ayushman Bharat- Pradhan Mantri Jan Arogya Yojana (AB-PMJAY)

- It is the flagship scheme of Government of India that provides a health cover of up to **Rs. ________ per family per year**, for secondary and tertiary hospitalizations.
- Launched on _____________________
- Cost bared by both _______________ **government.**

Benefits

- Health cover of up to Rs. 5,00,000 per family per year, for secondary and tertiary care hospitalization
- ___________ access to hospitalization services

- _________ on family size, age or gender
- **_____ pre-existing conditions are covered**
- Benefits are _______ **across the country** in all empanelled hospitals
- **1685 procedures** covering treatment, food, drugs and supplies, and diagnostics services
- PM-JAY covers cost of hospitalization, treatment, up to ______ **of pre-hospitalization** and _______ **of post hospitalization** follow-up care.

NOTES

Chapter-11: Nutrition and health

Energy Intake

Contributors to the total energy intake, in terms of energy provided, in an average Indian diet:

- **Protein**:
- **Fats:**
- **Carbohydrate:**

Proteins

Essential amino acids

- Out of 20, _____ are called **essential because the body cannot** synthesize them in amounts corresponding to its needs, and therefore, they **must be obtained from _______ proteins.**
- **Both essential and non-essential amino acids are needed for synthesis of ______ proteins,** the former must be supplemented through diet, whereas the latter can be synthesized by the body provided other building blocks are present.
- A protein is said to be **biologically complete** if it contains ____ **the EAA** in amounts corresponding to human needs.
- **When one or more of the EAA are lacking** the protein is said to be **biologically _____________.**

- **Animal proteins are rated __________ to vegetable proteins** because they are biologically complete.

Essential amino acids	Non-essential amino acids
PVT TIM HiLL	**Almost All Girls Go Prom Shopping**
• **P** = • **V** = • **T** = • **T** = • **I** = • **M** = • **Hi**= • **L** = • **L** =	• **Almost** = • **All** = • **Girls** = • **Go** = • **Prom** = • **Shopping** =

Amino acid	Biological Function
Tryptophan	
Methionine	
Cystine and Tyrosine	

Limiting amino acid:

- __________ **and** _________ in most of **cereal proteins**
- ___________ in most of **pulse proteins**

Functions of Proteins

- Body building
- Repair and maintenance of body tissues
- Maintenance of
- Synthesis of certain substances like ________, plasma proteins, __________, ________, __________ and coagulation factors.
- Proteins are connected with the ________ **mechanism of the body.**
- The _____________ immune response and the **bactericidal activity of leucocyte** become lowered in severe forms of protein energy malnutrition.
- Proteins **can also supply energy (__kcal per one gram)** when calorie intake is inadequate**, but this is not their primary function.**

Sources of Protein	
Animal Sources	**Vegetable Sources**
Milk, meat, eggs, cheese, fish and fowl	Found in pulses (legumes), cereals, beans, nuts, oil-seed cakes, etc
Egg proteins are considered to be the best because of their _____ biological value and	In India, cereals and pulses are the main sources of dietary protein because they are

digestibility and are used in nutrition studies as a "__________ protein".	cheap, easily available and consumed in bulk.

FATS

Fatty acids

- **Saturated Fatty Acids:** Such as _______, ________ and ________ acids, and mostly found in animal fats.

- **Unsaturated Fatty Acids:** Divided into monounsaturated (MUFA) (e.g., _____ acid) and polyunsaturated fatty acids (PUFA) (e.g., ________ acid and ___________ **acid**), mostly found in vegetable oils.

- **Exceptions:** Coconut and palm oils, although vegetable oils, have an extremely high percentage of saturated fatty acids. On the other hand, fish oils, although they are not vegetable oils, contain poly and mono-unsaturated fatty acids.

Essential Fatty acids

- Those that _______ be synthesized by humans, derived only from food.

- Most important essential fatty acid is _____________

- Other essential fatty acids are __________ acid, _________ acid and ___________________ acid.

Trans-fatty acids

- Trans-fatty acids are geometrical isomers of _________________ fatty acids.

- **Partial hydrogenation,** the process used to increase shelf-life of PUFAs, creates trans fatty acids and also **removes the critical __________ bonds in essential fatty acids.**

- Trans-fatty acids render the plasma lipid profile ___________ atherogenic than saturated fatty acids, by not only ____________ LDL cholesterol but also by ____________ HDL cholesterol.

- Intake of trans-fatty acids _________ **the risk of coronary heart disease.**

- It takes _______ for trans fatty acids to be flushed from the body.

Phrenoderma:

Deficiency of ________________ in the diet is associated with ________________ **skin,** a condition known as **Phrenoderma or ______ skin.**

Carbohydrates

- Main source of energy provide ___ **Kcals per gram.**
- Essential for **oxidation of fats** and **synthesis** of certain **non-essential amino acids.**
- Three main source of carbohydrates are starch, sugar and cellulose.

Dietary Fibre

- Dietary fibre is known to be associated with _______ **incidence of coronary heart disease.**
- Dietary Fibre have ____ **metabolic effects**

Vitamins

Vitamin A Deficiency

Ocular Manifestations

- ___________________
- _________________**:** This is the first clinical sign of vitamin A deficiency. The conjunctiva appears muddy and wrinkled. Described as "emerging like sand banks at receding tide" when the child ceases to cry.
- _____________**:** These are triangular, pearly-white or yellowish, foamy spots on the bulbar conjunctiva on either side of the cornea. They are frequently bilateral.
- ______________**:** The cornea appears dull, dry and non-wettable and eventually opaque.

- **__________________:** Keratomalacia or liquefaction of the cornea is a grave medical emergency. The cornea (a part or the whole) may become soft and may burst open.

Extra-Ocular Manifestations

- Follicular hyperkeratosis
- Anorexia
- Growth retardation.

Treatment

- Early stages of xerophthalmia can be reversed by administration of massive dose (__________ IU or 110 mg of retinol palmitate) orally on ______ successive days.

Prevention

- **Single** massive **dose** of __________ **IU** of vitamin A in oil (retinol palmitate) orally **every ___ months to preschool children (___ year to ___ years)**.
- Half that dose **(___________ IU)** to children between **___ months and ____ year of age.**

Prevalence criteria for determining xerophthalmia	
Criteria	**Prevalence in population at risk (6 months to 6 years)**
Night blindness	

Bitot spots	
Corneal xerosis/ corneal ulceration/ keratomalacia	
Corneal scar	
Serum retinol (<10 mcg/ dl)	

Vitamin D Deficiency

1) Rickets:

- Observed in **young children** between the age of ____ **months and ______ years.**
- There is _________ **calcification** of growing bones.
- The disease is characterized by growth failure, bone deformity, muscular hypotonia, tetany and convulsions due to hypocalcaemia.
- There is an __________ **concentration of alkaline phosphate** in the serum.
- The bony deformities include curved legs, deformed pelvis, _________ chest, ___________ **sulcus, __________ rosary,** kyphoscoliosis, etc.
- The milestones of development such as walking and teething are delayed.

2) Osteomalacia:

In adults, occurs mainly in women, especially during pregnancy and lactation when requirements of vitamin D are increased.

Vitamin K

- It occurs in atleast two major forms - vitamin K1 and K2.
- Vitamin K1, is abundant in ____________ particularly ____ ______ **ones**, while **vitamin K2** is synthesized by __________ __________.
- **Cow's milk is a ________ source** (60 mcg/L) of vitamin K than human milk (15 mcg/L).
- **Long-term administration of antibiotic** doses for more than a week suppress the normal intestinal flora, and cause **deficiency** of vitamin K.
- Vitamin K stimulates the **production and/ or the release of certain ____________ factors**, which are essential for blood clotting.
- In vitamin K deficiency, the **prothrombin** content of blood is markedly ___________ and the blood **clotting time** is considerably _____________.
- The daily requirement **is _______ mg/kg for the adult.**
- Newborn infants is deficient in vitamin K due to minimal stores of prothrombin at birth and lack of an established intestinal flora.
- Soon after birth, all infants or those at increased risk should receive a **single intramuscular dose of a vitamin K preparation** (0.1-0.2 mg of menadione sodium bisulphite or 0.5 mg of vitamin K,) by way of prophylaxis.

B Group of Vitamins

Vitamin	Deficiency state
Thiamine (B_1)	
Riboflavin (B_2)	
Niacin (B_3)	
Pantothenic acid (B_5)	
Folic acid (B_9)	
Cobalamin (B_{12})	

Thiamine (B1) Deficiency

1) Beriberi:

Occurs in three main forms:

- **Dry-** Characterized by
- **Wet-**
- **Infantile-**

2) Wernick's encephalopathy:

- Often seen in ________________
- Characterized by _____________, __________, _______ and _________ **deterioration.**

- Occurs occasionally in **people who _____.**
-

Minerals

Magnesium

- It is a constituent of ______, and is present in all body cells.
- Human adult body contains about ___ g of magnesium of which about half is found in the skeleton.
- It is essential for normal metabolism of ______ and _________.
- Magnesium deficiency may occur in chronic alcoholics, cirrhosis of liver, toxaemias of pregnancy, protein-energy malnutrition and malabsorption syndrome.
- Clinical features are irritability, tetany, hyper-reflexia and occasionally hypo-reflexia.
- Requirements are estimated to be about _____ mg/day for adults.

IRON

Iron losses

- The total daily iron loss of an adult is probably ____ mg.
- Iron loss is _____ mg per 28 days cycle in menstruating women.

Stages of Iron Deficiency

First stage

- **___________ storage of iron without any other detectable abnormalities.**

Intermediate stage ("latent iron deficiency")

- Iron stores are exhausted, but anaemia has not occurred yet.
- Recognition depends upon the measurement of serum ferritin levels.
- The **percentage saturation of transferrin falls** from a normal value of 30 % to _______ %.
- This stage is the most ______________ stage in India.

Third stage

- **________ iron deficiency** when there is a **_______ in the concentration of circulating haemoglobin** due to **________ haemoglobin synthesis.**
- End result is nutritional anaemia which is not a disease entity.
- Besides anaemia, there may be other functional disturbances such as impaired cell-mediated immunity, reduced resistance to infection, increased morbidity

and mortality and diminished work performance.

Evaluation Of Iron Status

Based on the following parameters:

(a) Haemoglobin concentration:

- Values below those given Haemoglobin concentration is a relatively **insensitive index** of nutrient depletion.
- This is because anaemia is a late manifestation of iron deficiency which can frequently occur without the manifestation of anaemia.

Cut-off points for the diagnos	
Group	**Hemoglobin (g/dl) (Venous blood)**
Adult males	13
Adult females, non-pregnant	12
Adult females, pregnant	11
Children (6 months to 6 years)	11
Children (6 to 14 years)	12

(b) Serum iron concentration:

- More useful index than haemoglobin concentration.
- Normal range is __________ mg/L
- Values ________ mg/L indicate iron deficiency.

(c) Serum ferritin:

- Single most ________ tool for evaluating the iron status.
- Most useful indicator of iron status in a population where the prevalence of iron deficiency is not high.
- Values ________ mcg/L indicate an absence of stored iron.

(d) Serum transferrin saturation:

- Should be above _____ %.
- Normal value is ____ %.

Iodine

- Adult **human body** contains about **50 mg of** iodine, and **blood level** is about **8-12 micrograms/dl.**
- Best source of iodine are __________
- Daily requirement of iodine for adults is ______ **micrograms**

Spectrum of iodine-deficiency disorders	
• Goitre • Hypothyroidism • Strabismus (squint) • Neuromuscular weakness • Endemic cretinism	• Subnormal intelligence • Mental deficiency • Intrauterine death (spontaneous abortion, miscarriage)

Epidemiological Assessment of iodine deficiency:

Urinary Iodine Excretion:

- Reflects __________ iodine intake
- Used for __________ control surveillance

Neonatal Hypothyroidism

- Most sensitive indicator of ____________ iodine deficiency

Serum Thyroxine (T4)

- More sensitive indicator of ______________ than T3

Fluorine

Endemic fluorosis

- Occurs when drinking water contains **excessive amounts** of **fluorine (_______ mg/L).**

Toxic manifestation of fluorosis:

(a) Dental fluorosis :

- Characterized by "_________" **of dental enamel**, at levels ____ **mg/L** intake.
- Teeth **lose their shiny appearance** and **chalk-white patches** develop on them, which is the early sign of dental fluorosis.
- Mottling is best seen on the ________ **of the _______ jaw**, almost entirely confined to permanent teeth.

(b) Skeletal fluorosis:

- Associated with lifetime **daily intake of _________ mg/L or more.**
- Concentration of >10 mg/L, __________ **fluorosis** can ensue which leads to permanent disability.

(c) Genu valgum:

- Syndrome was observed among people whose **staple was** _______________

Intervention:

(a) Changing the water source:

- Use water source with ________ **fluoride content (_________ mg/L).**
- Running surface water contains ________ quantities of fluorides than ground water sources (wells).

(b) Chemical treatment:

- The National Environmental Engineering Research Institute, Nagpur developed a technique for **removing fluoride** by chemical treatment. It is called __________ technique for defluoridation of water.
- It involves the addition of two chemicals **(viz. _____ and ____)** in sequence followed by ___________, ____________ and ________.

Zinc

- It is a component of _______ **enzymes in body.**
- It plays a role in **carbohydrate** and **protein** metabolism, ______ **synthesis** and ________ function.
- Average adult body **contains** ________ **g** of zinc.
- **Deficiency symptoms include:** Growth failure, sexual infantilism in adolescents,

loss of taste and delayed wound healing and low circulating zinc levels in clinical disorders such as liver disease, pernicious anaemia, thalassaemia and myocardial infarction.

- Maternal zinc deficiency has been associated with spontaneous abortion, congenital malformations like anencephaly, low birth weight, intrauterine growth retardation and preterm delivery.

- Zinc plays an important role as __________ agent.

- Suggested **daily intake for adults** is ___ mg per day for men, ___ mg per day for women, __ mg per day for children and __ mg for infants.

Milk

Nutritive value of milks compared (Value per 100 gm)				
	Buffalo	**Cow**	**Goat**	**Human**
Fat (g)	6.5	4.1	4.5	3.4
Protein (g)	4.3	3.2	3.3	1.1
Lactose (g)	5.1	4.4	4.6	7.4
Calcium (mg)	210	120	170	28
Iron (mg)	0.2	0.2	0.3	0.4-0.5

Vitamin C (mg)	1	2	1	3
Minerals (g)	0.8	0.8	0.8	0.1
Water (g)	81.0	87	86.8	88
Energy (kcal)	117	67	72	65

Reference Indian adult man and woman

"Reference man"

- Is aged between ________ **years**
- Weighs ______ **kg** with a height of ___ **metre** and BMI of ______
- Is _____ **from disease** and **physically** _____ for active work.
- On each working day, he is engaged in ___ **hours** of occupation which usually involves _________ **activity**
- While when not at work he spends 8 hours in bed, 4-6 hours in sitting and moving about, 2 hours in walking and in active recreation or household duties.

"Reference Woman"

- Is aged between __________ **years**
- _____________ **and** _______________ (NPNL)
- Weighs _____ **kg** with a height of _____ **metre** and BMI of _____
- _______ **from disease** and **physically** ___ for active work.

- On each working day she is engaged in ___ **hours** of occupation, which usually involves ___________ activity

- When not at work she spends 8 hours in bed, 4-6 hours in sitting and moving about, 2 hours in walking and in active recreation or household duties.

Energy requirement (kcal/d) as proposed by ICMR			
Age group	**Category**	**ICMR 2020**	**ICMR 2010**
Adult Men	Sedentary work		2320
	Moderate work		2730
	Heavy work		3490
	Sedentary work		1900
	Moderate work		2230
Adult Women	Heavy work		2850
	Pregnant		+350
	Lactating (0-6 m)		+600
	Lactating (7-12 m)		+520
Infants	0-6 months	550	550
	6-12 months	670	670
	1-3 y	1010	1060
Children	4-6 y	1360	1350
	7-9 y	1700	1690

Source of energy

- **Protein**:
- **Fat**:
- **Carbohydrate**:

- **Dietary fibre:**

Food Toxicants			
Food	**Adulterant**	**Toxin**	**Disease**
Dal	Khesari dal (Lathyrus sativus)	BOAA	
Oil	Argemone Mexicana (oil)	Sanguinarine	
Millet	Crotalaria seeds (Jhunjhunia)	Pyrrolizidine alkaloids (Hepatotoxic)	
Bajra, wheat	Claviceps purpurea	Clavine alkaloids	

Nutrition programmes in india	
Programme	**Ministry**
▪ Vitamin A prophylaxis programme ▪ Prophylaxis against nutritional anaemia ▪ Iodine deficiency disorder control programme	
▪ Special nutrition programme ▪ Balwadi nutrition programme ▪ ICDS programme	

Mid-day meal programme	
Mid-day meal scheme	

Mid-day meal programme

- Also known as _______________ **Programme.**
- **Objective:** Attract more children for __________ to schools and _______ them so that __________ can be improved.

Broad principles:

- Meal should be __________ **not a substitute** to home diet.
- Meal should supply atleast _______ of total **energy** requirement and _____ **of protein** need.
- **Cost** of meal should be reasonably ________
- Should be prepared _______ **in schools;** no complicated cooking process should be involved.
- _______________ **foods** should be used as it will reduce cost of meal
- **Menu** should be ________________ to avoid monotony.

Mid-day meal Scheme

- Is now known as __________ **Scheme** is also known as **National Programme of**

Nutritional Support to Primary Education.

- It was launched as a __________ **sponsored scheme** on 15th August 1995 and revised in _______.
- Its **objective** being universalization of primary education by increasing enrolment, retention and attendance and simultaneously impacting on nutrition of students in primary classes.
- The programme originally covered children of primary stage (**classes _________** in government, local body and government aided schools.
- It was extended in Oct. 2002, to cover children studying in **Education __________ Scheme** and Alternative and **_____________ Education Centres** also.
- The central assistance provided to states under the programme is by way of **free supply of ________** from nearest Food Corporation of India godown at the rate of 100 g. per student per day and subsidy for transport of food grain.
- To achieve the objective, a cooked mid-day meal with minimum **_____ Calories** and **_________ grammes of protein** content will be provided to all the children in class _________

NOTES

Chapter-12: Medicine and social sciences

Acculturation

- Acculturation means ______________
- When there is contact between two people with different types of culture, there is diffusion of culture both ways.

Standard of living

The standard of living in a country depends upon:

Errors of perception

- **Imperception:**
- **Illusion**:
- **Hallucination**:

Attitudes

An attitude includes three components:

Types of Learning

There are 3 types of learning:

Defence mechanisms

When an individual is faced with problems, difficulties or failures, he employs certain ways or devices to achieve health, happiness or success. These are called defence mechanisms.

Psychologists have identified a number of such defence mechanisms, which "include the following :

1. _____________: Instead of accepting failure and correcting himself, the individual tries to make excuses and justifies his behaviour. It is like the proverbial fox declaring that the grapes were sour, when it could not reach them. This is called rationalization. It is a face-saving device.

2. ____________: Sometimes the individual blames others for his mistakes or failures. It is just like the student saying that he could not score good marks in the examination because, his teacher did not like him.

3. ____________: The familiar example is that the student who is not good in his studies may distinguish himself in sports or dramatics, music or other activities.

4. ______________: Some students pretend illness and do not appear for examinations. This is an escape phenomenon.

5. _______________: An office clerk badly snubbed by his superior takes it out on his wife and children on reaching home. This is like a rebound phenomenon. It is trying to escape from one situation and fixing blame on another situation.

6. ____________: Some people resort to childhood practices (e.g., weeping when something goes wrong) as a mode of adjustment.

IQ = Intelligence Quotient

Levels of Intelligence	IQ Range
Idiot	0-24
Imbecile	25-49
Moron	50-69
Borderline	70-79
Low normal	80-89
Normal	90-109
Superior	110-119
Very superior	120-139
Near genius	140 and over

Family Life cycle

Basic Model of a Nuclear Family Life Cycle		
Phases of family life cycle	Events characterizing	
No. Description	Beginning of phase	End of phase
I.		
II.		
III.		
IV.		
V.		
VI.		

Socio-economic status scale

- **Kuppuswamy** in India prepared a scale **based on __________, ___________ and** _________ which are the three major variables contributing to socio-economic status in _______ **areas.**
- ________ **and** _______________ scale for use in **rural and urban areas.**

Wealth Index

- ______________________________ use an index of the economic status of households called the **wealth index.**
- It is an indicator of the level of wealth that is consistent with expenditure and income measures.
- The economic index was constructed using the household asset data and housing characteristics
- The sample is then divided into _________ i.e..________ groups with an equal number of individuals in each.

Consumer Protection Act

COPRA is a piece of comprehensive legislation and recognizes six rights of the consumer, namely;

- Right to ________
- Right to be _________
- Right to ________
- Right to be _______
- Right to _____________
- Right to ________________

The **monetary limits** of the compensation that can be granted by the consumer courts are as follows:

- **District consumer** – upto Rs _____________
- **State Commission-** Rs ______________
- **National commission-** above Rs ___________

Global Hunger Index (GHI)

- Designed to comprehensively measure and track hunger globally, by region and country.
- It highlights successes and failures in hunger reduction.
- It is calculated each year by the International Food Policy Research Institute. GHI for the year 2014 is ________
- **GHI combines three equally weighted indicators into one index:**

Formula:

GHI = Proportion of _____________ population (PNU) + Children _____________ (CUW) + ______________ in per centage (CM)

The calculations result in a 100-point scale on which zero is the _____ score (no hunger) and 100 the _______, although neither of these extremes is reached in practice.

Social Security

It is defined as security that society furnishes through appropriate organization, against certain risks to which its members are exposed.

Risks covered are:

NOTES

Chapter 13: Tribal health in India

Health care infrastructure:

As per the present norms, tribal and hilly areas should have:

- One Health Sub-centre (HSC) per ______ population
- One Primary Health Centre (PHC) per _________ population
- Community Health Centre-(CHC) per _________ population

[“This chapter has never been asked before and doesn’t hold exam-relevant content. Don’t waste time here — you can skip it and focus on high-yield areas.”]

Chapter 14: Sustainable Development Goals

Specific plan-main targets 2030	
Ending epidemic of AIDS	Reduce the annual number newly infected with HIV by ____% and the annual number of people dying from AIDS-related causes by ____% (compared with 2010)
Ending epidemic of TB	____% reduction in TB deaths, ____% reduction in TB incidence rate (to less than 20 per 100,000 population), _____ TB-affected families facing catastrophic costs due to TB
Ending epidemic of malaria	____% reduction in global malaria mortality rate, ___% reduction in global malaria case incidence, Malaria eliminated from at least ___ countries, Re-establishment of malaria prevented in all countries identified as malaria-_____
Ending epidemic of NTDs	____% reduction in the number of people requiring interventions against NTDs

Control hepatitis	____% decline in new cases of chronic HBV infection between 2010 and 2030; ___% reduction in new cases of chronic HCV infection over the same period; ___% reduction in HBV- and HCV-related deaths.
Combat Water-borne diseases	No one practices open ________ (by 2025), Everyone uses a basic drinking-water supply and handwashing facilities at home, Everyone uses adequate ________ when at home (by 2040), All drinking-water supply, sanitation and hygiene services are delivered in progressively affordable, accountable and financially and environmentally sustainable manner

Type of indicator	SDG target	Proposed indicator
Impact	3.4	NCD mortality
	3.4	Suicide mortality
	3.9	Mortality due to air pollution (household and ambient)
Coverage/ risk factors	3.8	UHC: NCDs tracers (hypertension treatment coverage diabetes treatment coverage, cervical cancer screening; tobacco use)
	3.a	Tobacco use
	3.5	Substance abuse (harmful use of alcohol)
Risk factors/ determinants	7.1	Clean household energy
	11.6	Ambient air pollution
	Other	Part of targets in goals on poverty, education, cities, etc.

NOTES

Chapter-15: Environment and Health

Water

- The basic physiological requirements for drinking water **is __ litres per head per day**. This is just for survival.

- Daily supply of _______ **litres** per capita is considered as an adequate supply to meet needs for **urban** domestic purposes while __ **litres** of water supply per capita per day in **rural** areas.

Sources of water supply:

- ______ is the prime source of all water and it is purest in nature.

- __________ is the main source of water supply in many areas. It is never safe for human consumption unless subjected to sanitary protection and purification before use.

- ___________ is the cheapest and most practical means of providing water to small communities.

Advantages of ground water:

- Likely to be free from pathogenic agents
- Usually requires no treatment
- Supply is likely to be certain even during dry season
- Less subject to contamination than surface water

Disadvantages of ground water:

- It is high in mineral content

- Requires pumping or some arrangement to lift the water

Feature	Shallow Well	Deep Well
Definition		
Chemical Quality		
Bacteriological Quality		
Yield		

Improvement of dug wells :

The unlined katcha wells may be made sanitary by:

-
-
- And filling the well with coarse sand __________ water level, and with clay ___________ level.

Masonry well improvement consists of making the:

- ___________________ of the lining water-tight
- Raising the lining ________ above the ground
- Providing a reinforced _________________ at the top.

Water related Disease	
Chemical	
Dental health	• Fluoride _____ mg/litre in drinking water protect against dental caries • High levels of fluoride cause _________ of the dental enamel
Cyanosis in infant	• High nitrate content of water is associated with __________________
Cardiovascular diseases	Hardness of water appears to have a __________ effect against cardiovascular diseases

Diseases transmitted because of inadequate use of water: ____ ________________
Diseases related to disease carrying insects breeding in or near water: Malaria, Filaria, Arboviruses, Onchocerciasis and African trypanosomiasis

Slow sand or Biological filters

- The most important part of the filter is the __________
- Surface of the sand bed gets covered with a slimy growth known as " ____________", ______ **layer,** _______ **layer or** ___________ **layer.**

- This layer is slimy and gelatinous and consists of ______________ and numerous forms of life including **plankton, diatoms and bacteria.**

- This formation of vital layer is known as ________ of the filter.

- The ___________ is the heart of the slow sand filter.

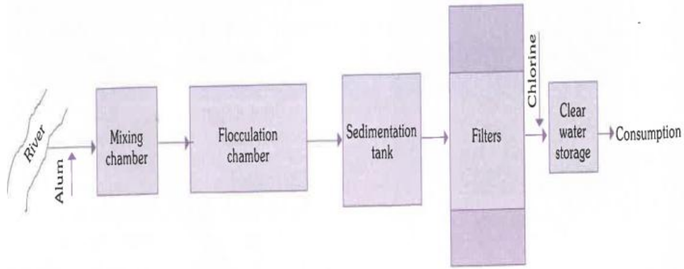


FIG. 6
Flow diagram of a rapid sand filtration plant

Rapid sand or Mechanical filters

Rapid sand filters are of two types, the gravity type (__________ **filter**) and the pressure types (________ **filter**).

	Rapid sand filter	Slow sand filter
Space		
Rate of filtration	200 m.g.a.d	2-3 m.g.a.d
Effective size of sand	0.4-0.7 mm	0.2-0.3 mm

Preliminary treatment		
Washing		
Operation		
Loss of head allowed	6-8 feet (2-2.5 m)	4 feet (1.5 m)
Removal of turbidity		
Removal of colour		
Removal of bacteria	98-99 per cent	99.9-99.99 per cent

Chlorination

- The **disinfecting** action of chlorine is **mainly due** to the ____________ and to a small extent due to the hypochlorite ions.
- The __________________ is the most effective form of chlorine for water disinfection.
- It is **more effective than the _______________ ion.**
- Chlorine acts best as a disinfectant when the **pH of water is around** ___ because of the predominance of hypochlorous acid.
- The point at which the chlorine demand of the water is met is called the _______________

- The presence of free residual chlorine for a **contact period of atleast ________** is essential to kill bacteria and viruses.

- The minimum recommended concentration of free chlorine is _____ mg/L for _________

Orthotolidine (OT) test

- **Purpose:** Used to determine ____________________ in water with speed and accuracy.
- **Reagent:** It is dissolved in 10% hydrochloric acid to create the reagent.
- **Reaction:** When reagent is added to chlorine containing water, the solution turns ________. Intensity of _________ colour corresponds to concentration of chlorine.

Orthotolidine- Arsenite (OTA) test

- Determine the **free and combined chlorine residuals _____________**
- Errors caused by the presence of interfering substances such as nitrites, iron and manganese all of which produce a yellow colour with Orthotolidine, are overcome by the OTA test.

Disinfection of wells

- The most effective and cheapest method of disinfecting wells is by _______________
- Estimate the chlorine demand of the well water by __________ apparatus and

calculate the amount of bleaching powder required to disinfect the well.

Water quality – Criteria and Standards

Constituents	Levels likely to give rise to consumer complaints
Colour	____ TCU
Turbidity	__ NTU
Chloride	____ mg/L

Bacteriological indicators

Coliform organisms:

The coliform group includes both faecal an non-faecal organisms. Typical example of the faecal group is ________ and of the non-faecal group, ________________

There are several reasons why coliform organisms are chosen as indicators of faecal pollution rather than the water-borne pathogens directly :

- The coliform organisms are __________ present in ______ abundance in the human intestine. It is estimated that an average person excretes 200-400 billion of these organisms per day. These organisms are foreign to potable waters, and hence their presence in water is looked upon as evidence of faecal contamination;

- They are ___________________________ - as small as one bacteria in 100 ml of water, whereas the methods for detecting the pathogenic organisms are complicated and time-consuming;

- They survive _________ than the pathogens, which tend to die out more rapidly than coliform bacilli; and

- The coliform bacilli have __________ resistance to the forces of natural purification than the water borne pathogens. If the coliform organisms are present in a water sample, the assumption is the probable presence of intestinal pathogens.

Faecal streptococci:

Presence of faecal streptococci in water is regarded as important confirmatory evidence of _________ faecal pollution of water.

Clostridium Perfringens:

The presence of spores of C. perfringens in a natural water suggests that faecal contamination has occurred, and their presence , in the absence of the coliform group suggests that faecal contamination occurred at some _____________

Surveillance of drinking water quality

Presumptive Coliform Test

1) Multiple tube method

Estimates the __________________ **(MPN)** of coliform organisms in 100 ml of water.

Procedure:

- Inoculate measured quantities of the sample water (0.1, 1.0, 10, 50 ml) into tubes of McConkey's Lactose Bile Salt Broth with _________________ as an indicator.

- The tubes are incubated for ___ hours.

- Presence of acid and gas, indicates a positive reaction and is known as "presumptive coliform count".

Confirmatory tests

Confirms the presence of coliform organisms, especially crucial for chlorinated water.

Procedure:

- Subculturing each presumptive positive tube in 2 tubes of brilliant green bile broth.

- Further confirmation of the presence of E. coli, if desired, can be obtained by testing for indol production at 44°C.

2) Membrane filtration technique

Used as a standard procedure to test for the presence of coliform organisms.

Procedure:

- Filter a measured volume of the sample through a ___________ membrane.

- Bacteria present in water are retained on membrane surface.

- Inoculate the membrane face upwards on suitable media

- Count colonies to obtain results within __________, compared to __________ hours required for multiple tube technique.

Hardness of water

Temporary hardness	Permanent hardness
Due to presence of	Due to presence of

- Drinking water should be _____________ (_____ mEq./litre or ________ mg/L).

- Softening of water is recommended when the hardness exceeds ____ **mEq/l (_____ mg per litre).**

- Several countries reports have shown an __________ statistical association between the hardness of drinking water and the death rate from cardiovascular diseases.

Removal of hardness	
Temporary hardness	**Permanent hardness**
• • Addition of • Addition of •	• Addition of •

Swimming Pool Sanitation

Chlorination with a continuous maintenance of _____ mg/litre (___ ppm) of free chlorine residual provides adequate protection against bacterial and viral agents.

Horrock's Apparatus

- It is designed to find out the _______ **bleaching powder required** for disinfection of water.
- Development of ______ colour indicates the presence of free residual chlorine.
- Note the first cup which shows distinct _____ colour. Supposing the 3rd cup shows blue colour, then 3 level spoonfuls or 6 grams of bleaching powder would be required to disinfect 455 litres of water.

Indices of thermal comfort

- **Kata thermometer** is used to measure the __________. A dry kata reading of ___ and above, and a wet kata reading of ___ and above, were regarded as indices of thermal comfort.
- **Effective temperature:** It is an arbitrary index which combines into a single value the effect of __________, ___________ and movement of the __________ on the sensation of warmth or cold felt by the human body.
- **Corrected effective temperature:**

- This index is an ___________ over the effective temperature index.
- Instead of dry bulb temperature, the reading of _______ ____________ is used to allow for radiant heat.
- That is C.E.T scales deal with all the four factors namely, ____ _________, ________, _________ and ________________

Note: The best indicators of air pollution are _____________, ______ and _______________

Mechanical Ventilation	
	In this system, air is extracted or exhausted to the outside by exhaust fans usually driven by electricity.
	In this system fresh air is blown into the room by centrifugal fans so as to create a positive pressure, and displace the vitiated air.
	Combination of exhaust and plenum systems of ventilation.
	Defined as "the simultaneous control of all, or at least the first three of those factors affecting both the physical and chemical conditions of the atmosphere within any confined space or room. These factors include temperature, humidity. air movement, distribution, dust, bacteria, odours and toxic gases, most of which affect in greater or lesser degree the human health and comfort".

Light

- An illumination of ________ foot candles is accepted as a basic minimum for satisfactory vison.

Light measurement units			
Description	**Quantity measured**	**Recommended Unit**	**Other Units**
Brightness of point source	Luminous intensity		
Flow of light	Luminous flux		
Amount of light reaching surface	Illumination Illuminance		
Amount of light re-emitted by surface	Brightness Luminance		

Recommended illumination (the IES Code)	
Visual task	**Illumination (lux)**
Casual reading	
General office work	
Fine assembly	
Very severe tasks	
Watch making	

Noise

A daily exposure up to _____ dB is about the limit people can tolerate without substantial damage to their hearing.

- **Normal conversation produce noise of :** _______ dB
- **Whispering:** _______ dB
- **Heavy street traffic:** _______ dB
- **Boiler factories:** _____ dB

Exposure to noise **above _____ dB** may rupture the tympanic membrane and cause **permanent loss of hearing.**

Effects of atmospheric pressure on health

High altitudes: ______________ and ______________ _________________

Low altitudes: ____________ (Persons working in diving bells and compressed air chambers).

Effects of heat stress

	Characterized by very high body temperature which may rise to 110^0F (43.3^0C) and profound disturbances including delirium, convulsions and partial or complete loss of consciousness
	Temperature above 106^oF
	Caused primarily by the imbalance or inadequate replacement of water and salts lost in perspiration due to thermal stress
	Occur in persons who are doing heavy muscular work in high temperature and humidity.

	Condition results from pooling of blood in lower limbs due to dilatation of blood vessels, with the result that the amount of blood returning to the heart is reduced, which in turn responsible for lowering of blood pressure and lack of blood supply to the brain.

Effects of cold stress

- **Temperature above freezing (wet cold condition): __________ or ___________**
- **Temperature below freezing (dry cold condition): _________.**

Air velocity

- The air velocity is measured by an instrument called the _____________.

- The wind direction is observed by an instrument called the ____________

- For proper lighting and ventilation, there should be an open space all around the house- this is called ________

Note: Overcrowding is considered to exist if ___ **persons over ___ yrs** of age , not husband and wife, of _________ sexes are obliged to sleep in the _______ room.

Methods of Disposal

1. ____________

- Used for land reclamation or general waste disposal.

- Most insanitary method of disposal according to WHO.

2. ___________________________

- The **most satisfactory method** for waste disposal in suitable lands.

- **Methods**: ______ Method, ______ Method and _____ Method.

3. _____________

- It is the method of choice where suitable land is not available.

- Hospital refuse which is particularly dangerous is best disposed of by incineration.

4. ______________

- It is a method of combined disposal of refuse and nightsoil or sludge.
- **Methods:** Bangalore method (____________ method) and Mechanical composting (___________ method)

5. _______________

- Common in rural areas where organic waste is composted into manure for agricultural use.

6. ________

- Suitable method for small camps.

Septic Tank

- Treats household sewage by ___________ **digestion** and ______________.
- Suitable for individual houses or small groups without public sewerage access.

Design Features:

- Septic tanks are not recommended for ________ communities.
- There should be a minimum air space of ____________ between the level of liquid in the tank and the undersurface of the cover.

- The bottom is ___________ towards the inlet end.
- There is inlet and outlet pipe which are ____________.
- Retention period is of __________

Working of a septic tank:

- The solids settle down in the tank, to form "________", while the lighter solids including grease and fat rise to the surface to form "__________".
- The solids are attacked by the anaerobic bacteria and fungi and are broken down into simpler chemical compounds. This is the first stage of purification, called ____________ **digestion.**
- The liquid which passes out of the outlet pipe from time to time is called the __________.
- It _____________ numerous bacteria, cysts, helminthic ova and organic matter in solution or fine suspension.
- Second stage, _________ **oxidation** takes place outside the septic tank, in the sub-soil.

Sewage treatment

The strength of the sewage is expressed in terms of :

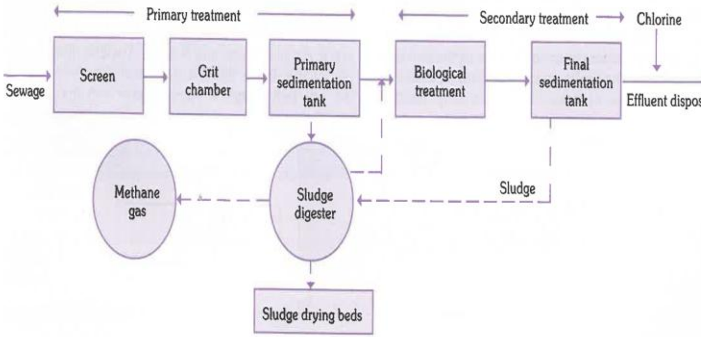


FIG. 11
Flow diagram of a modern sewage treatment plant

- The heart of the activated sludge process is the ____________
- The mixture is subjected to aeration in the aeration chamber for about ____________

Arthropod-borne diseases	
Arthropod	**Diseases transmitted**
Mosquito	Malaria, filaria, viral encephalitis (e.g., Japanese encephalitis), viral fevers (e.g., dengue, West Nile, viral haemorrhagic fevers (e.g., yellow fever, dengue haemorrhagic fever)
Housefly	Typhoid and paratyphoid fever, diarrhoea, dysentery, cholera, gastroenteritis, amoebiasis, helminthic infestations, poliomyelitis, conjunctivitis, trachoma, anthrax, yaws, etc.
Sandfly	
Tsetse fly	

Louse	Epidemic typhus, relapsing fever, trench fever, pediculosis
Rat flea	Bubonic plague, endemic typhus, chiggerosis, hymenolepis diminuta.
Blackfly	
Reduviid bug	
Hard tick	
Soft tick	
Trombiculid mite	
Itch-mite	

Cyclops	
Cockroaches	

Transmission of arthropod borne disease

Three types of transmission cycles are involved in the spread of arthropod-borne disease:

(1) Direct Contact: The arthropods are directly transferred from man to man through close contact, e.g., ______________________

(2) Mechanical Transmission: The disease agent is transmitted mechanically by the arthropod. The transmission of diarrhoea, dysentery, typhoid, food poisoning and trachoma by the ________.

(3) Biological Transmission: When the disease agent multiplies or undergoes some developmental change with or without multiplication in the arthropod host, it is called biological transmission.

This may be of three types :

a) ___________: When the disease agent undergoes no cyclical change, but multiplies in the body of the vector, transmission is said to be propagative, e.g., plague bacilli in rat fleas

b) _______________: The disease agent undergoes cyclical change, and multiplies in the body of the arthropod, e.g., malaria parasite in anopheline mosquito

c) ________________: When the disease agent undergoes cyclical change but does not multiply in the body of the arthropod, e.g., filarial parasite in culex mosquito and Guineaworm embryo in cyclops.

	Anopheles	Culex	Aedes	Mansonia
Eggs	Laid ______. ______-shaped with lateral floats.	Laid in small clusters or rafts Do not possess latera	Laid ______ – ______ shaped Do not possess lateral floats	Laid in _____ shaped clusters attached to undersurface of leaves of certain aquatic plants.

		l floats		
Larvae	Rest parallel to water surface No siphon tube Palmate hairs present on	• Suspended at an angle with head downward • Siphon tube present. • No palmate hairs		

	abdominal segments	
Pupae	Siphon tube is broad and short.	Siphon tube is long narrow
Adults	Inclined at an angle to surface during rest Wings spotted Palpi long in both sexes	• Hunched back appearance on rest • Wings unspotted • Palpi short in female.

- Aedes aegypti index is kept _______ **at all ports.**

- The period that elapses from the moment a blood meal is taken until the eggs are laid is called _____________ cycle; it is about 48 hrs in hot and humid tropical areas.

Aedes

- Aedes mosquitoes have white stripes on a black body known as _______ **mosquitoes.**

- The ________ are **fearless biters,** and they bite chiefly during the day.

- They do not fly over long distances usually less than 100 metres.

Type of Mosquito	Associated Diseases
Anopheles	
Culex	
Aedes	
Mansonoides	

Disease transmitted	
Hard tick	**Soft tick**
• Tick typhus (Rocky mountain spotted fever) • Viral encephalitis (e.g., Russian spring-summer encephalitis) • Viral fevers (e.g., Colorado tick fever) • Viral haemorrhagic fevers (e.g. KFD in India) • Tularaemia • Tick paralysis • Human babesiosis	

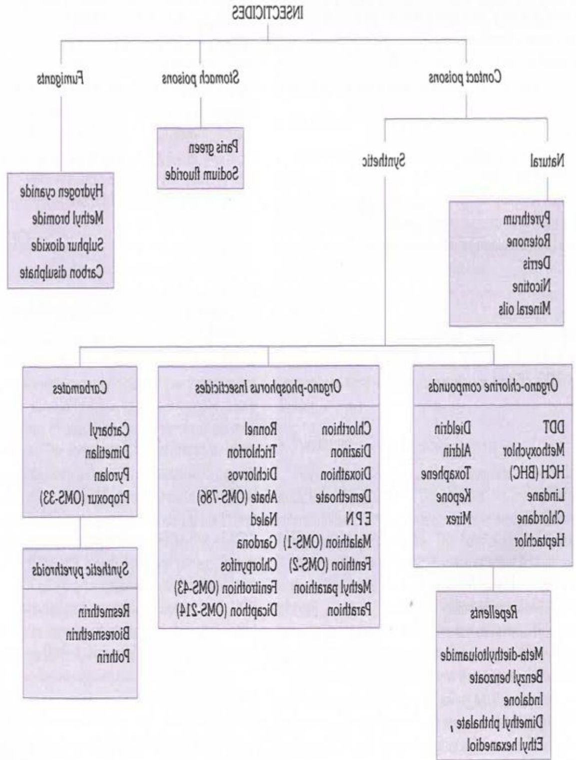


FIG. 17
Chemical control of arthropods of public health importance

NOTES

Chapter-16: Hospital waste management

__________ is the method of choice for most hazardous healthcare wastes

Category	Type of waste
Yellow	
Red	
White (Translucent)	

Blue	

Chapter-17: Disaster management

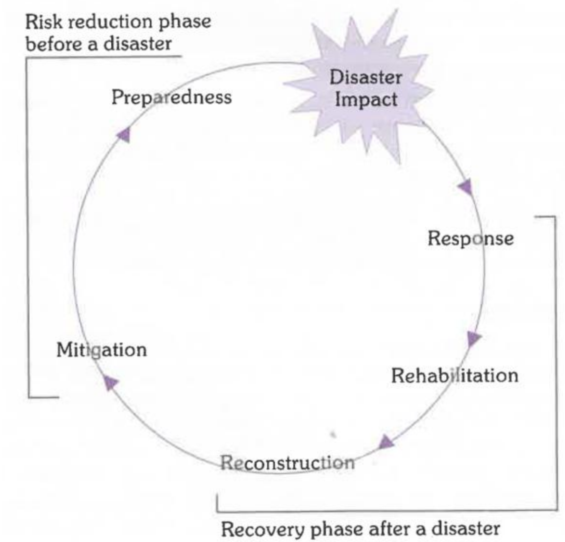


FIG. 1
Management sequence of a sudden-onset disaster

There are three fundamental aspects of disaster management:

- Disaster response
- Disaster preparedness
- Disaster mitigation

Triage

Triage consists of rapidly classifying the injured on the basis of severity of their injuries and the likelihood of their survival with prompt medical intervention.

- ______ indicates high priority treatment or transfer
- ________ signals medium priority

- _______ indicates ambulatory patients
- _________ for dead or moribund patients

Vaccination

- The WHO _________ **recommend typhoid and cholera vaccines** in routine use in endemic areas.
- Supplying safe drinking water and proper disposal of excreta continue to be the most practical and effective strategy.
- Significant increase in tetanus incidence have not occurred after natural disasters.
- Mass vaccination of population against tetanus is usually unnecessary.

Chapter 18: Occupational Health

Ergonomics:

The object of ergonomics is "to achieve the best mutual adjustment of ____ and his ______, for the improvement of human _______ and ___________".

Occupational hazards

Physical hazards:

- Direct effects of **heat exposure** are ______, heat ________, heat __________, heat ________, heat ______ and heat ______.
- Hazards associated with **cold work** are ________, ________, ____________, _______ foot and __________ foot.
- Hazards associated with **pressure** are ________ disease, ____ embolism and _______.
- The chronic effect of **light** on health include ______________.
- Exposure to **vibration** leads to ___________
- **Ultraviolet radiation** exposure leads to ______ (_______ flash)
- International commission of radiological protection has set the **maximum permissible level** of occupational exposure at ___ rem per year to the whole body.

Chemical hazards:

- Particles smaller than ___ microns are directly inhaled into lungs and are retained there. This fraction of dust is called respirable dust and mainly responsible for pneumoconiosis.

- Exposure to gases is a common hazard in industries. Gases are sometimes classified as:

    - **Simple gases** (e.g., _______, __________)
    - **Asphyxiating gases** (e.g. ____________, ________ gas, ______________, __________)
    - **Anaesthetic gases** (e.g., _________, ________, ______________).

Pneumoconiosis

Dust within the size range of _________**micron**, is a health hazard producing, after a variable period of exposure, a lung disease known as pneumoconiosis.

Dusts (Pneumoconiosis)			
Inorganic dusts		**Organic (vegetable) dusts**	
Coal dust		**Cane fibre**	
Silica		**Cotton dust**	
Asbestos		**Tobacco**	
Iron		**Hay or grain dust**	

Silicosis

- Silicosis is the major cause of ________________ and ________ among occupational diseases.
- It is caused by inhalation of dust containing ________ **or** _____ ________ (______).
- By exposure to industries, like _______ industry, _______ and _______ industry, _____ **blasting**, metal grinding, building and construction work, _____ mining, iron and steel industry.
- X-ray of chest shows ____________ **appearance.**
- The radiological evidence in the two conditions is so similar that one is apt to mistake a case of silicotic to be a case of ____________ of lungs.

- Silicosis was made a __________ **disease** under the factories act and mines act.

Anthracosis

Also known as coal miners pneumoconiosis has two general phases:

- The **first phase** is labelled as ________ **pneumoconiosis** associated with little ventilatory impairment.
- The **second phase** is characterized by ________________ _________ (_____) this causes severe respiratory disability and frequently result in premature death.

Byssinosis

- Due to inhalation of ______ **fibres** dust over long period. Symptoms are chronic cough and progressive dyspnoea.

Bagassosis

- Due to inhalation of __________ **or __________ dust.**
- It was first reported in _____________ **manufacturing firm.**
- Sugarcane fibre are now utilized in manufacturing of paper, cardboard and rayon.
- Due to __________ actinomycete, which ________________ __________ was suggested.
- Skiagram may show __________ **in lungs.**

Asbestosis

- Asbestos is of two types- __________ **or ________ variety and __________ type.**
- **90%** of world production of asbestos is of __________ **variety,** which is hydrated magnesium silicate while amphibole types contain little magnesium.
- The amphibole types occurs in different varieties, e.g., ________ (blue), _______ (brown), and ____________ (white)
- Asbestos enter by inhalation and causes pulmonary fibrosis, respiratory insufficiency, carcinoma bronchus, mesothelioma and cancer of GI tract.
- ____________ is a rare form of cancer of pleura and peritoneum, has shown to have a strong association with the __________ variety of asbestos.

- The sputum shows ________ **bodies** while X-ray shows ______ _______ **appearance.**

Farmer's lung

- Due to inhalation of _________ **or** ______ **dust.**
- _________________ is the main cause of farmer's lung.

Lead Poisoning

- Lead is **used widely in industries because** _____ boiling point, _______ with other metals easily to form alloys, easily _______ and ___________.
- All lead compounds are toxic - **lead _______, lead _____** and **lead _________ are the most dangerous**; lead ________ is the least toxic.
- Increase from _____ **to _____/100 ml blood** is associated with clinical symptoms.

Clinical picture:

- Clinical picture of **lead poisoning or** _________ is different in inorganic and organic lead exposures.
- Toxic effects of **inorganic lead exposure** are abdominal _____, obstinate constipation, ____ of appetite, ________on gums, _____________ of red cells, anaemia, wrist & foot drop.

- Toxic effect of **organic lead exposure** are insomnia, headache, mental confusion and delirium.

Diagnosis:

Test	Indication of Lead Exposure
Coproporphyrin in urine (CPU): Useful screening test	
Amino levulinic acid in urine (ALAU)	
Lead in blood	
Lead in urine	
Basophilic stippling of RBC	

Preventive Measures:

- **Substitution**: Lead compounds substituted by less toxic materials.
- Isolation
- Local exhaust ventilation
- Personal protection
- Good house-keeping
- **Periodic examination of workers:** All workers must be given periodical medical examination.
- **Personal hygiene:** Handwashing before eating is an important measure of personal hygiene.
- Health education

Management:

- ________________, like _________, is a chelating agent and works by promoting lead excretion in urine.

Occupational Cancer

1. Skin cancer

- ______________ was first to draw attention to **cancer of** _________ **in** ____________ **sweeps** in 1775.
- It was later discovered that ________, _________, **certain** _____ **and** ______ were major risk factors.
- Nearly _____ **%** of occupational cancers are skin cancer.
- **High risk occupations:** _____ workers, coke oven workers, ____ distillers, oil refiners, dye-stuff makers, ______ makers and in industries associated with the use of mineral oil, pitch, tar and related compounds.

2. Lung cancer

- Lung cancer is a hazard **in** _____ **industry, asbestos industry,** _______ and ____________ **work,** __________ **roasting plants** and in the **mining of radio-active substances (e.g.,** __________**).**

3. Cancer bladder

- First reported in ________ industry and later observed in the ________ industry.
- Caused by exposure to ____________, which are metabolized in the body and excreted in the urine.
- **High risk industries:** _________ and _______ industry, rubber, gas and the _______________ **industries.**

4. Leukaemia

- Caused by exposure to _________, ___________ rays and radio-active substances.

Protection for women workers

- Expectant mothers are given maternity's leave for ____ weeks
- Provision of _____ antenatal, natal and postnatal services
- The factories act (Section 66) prohibits night work between ___ p.m. and ___ a.m.;
- The Indian Mines Act (1923) prohibits work __________.
- Maternity Benefit Act provides creches in factories where more than _____ women workers are employed

Regarding protection of children, the Constitution of India declared: "No child below the age of _________ shall be employed to work in any factory or mine or engaged in any other hazardous employment.

The Factories Act, 1948

- The Act defines factory as an establishment **employing ____ or more workers where power is used an _____ or more workers where power is not used.**
- There is no distinction between perennial and seasonal factories.

Safety officer	
Canteen	
Creches	
Welfare officer	

Employment of young persons:

- Prohibits employment of children below the age of ____ yrs and declares persons between the ages ___ and __ to be adolescents.
- Adolescent employee is allowed to work only between __ am and ___ pm.

Hours of work:

- Act prescribed a maximum of ____ working hours per week, not exceeding ___ hours per day with rest for atleast ____ hour after ___ hrs of continuous work.
- For adolescent the hours of work have been reduced from ___ to ____ hrs per day.

- The total number of hours of work in a week including overtime shall not exceed ____.

Leave with wages:

- Every worker will be entitled to leave with wages after ____ months continuous service at the following rate;
  - Adult- one day for every ____ days of work
  - Children- one day for every ___ days of work
- The leave can be accumulated upto ____ days in case of adults and ____ days in case of children.

The Employees state insurance Act, 1948

- The scheme is run by contributions by employees and employers and grants from ______ and ______ Governments.
- The **employer** contributes ______ **per cent** of total wage bill; the **employee** contributes ______ **per cent** of wages.
- Employees getting daily wages below **Rs.**_____ are exempted from payment of contribution.
- The State Government's share of expenditure on medical care is ______ of total cost of medical care
- The ESI Corporation's share of expenditure on medical care is ______ of total cost of medical care.

BENEFITS TO EMPLOYEES

The Act has made provision for the following benefits to insured persons or, to other dependants as the case may be:

- ________ benefit
- _________ benefit
- __________ benefit
- ___________ benefit
- __________ benefit
- ________ expenses
- _____________ allowance.

Sickness benefit

- It consists of **periodical _____ payment to an insured person** in case of sickness, if his sickness is duly certified by an Insurance Medical Officer or Insurance Medical Practitioner.
- The benefit is payable for a **maximum period of** ____ days, in any continuous period of 365 days, the daily rate being about **_____%** of the average daily wages.
- In order to qualify the insured person is required to contribute for ____ days in a contribution period of ___ months.

Extended Sickness Benefit

In addition to ____ days of sickness benefit, insured persons suffering from certain long-term diseases are entitled to this benefit for maximum period of _____ years.

List of diseases for extended sickness benefit:

- **Infectious Diseases:** ____, ________, Chronic ________ and ________.
- **Neoplasms:** Malignant diseases.
- **Endocrine, Nutritional and Metabolic Disorders:** Diabetes mellitus with proliferative retinopathy/ Diabetic foot/Nephropathy.
- **Disorders of Nervous Systems:** Monoplegia, Hemiplegia, Paraplegia, Hemiparesis, Intracranial space occupying lesion, Spinal Cord compression, Parkinson's disease, Myasthenia Gravis/Neuromuscular dystrophy, Immature cataract with vision 6/60 or less, Detachment of retina, Glaucoma.
- **Diseases of Cardiovascular System:** CAD (Unstable angina, MI with ejection less than 45%)**,** CHF- Left, Right, Cardiac valvular diseases with failure/complications, Cardiomyopathies, Heart disease with surgical intervention along with complications.
- **Chest Diseases:** Bronchiectasis, ILD, COPD with CHF (Cor Pulmonale).
- **Diseases of the Digestive System:** Cirrhosis of liver with ascities/chronic active hepatitis.
- **Orthopaedic Diseases:** Dislocation of vertebra/prolapse of intervertebral disc, Non union or delayed union of fracture, Post Traumatic Surgical amputation of

lower extremity, Compound fracture with chronic osteomyelitis.

- **Psychosis:** Schizophrenia, Endogenous depression, Manic Depressive Psychosis (MDP), Dementia

- **Others:** More than 20% burns with infection/complication, Chronic renal failure, Reynaud's disease/Burger's disease.

Maternity benefit

- The duration of benefit is ___ weeks, for miscarriage ___ weeks and for sickness arising out of confinement is ____ days.

- Benefit is allowed at _____ wages

- The rate of confinement expense has been increased from Rs ______ to Rs ______ per confinement.

NOTES

Chapter-19: Genetics and Health

Autosomal dominant		Autosomal recessive	
HARM P³ BARS		CAT GAPS FM²	
H		C	
A		A	
R		T	
M		G	
P³		A	
B		P	
A		S	
N		F	
S		M²	

Recessive sex-linked traits		Dominant X-linked traits	
"He Didn't Come Home — Girls Ran Away"		VFB	
He		V	
D			
C		F	
H			
G		B	
R			
A			

Erythroblastosis foetalis

- If the foetus is **Rh-_________ and the mother Rh-_________**, certain consequences are likely.
- Some of the foetal red cells cross the placenta and enter the maternal circulation where they act as foreign antigen and the production of Rh antibodies.
- The **weak or albumin antibodies are small ____ gammaglobulins** which cross the placental barrier and pass back into the blood circulation of the foetus.
- When this happens, the RBC of the foetus are destroyed. If the damage is severe, the foetus is killed in utero and results in miscarriage; if the damage is less severe, the infant may be born with jaundice, anaemia and oedema.
- This symptom complex is known as Rh-haemolytic disease or erythroblastosis foetalis.

Sickle cell anaemia

- It is an _________________ disorder in which an abnormal haemoglobin leads to chronic haemolytic anaemia.
- The disorder is a classic example of _____ **mutation in DNA.**
- Individuals with ____ gene of this disease are clinically healthy, but their RBC look abnormal under the microscope.
- Persons with _____ genes (homozygous) of this disease suffer from acute anaemia and in most cases die before puberty.

- The rate of sickling is influenced, most importantly by ________________ in the individual red blood cells.

Thalassemia

- Is a hereditary disorder characterised by ________ **in synthesis of globin chain (alpha or beta).**
- Which leads to reduced haemoglobin synthesis and eventually produces a ________________ **anaemia** because of defective haemoglobinization of red cells.

- **Alpha thalassaemia is due to** ___________, causing reduced α-globin chain synthesis.
- **Beta thalassaemia** are usually caused by ____________ rather than large deletions.
- Signs of thalassaemia develop after ________ of age, because this is the time when haemoglobin synthesis switches from haemoglobin ____ to haemoglobin _____.
- Asian couples whose parents on both sides have the alpha thalassaemia trait are at risk of producing an infant with ________________
- Mediterranean people with both parents heterozygous for beta thalassaemia, are at risk of producing _________ child.

Haemophilia

- It is a ___________________ disorder.
- _______ incidence in all ethnic groups and geographical areas.
- While the disorder affects _______, it is carried by _______, who are only occasionally affected, usually _______.
- The disorder concerns the **absence, decrease or deficient function of blood coagulating factor**, leading to excessive, prolonged or delayed bleeding.
- In severe cases it most commonly occurs in the ______ **joints of the limbs.**
- Unless such bleeding is controlled promptly by infusion of the deficient factor, there is **progressive joint disease and muscle atrophy,** leading to serious physical, psychological and social handicaps.
- Until recently, the foremost **cause of death was ___________**, especially in the skull.
- In countries with highly developed haemophilia care programmes, **therapy with _________ derivatives** has reduced mortality.
- Survival in patients without these infections is almost the same as that of the general population.

Cystic fibrosis

- Cystic fibrosis is a genetic disease which affects the respiratory and gastrointestinal tracts and sweat glands.
- It is an ________________ trait.

Phenylketonuria

- Phenylketonuria is an ______________ disorder
- Resulting in a deficiency of the liver enzyme ____________ _____________ which converts phenylalanine to tyrosine.
- The name PKU is derived from the build-up of ____________ _______ **in the urine**, a characteristic of the disease**.**
- Phenylalanine accumulates in the blood and tissues and has a toxic effect on the brain leading to _________________
- Testing of bottle-fed infants should be done no sooner than ____ hours after the first successful formula feeding.
- Breast-fed babies, however, are tested at ___ **days**, since breast milk often has little protein content before the 5th day.

Health promotional measures	
EUGENICS	
The science which aims to improve the genetic endowment of human population.	
__________ eugenics	**_________ eugenics**
Aim is to reduce frequency of hereditary disease and disability in the community to as low as possible.	This is a more ambitious programme than negative eugenics. It seeks to improve the genetic composition of the population by encouraging the carriers of desirable genotypes to assume the burden of parenthood.

EUTHENICS
• Throughout the course of history, man has been adapting environment to his genes more than adapting his genes to the environment. • Studies with mentally retarded (mild) children indicated that exposure to environmental stimulation improved their IQ. • Thus the solution of improving the human race does not lie in contrasting heredity and environment, but rather in the mutual interaction of heredity and environmental factors.

Amniocentesis

This procedure can be used as early as ____ week of pregnancy when abortion of affected fetus is still feasible.

Amniocentesis is called for in the following circumstances if the parents are prepared to consider abortion:

- A mother aged ______ years or more
- Patient who have had a child with ________ syndrome or other chromosomal anomalies:
- Parents who are known to have chromosomal ______________
- Parents who have had a child with a _____________ detectable by amniocentesis. The most common are defects of neural tube, anencephaly and spina bifida which can be detected by an elevation of _______________ in the amniotic fluid.
- When determination of sex is warranted, given a ____________ of a sex linked genetic disease e.g., certain muscular dystrophies.

Inborn Error of Metabolism	Urine odor
Maple syrup urine disease	Maple syrup
Isovaleric acidemia Glutaric acidemia type 2	Sweaty feet
Tyrosinemia	Boiled cabbage
Multiple carboxylase deficiency	Tomcat urine
Phenylketonuria	Mousy

NOTES

Chapter 20: Mental health

"This chapter has never been asked before and doesn't hold exam-relevant content. Don't waste time here — you can skip it and focus on high-yield areas."

Chapter 21: Health information and basic medical statistics

Sources of Health Information

1) Census

- Taken at regular intervals, usually of ____ years.
- **Defined as** "the total process of _______, ________ and _________ demographic, economic and social data pertaining at a specified time or times, to all persons in a country or delimited territory".
- **Drawback** is that it needs considerable organization, a vast preparation so the full results are ___________ quickly.
- The last census was held in __________. It is ___________ **counting of population.**
- It is conducted at the end of the _____ quarter of the ____ year in each decade, the reason being, most people are usually _______ in their own homes during that period.
- The legal basis of the census is provided by the Census Act of 1948. The supreme officer who directs, guides and operates the census is the _____________________.
- Without census data, it is not possible to obtain quantified health, demographic and socio-economic indicators.

2) Registration of vital events

- If registration of vital events is _______ and ________, it can serve as a reliable source of health information.

- **Defined as** "legal registration, statistical recording and reporting of the occurrence of, and the collection, compilation, presentation, analysis and distribution of statistics pertaining to vital events, i.e., ________, _______, ________ deaths, ________, ________, _________, _________, recognitions, annulments and legal separations".

The Central Births and Deaths Registration Act, 1969

- The Act provides for compulsory registration of births and deaths throughout the country, and compilation of vital statistics in the States so as to ensure uniformity and comparability of data.

- The time limit for registering the event of births and that of deaths is ________. In case of default a late fee can be imposed.

Lay reporting

- __________ health workers are employed to record the _____ and ________ in countries that have made ____ progress in the development of a comprehensive vital registration system.

- It is defined as the collection of information, its use, and its transmission to other levels of the health system by non-professional health workers.

3. Sample Registration System (SRS)

- It is a __________ **system**, consisting of continuous enumeration of births and deaths at national and state levels by an __________ and an **independent survey** every ___ months by an ____________________

4. Notification of diseases

- At international level, the following diseases are notifiable to WHO in Geneva under the international health regulations, viz. _________, _________ and ____________

5. Hospital records

Main drawbacks of hospital data are:

a) **They constitute only tip of iceberg-** ______ cases may not attend hospitals; ___________ cases are always missed.

b) The _____________ may vary from hospital to hospital

c) Population served by a hospital (population at risk) __________ defined. There are no precise boundaries to the catchment area of a hospital. In effect, hospital statistics provide only the numerator, not the denominator.

A study of hospital data provides information on the following aspects:

a) ___________ sources of patients

b) __________ distribution of different diseases and duration of hospital stay

c) Distribution of ___________

d) Association between ___________ diseases

e) the period between ________ and __________ admission

f) the distribution of patients according to different social and biologic characteristics

g) the ______ of hospital care

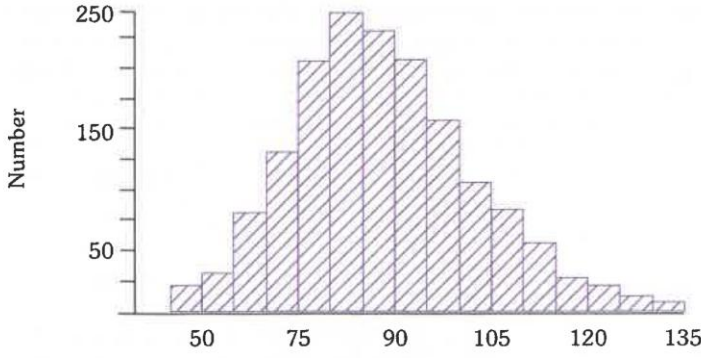
6. Record linkage

- The term **record linkage** is used to describe the process of bringing together records relating to _____________ (or to _____ ________), the records originating in different times or places.
- The term **medical record linkage** implies the assembly and maintenance for each individual in a population, of a file of the more important records relating to his health. Events commonly recorded are _______, _________, _______, ________ admission and discharge.

Charts and diagrams

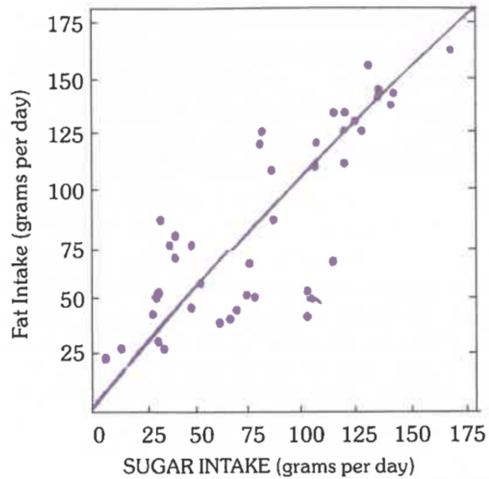
Histogram:

- It is a __________ diagram of _________ distribution.
- It consists of ________ of blocks.
- The _____________ are given along horizontal axis and ____________ along vertical axis.
- The area of each block or rectangle is proportional to _________.



Frequency distribution of diastolic blood pressure in females aged 45–64 years

Scatter diagram



Relation between average fat intake and sugar intake in 41 countries

Line diagram:

- Line diagrams are used to show the ________ of events with the passage of ______.

- It shows the **relationship between** ______________

- E.g., Fig. shows a positive correlation between the intakes of fat and sugar in the average diets of 41 countries.
- Populations with more income are known to consume more protein, fat and also sugar.
- If the **dots cluster round a straight line**, it shows _________ **relationship** of a linear nature.
- If there is **no such cluster,** there is probably ____ **relationship** between the variables.

Statistical averages

The Mean:

- To obtain the mean, the individual observations are first ______ together, and then _______ by the number of observations.
- The mean is calculated thus : the diastolic blood pressure of 10 individuals was 83, 75, 81 , 79, 71, 95, 75, 77, 84, 90. The total was 810. The mean is 810 divided by 10 which is 81.0

The Median:

- The median is an average of a different kind, which does not depend upon the total and number of items.
- To obtain the median, the data is first arranged in an ascending or descending order of magnitude, and then the value of the middle observation is located, which is called the median.

The Mode:

- The mode is the commonly occurring value in a distribution of data.
- It is the most frequent item or the most "fashionable" value in a series of observations.
- For example, the diastolic blood pressure of 20 individuals was : 85, 75, 81, 79, 71, 95, 75, 77, 75, 90, 71, 75, 79, 95, 75, 77, 84, 75, 81 , 75. The mode or the most frequently occurring value is 75.
- Therefore, mode is not often used in biological or medical statistics.

Measure of dispersion

The Range:

- The range is by far the __________ measure of dispersion.
- It is defined as the difference between the _________ and _______ figures in a given sample.
- For example, from the following record of diastolic blood pressure of 10 individuals - 83, 75, 81 , 79, 71, 90, 75, 95, 77, 94. It can be seen that the highest value was 95 and the lowest 71.
- The range is expressed as 71 to 95 or by the actual difference.

The Standard deviation:

- The standard deviation is the most _________ used measure of deviation.

- In simple terms, it is defined as "Root - Means - Square – Deviation.

Example : The diastolic blood pressure of 10 individuals was as follows : 83, 75, 81, 79, 71, 95, 75, 77, 84, 90. Calculate the standard deviation.

Answer

x	$(x-\bar{x})$	$(x-\bar{x})^2$
83	2	4
75	–6	36
81	0	–
79	–2	4
71	–10	100
95	14	196
75	–6	36
77	–4	16
84	3	9
90	9	81
$\bar{x} = 81 \quad \eta = 10$		Total = 482

$$\text{S.D.} = \sqrt{\frac{\Sigma(x-\bar{x})^2}{\eta-1}} = \sqrt{\frac{482}{10-1}} = \sqrt{\frac{482}{9}}$$

$$= \sqrt{53.55} = 7.31$$

Normal Distribution

- The standard normal curve is a _________, __________, perfectly ________________ curve, based on an _________ large number of observations.

- The total area of the curve is ___ ; its mean is _______; and its standard deviation is ______.

- The mean, median and mode all __________.

$$\text{Standard Error of Proportion} = \sqrt{\frac{pq}{n}}$$

where p = proportion of males; q = proportion of females and n = size of the sample. Substituting the values, we get :

$$\text{S.E. (Proportion)} = \sqrt{\frac{52 \times 48}{100}} = 5.0$$

Degree of freedom

$$\text{d.f.} = (c - 1)(r - 1)$$

c = Number of columns

r = Number of rows

In our table (cited above), there are two rows (attacked and not attacked) and two columns (vaccine A and vaccine B). It is a 2×2 contingency table. The degree of freedom will be :

$$\text{d.f.} = (c - 1)(r - 1) = (2 - 1)(2 - 1) = 1$$

Correlation And Regression

Meaning of Correlation

- In order to find out whether there is ________________ or not between two variables (we may call them x and y), we calculate what is known as Co-efficient of Correlation.
- Correlation is a statistical measure that describes the ________ and __________ of a relationship between two variables, but it does not imply causation or specify which variable is independent or dependent.
- The correlation coefficient, denoted by "r" measures the strength and direction of a linear relationship between two variables. It ranges from ___ (strong negative association) to __ (strong positive association), with ___ indicating no association between x and y.

- Correlation does not necessarily prove ___________.

Meaning of Regression

- If we wish to know in an individual case the value of one variable, knowing the value of the other, we calculate what is known as the regression coefficient of one measurement to the other.

- It is customary to denote the _____________ variate by x and the _____________ variate by y.

NOTES

Chapter 23: Health Planning and Management

Health Planning

Objective (Point):

- It is _______ - it is either ________ or _____________
- It is a planned __________ of all activities.
- Objectives are concerned ________ with the problem itself.
- It is constrained by _______.

Target:

- Refers to a ________ activity, such as the number of blood films collected or vasectomies done.
- It permits the concept of degree of ____________.
- Targets are thus concerned with the factors involved in a problem.
- It is constrained by _______.

Goal:

- It is defined as the ultimate desired state towards which objectives and resources are directed.
- Goals are _____ constrained by time or existing resources, nor are they necessarily attainable.

Plan:

- It consist of five major elements;
  __________, __________, ______________, ___________ and __________.

Steps of Planning Cycle		
Mnemonic: **"Great Teachers Form Plans, Defining Solutions So Best Ideas Get Proper Implementation, Orientation, Execution, And Evaluation."**		
1.	**Great**	
2.	**Teachers**	
3.	**Form**	
4.	**Plans**	
5.	**Defining**	
6.	**Solutions**	
7.	**So**	
8.	**Best**	
9.	**Ideas**	
10.	**Get**	
11.	**Proper**	

12.	**Implementation**	
13.	**Orientation**	
14.	**Execution**	
15.	**And Evaluation**	

Health Management Methods and Techniques

Methods based on behavioral sciences	Quantitative methods
1. 2.	1. 2. 3. 4.

3. 4. 5.	5. 6. 7. 8. 9. 10.

Management by objectives (MBO)

- Objectives are set forth for __________ units and subunits, each of which prepares its _____ plan of action- usually on a _______ term basis.

- This helps in achieving the results more effectively and smoothly

Quantitative methods

Cost-Benefit Analysis

- The ________ **benefit of programme is compared with the** ______ of that programme.
- The **benefits are expressed in** ___________ **terms** to determine whether a given programme is economically sound, and to select the best out of several alternate programmes.
- **Main drawback:** Benefits in the health field _______ **always be expressed in** __________ **terms.**

Cost-Effective Analysis

- It is _______ **promising tool** than cost-benefit analysis.
- **Expressed in terms of ________ achieved**, e.g., number of lives saved or the number of days free from disease.

Cost-Accounting

- It provides **basic data on _____ structure** of any programme.
- **Purposes in health services:** Cost control, Planning and allocation of people and financial resource and pricing of cost reimbursement.

Input-Output Analysis

- “_________" refers to all health service activities which consume resources (manpower, money, materials and time)
- "_________" refers to such useful outcomes as cases treated, lives saved, or inoculations performed.

Network Analysis

- It is a ________ plan of all events and activities to be completed to reach an _____ objective.

Types

(A) PERT (Programme Evaluation and Review Technique)

- It makes possible more detailed ________ and more comprehensive supervision.
- Essence is to construct an _______ Diagram and represents the ________ sequence in which events must take place.
- By this u can calculate the _____ by which each activity must be completed, and to identify those activities that are ________.
- It aids in _________, __________ and _________ the project.

(B) Critical Path Method (CPM):

- The ________ path of the network is called "critical path".
- If any activity along the critical path is ________, the entire project will be ___________.

Planning-Programming- Budgeting System (PPBS)

- ______________ **Approach',** i.e., all budgets start at _____ and no one gets any budget that he cannot specifically justify on a year-to-year basis.

Work sampling

- It helps in _____________ the methods of performing jobs and determining the manpower needs in any organization.

NATIONAL HEALTH POLICY-2017

A. Health status and programme impact

Life expectancy and healthy life

- _________ life expectancy at birth from _____ to ____ by 2025.
- Establish regular tracking of **DALY index** as a measure of disease burden **by 2022.**
- __________ of **TFR to _____** at national and sub-national level **by 2025.**

Mortality by age and/or cause

- ______ **under-five mortalities to __ by 2025** and **MMR to 100 by 2020.**
- Reduce infant mortality rate **to 28 by 2019.**
- _________ neonatal mortality to _____ and stillbirth rate to a "_____ digit" by 2025.

Reduction of disease prevalence/ incidence

- Achieve global target of 2020, termed as target of 90:90:90, for HIV/AIDS i.e.

- Elimination status of leprosy by 2018, Kala-azar by 2017 and lymphatic filariasis in endemic pockets by 2017.

- Cure rate of ______ in new sputum positive patients for **TB** and **_______ incidence** of **new cases,** to reach elimination status **by 2025.**

- ________ the **prevalence of blindness to __________ by 2025** and disease burden by one third from current levels.

- ________ premature mortality from CVD, cancer, diabetes or chronic respiratory diseases **by _____ by 2025.**

B. Health systems performance

Coverage of health services

- ________ utilisation of public health facilities by **_____ from current levels by 2025.**

- Antenatal care coverage to be sustained above ____% and skilled attendance at birth above ____% by 2025.

- More than ____% of newborn are fully immunized by one year of age by 2025.

- Meet need of family planning **above ______ at national and sub-national level by 2025.**

- **_____ of known hypertensive** and **diabetic** individuals at household level maintain **__________ disease status' by 2025.**

Cross-sectoral goals related to health

- Reduction in **prevalence of current tobacco** uses by ____% by 2020 and **____% by 2025.**
- **Reduction of ____% in prevalence** of stunting of **under- five children by 2025**.
- Access to safe water and sanitation to all by 2020 (Swachh Bharat Mission).
- Reduction of occupational injury by half from current levels of 334 per lakh agricultural workers by 2020.

C. Health systems strengthening

Health finance

- Increase health expenditure by the Government as a percentage of GDP from the existing **_____% to ____% by 2025.**
- Increase state sector health spending to > ___% of their budget by 2020.
- Decrease in proportion of households facing catastrophic health expenditure from the current levels **by ____% by 2025.**

Health infrastructure and human resource

- Ensure availability of paramedics and doctors as per Indian Public Health

Standard (IPHS) norm in high priority districts by 2020.

- Increase community health volunteers to population ratio as per IPHS norm, in high priority districts by 2025.
- Establish primary and secondary care facility as per norms in high priority districts (population as well as time to reach norms) by 2025.

Health management information

- Ensure district-level electronic database of information on health system components by 2020.
- Strengthen the health surveillance system and establish registries for diseases of public health importance by 2020.
- Establish federated integrated health information architecture, health information exchanges and national health information network by 2025.

NITI AAYOG

- Government of India has established NITI Aayog (**National Institution for ___________ India**) to **replace ___________ Commission** on 1st January 2015.
- It will seek to provide a **critical _________ and _______ input into the development process.**

- NITI Aayog will emerge as a **"__________"** that will provide Governments at the ________ **and** _______ **levels** with relevant strategic and technical advice across the spectrum of key elements of policy.

- In addition, the NITI Aayog will monitor and evaluate the implementation of programmes, and focus on ___________ **upgradation** and _________ **building.**

Directorate General of Health Service

Functions:

- All the _______________ in the country and international air ports are directly controlled by the Directorate General of Health Services.

- **Control of _______ standards:** The Drugs Control Organization is part of the Directorate General of Health Services, and is headed by the Drugs Controller. Its primary function is to lay down and enforce standards and control the manufacture and distribution of drugs through both ______ and _______ Government Officers.

- ________ store depots
- ___________ training to different categories of health personnel
- Medical __________
- Medical __________
- _________ govt. health scheme
- __________ health programmes

Panchayati raj

The Panchayati Raj is a 3-tier structure of rural local self-government in India, linking the village to the district. The three institutions are:

(1) ___________ - at the village level;
(2) ________________ - at the block level; and
(3) _______________ - at the district level.

The Panchayati raj at the village level consists of:

-
-
-

Elements of evaluation

- R
- A
- A
- A
- E
- E
- I

NOTES

Chapter-22: Communication for Health Education

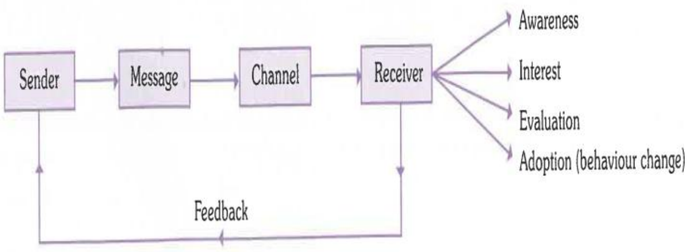


FIG. 1

Communication process

Health Communication

Suggests __________ **and** __________ communication of knowledge.

- One way communication (__________ **method**)
- Two-way communication (___________ **method**)

Functions of health communication:	
• I • E • M • P	• C • R • H • O

___________**:** A conscious attempt by one individual to change or influence the general beliefs, understanding, values and behaviour of another individual or group of individuals in some desired way.

Health Education	**Propaganda or publicity**
	Knowledge instilled in the minds of people.
	Prevents or discourages thinking by ready-made slogans.
	Arouses and stimulates primitive desires.
	Develops reflexive behaviour, aims at impulsive actions
	Appeals to emotion
	Develops a standard pattern of attitudes and behaviours according to the mould used.
	Knowledge is spoon fed and passively received.
	The process is information centered - no change of attitude or behavior designed.

	Models Of Health Education
	• This model is primarily interested in the recognition and treatment of disease (curing) and technological advances to facilitate the process. • It is concerned with disease or opposed to illness. • In this model social, cultural and psychological factors were thought to be of little or no importance. • The medical model did not bridge the gap between knowledge and behaviour.

	Adoption Model: **1. Awareness:** Interest **2. Motivation:** Evaluation and Decision-making **3. Action:** Adoption or acceptance
	An effective health education model is based on precise knowledge of human ecology and understanding of the interaction between the cultural, biological, physical and social environmental factors.

Principles of Health Education	
• C • I • P • M • C • R	• L • K • S • G • F • L

Methods in health communication		
• Personal contact • Home visits • Personal letters	**Discussion methods** - - - - - - -	• Television • Radio • News paper • Printed material • Direct mailing • Posters • Health museums and exhibitions • Folk methods • Internet

NOTES

Chapter-25: International Health

World Health Organization

- It is a specialized, non-political health agency of united nations
- Headquarters at ________, Switzerland.
- **Objective:** Attainment by all people of the highest level of health.

Work of WHO:

- Prevention and control of specific disease
- **Development of comprehensive health services:** Appropriate technology for health (ATH) programme launched by WHO.
- Family health
- **Environmental health:** Programmes launched such as "WHO Environmental health ________ programme" and "WHO environmental health __________ programme."
- **Health statistics:** The data is published in

    1.
    2.
    3.

- Biomedical Research
- Health literature and information
- Cooperation with other organizations

UNICEF

- The **headquarter** is at united nations, ___________
- UNICEF is currently interested in ' _________________________________ **approach** to meet the needs of children as an integral part of the country's development effort.
- The UNICEF is promoting a campaign known as **GOBI campaign** to encourage 4 strategies for a "child health revolution":
    - **G**
    - **O**
    - **B**
    -

UNDP

- United nations ___________ programme
- **Objective:** Help ______ nations develop their human and natural resources more fully.

UNFPA

- United nations fund for __________ activities.
- Designed to develop national capability for the manufacture of contraceptives, to develop population education programmes, to undertake organized sector projects, to strengthen programme management as well as to improve output of grass-root level health workers and introduction of innovative approaches to family planning and MCH care.

FAO

The **Food and agriculture organization (FAO)** with headquarters in __________.

ILO

The **International Labour organization (ILO)** with headquarters in _______________

SIDA

- The Swedish International Development Agency is assisting the ______________________ Programme since 1979.
- The SIDA assistance is usually spent on procurement of supplies like __________________________________ _

____

DANIDA

The Government of Denmark is providing assistance for the development of services under ______________________ Programme since 1978.

Rockefeller Foundation

Control of ____________ **disease** and Establishment of all India institute of __________________ **health**

Ford Foundation

The Ford Foundation has helped India in the following projects :

1) **Orientation training centres :** The centres provide training courses in public health for medical and paramedical personnel from all over India.

2) ________________ **projects :** Aimed at solving basic problems in ____________________, e.g., designing and construction of hand-flushed acceptable ______________ in rural areas

3) **Pilot project in rural health services:** Developed coordinated rural health services for a population of 100,000 to serve as a model for health administrators.

4) **Establishment of __________:** Ford Foundation has supported the establishment of the National Institute of Health Administration and Education at Delhi.

5) **Calcutta ______________________ scheme :** The Foundation has helped in the preparation of a master plan for water supply, sewerage and drainage.

6) __________________ **programme :** The Foundation is supporting research in reproductive biology and in the family planning fellowship programmes.

CARE

- Cooperative for assistance and relief everywhere
- Primary objective **provide food for children in the age group of ________ years.**
- **It is helping in the following projects :**
    - Integrated Nutrition and Health Project
    - Better Health and Nutrition Project
    - Anaemia Control Project
    - Improving Women's Health Project
    - Improved Health Care for Adolescent Girl's Project
    - Child Survival Project
    - Improving Women's Reproductive Health and Family Spacing Project
    - Konkan Integrated Development Project etc.